

THE ENERGY BALANCE PROJECT

Mathematics, Implementation and Evidence

Part II

An integrated explanatory edition

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From an exact physical account to a scientific claim that can be checked.

Author-review manuscript. New Gaussian exposition and explicitly scoped historical readings. No new scientific experiment is reported by this revision.

Author, rights and review status

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All new numerical examples are illustrative mathematics on explicitly supplied states, not measured data, operating recommendations, registered parameters or scientific trajectories. Historical readings retain their own experimental status; no new study is executed by reproducing them.

Why these two books now belong together

Part I introduced a question: can an account of physical action describe changes in declared functional deviation without confusing those changes with prices, ownership or a promise of recovery? It developed the objects needed to ask that question and finished with exact finite EBU. This volume starts where that introduction ends. Its task is to show how an equation becomes a precise claim, how software can preserve or corrupt that claim, and how evidence can support only some of the conclusions a reader might be tempted to draw.

The former Parts II and III separated mathematical structure from code and evidence. That separation made sense as a classification of subjects, but it also interrupted the argument. Readers encountered a guarantee in one book and the implementation assumption on which it depended in another. The combined volume repairs that connection. Mathematical definitions come first; group and path accounting precede any discussion of persistent capacity; physical permission precedes execution order; tests precede interpretation. The historical experiments then become a worked lesson in asking questions, not a collection of success claims placed after the mathematics.

This is not a short digest of the two originals. The new explanatory spine is joined to complete, identified historical readings at the point where they help the argument. In particular, the detailed dissipation and numerical proofs, preservation and route reasoning, software safeguards, experimental sequence, flat outcomes and reproduction discipline remain available. Old assumptions are kept visible because concealing them would make the record less intelligible, not more modern.

One continuous course, two kinds of reading

Main chapters continue Part I's numbering: 33 through 62. These chapters use the prospective local Gaussian domain unless a passage declares another model. References and scales are fixed; the first physical domain is lossless and conserves one homogeneous scalar. A controller is not smuggled into the definition of the potential.

Historical readings are marked H44, H45 and so on, after the current chapter that contains them. Each page carries a historical banner and identifies its original part and PDF page. Its body, equations and internal numbering are preserved. A reference to original Equation 38.4 inside a reading is therefore an original-source reference, not a reference to the new Chapter 38. New running page numbers belong to this integrated volume. Original cross-links are not copied into destinations where they would become misleading.

Read a historical proof with a pencil beside its assumptions. Some steps, such as endpoint cancellation, apply to many potential families. Other steps require a specified response law, a regenerative source classification, a particular numerical certificate or a frozen experiment. The surrounding current text explains those boundaries. “Historical” does not mean false. It means that the object and authority of the claim remain identifiable.

The supplied original PDFs remain unchanged. Their editable manuscripts were not recovered. New chapters are editable source; historical readings retain their original typesetting with smaller displayed chapter titles and new margin furniture. The source map records what was reconstructed, what was retained as a reading, and what remains only in the original. In particular, the old threshold laboratory is replaced by a Gaussian worked laboratory; its numerical outcomes are not converted into Gaussian findings.

The dependency route

Chapters 33–39 establish the domain, units, geometry, physical graph, local support, force and finite quote. The distinction between force and flux is explained here, including why Onsager’s modelling pattern was useful and why borrowing it is not a thermodynamic proof about an economy.

Chapters 40–43 compose actions without double counting. A group value is calculated before a child attribution; attribution is defined before an owner account; external changes are separated before an invariant is written. This order prevents a correct sum identity from appearing to prove an unannounced settlement rule.

Chapters 44–46 examine the additional assumptions needed for dynamics, permission, routes and long-run reasoning, then work through a detailed Gaussian laboratory. The laboratory is fixed-state arithmetic, not an executed scientific trajectory or a chosen campaign configuration.

Chapters 47–56 translate those distinctions into software responsibilities: schemas, permission, chronology, menus, evaluation, accounts, randomness, tests and provenance. Chapters 57–60 recover the historical engine and evidence in enough detail to understand both informative results and limits. Chapters 61–62 end with what a future study could establish and an audit that makes missing authority visible.

Four statements that must never collapse into one

A conditional identity is a mathematical result. A faithful implementation is a software achievement. A registered design is a plan. An observed scientific outcome requires an actual authorized run and its preserved evidence. Passing from one to the next needs additional work; a book’s chapter order is not that authorization.

How this volume changes the series

The current series has eight parts, not nine. Former Parts II and III become this Part II. Former Parts IV–IX shift down one number to Parts III–VIII. Part I keeps its role and chapter sequence. Old editions, citations, commit records and historical chapter names retain their original identifiers; renumbering a forward plan is not rewriting a scientific source.

Only the first two parts are being delivered in this revision. The later six remain a plan whose subjects depend on the claims they need, not on an assumption that every earlier experiment will succeed. The closing series map gives both the new numbering and its old equivalent.

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From a physical story to a mathematical claim

A store has more water than its declared reference and a clinic has less. Moving water from one to the other seems helpful. Mathematics begins by asking what that sentence has left unspecified. How much water? Which reference? Which route? What else changes? Does the action remain helpful when it is large? Who is permitted to perform it? The value of a formal model is not that it makes these questions disappear. It gives each question an object precise enough to inspect.

33.1 The declared world

Let there be a finite number n of accounting cells, with state

$$x = (x_1, \dots, x_n)^T \in \mathcal{X}_M, \quad \mathcal{X}_M = \{x \in \mathbb{R}^n : x_i \geq 0, \sum_i x_i = M\}. \quad (33.1)$$

Every x_i measures the same homogeneous resource. The superscript T turns a written row into a column. The set $\mathcal{X}_M$ is the nonnegative fixed-mass simplex: all ways of distributing the same total M among the cells. For two cells it is a line segment, not the whole plane. For three it is a triangle embedded in three-dimensional space.

The programme declares a reference $x^* \in \mathcal{X}_M$ and scales $\sigma_i > 0$. Membership of the reference in the same simplex matters. If two cells contain twenty units in total, references requiring twelve units in each cannot be reached by closed internal transfer. That mismatch is not an actor failure. It is a different modelling problem.

Compatibility is necessary, not sufficient, for reachability. A graph with no route to the clinic cannot deliver water there. A menu containing only four-unit transfers may be unable to remove a one-unit discrepancy exactly. The geometry says which states are allowed by total mass; the action system says which transitions are available.

33.2 Definition, proposition and evidence

We define a potential and derive its consequences. A definition is a choice about the model, not a discovery that nature uses that formula. A proposition may then be true for every state in the declared domain. A worked example can help a reader follow the proof but cannot replace it. An experiment would examine a specified transition process under registered conditions.

The programme document [3] makes this distinction explicit. The active model is prospective. Its exact quadratic identities can be proved now. Whether a random-affordable actor process produces useful recovery, distribution or long-run behavior is a different question.

Standing assumptions for active examples

One homogeneous conserved scalar; a finite number of cells; nonnegative stocks; fixed compatible references; fixed positive scales; lossless internal actions; additive declared increments. Unless a section explicitly says otherwise, separate process burden is zero. No restoring dynamics or optimizing actor is hidden in these assumptions.

33.3 What the state leaves outside

A simple water stock model does not automatically represent water quality, pumping energy, soil damage or a patient's health. These omissions limit the claim. They cannot be repaired by saying the scalar potential is holistic. If quality is consequential, it needs a represented coordinate or an explicitly scoped limitation. If pumping consumes electricity, a broader physical account must link that transformation to an energy carrier.

Nor does the model identify a reference with a moral optimum. A reference may later be informed by functional evidence and public choices, but the proof that follows only knows its declared number. Keeping this boundary visible lets mathematical work proceed without pretending that calibration or governance has already been solved.

33.4 A useful proof-reading habit

Before accepting a theorem, write its domain in words. When we later prove that V is bounded, the reason will be finite mass and fixed scales, not an EBU controller. When we prove that a sequence telescopes, the reason will be consecutive endpoint cancellation, not benevolent actors. When we prove

that path contributions sum to the group value, the reason will be linearity and the fundamental theorem of calculus, not a fair ownership rule.

This habit prevents a correct sentence from acquiring an incorrect sequel. “The accounts close” does not imply “the world recovers.” “The action has positive value” does not imply “the action is physically possible.” Each sequel needs its own premises.

Exercises and discussion

For a three-cell world with mass twelve, decide whether $(4, 4, 4)$, $(5, 5, 5)$ and $(12, 0, 0)$ are compatible references. The first and third are; the second is not. Compatibility of the third still says nothing about whether the graph can move all stock to cell one.

Explain why the phrase “Gaussian equilibrium” is potentially misleading. The Gaussian construction defines reference geometry. To call the reference an equilibrium of a process, one must specify that process and show that it leaves the reference unchanged. To call it attracting requires more still.

Chapter recap

A model needs a domain, a state, a reference, a transformation and a claim. The rest of this volume makes those objects explicit. Its mathematical results are conditional statements about the declared objects, not a shortcut around physical measurement or scientific testing.

Units, vectors and conservation

Many apparent disagreements about EBU are disagreements about units. One author writes q for a rate, another for a finite quantity; both use the same symbol in a correct-looking equation. The error appears only when the duration differs from one. We will therefore identify quantity before performing algebra.

34.1 A dimensional dictionary

Write X for the unit of physical stock and T for time. In the active Gaussian normalization, V is dimensionless, as are its finite differences. Calling that numerical accounting unit EBU does not make it a joule, a litre or a currency unit. A later calibrated conversion must be declared rather than inferred from the name.

Object	Unit and meaning
x_i, x_i^*, σ_i, q	X : stock, reference, scale, finite withdrawal
j	X/T : a rate
$\mu_i = \partial V / \partial x_i$	$1/X$: potential change per unit stock
$H_{ii} = 1/\sigma_i^2$	$1/X^2$: curvature
$q\mu_i, q^2 H_{ii}$	dimensionless finite-account terms

If a device operates at constant rate j for duration Δt , its quantity is $q = \Delta t j$. Thus $\delta = Sq$ and $\delta = \Delta t S j$ are equivalent descriptions, not two steps to apply successively. The historical foundation uses rate notation [4]; this edition uses finite quantities in its active examples.

34.2 Reading a matrix multiplication

For a lossless chain $1 \rightarrow 2 \rightarrow 3$, let

$$S = \begin{pmatrix} -1 & 0 \\ 1 & -1 \\ 0 & 1 \end{pmatrix}, \quad q = \begin{pmatrix} a \\ b \end{pmatrix}. \quad (34.1)$$

Then

$$Sq = \begin{pmatrix} -a \\ a - b \\ b \end{pmatrix}. \quad (34.2)$$

The first cell loses a , the middle receives a and sends b , and the last receives b . The middle coordinate is not a transport label that may be omitted: it is a physical stock with its own predecessor and successor.

Every column of S sums to zero. If $\mathbf{1}$ is the vector of ones, $\mathbf{1}^T S = 0$, so

$$\mathbf{1}^T(x + Sq) = \mathbf{1}^T x. \quad (34.3)$$

This proves conservation for the declared lossless increments. It does not prove nonnegativity: a request $a > x_1$ still conserves the algebraic total while making the source negative. Conservation and admissibility must be checked separately.

34.3 Loss is a physical account, not a vanished number

For a lossy transfer, a column may have -1 at the source and η at the destination. Its sum is $-(1 - \eta)$. To close a broader physical account, the missing quantity must enter a declared loss carrier or cross an explicit boundary. For instance an added coordinate ℓ can receive $(1 - \eta)q$. That is a conservation declaration, not yet a valuation of ℓ or permission to charge a second process cost.

The active first domain sets $\eta = 1$. This does not dismiss loss. It avoids claiming that an unresolved loss-aware runtime and valuation convention has already been accepted. Chapter 60 returns to the old Definition 6.4 wording: adding a loss coordinate changes what can honestly be said about a carrier being absent from state.

34.4 Changing the physical unit

Suppose litres are replaced by millilitres. Multiply all stocks, references, scales and finite quantities by $c = 1000$. Each standardized deviation remains unchanged:

$$\frac{cx_i - cx_i^*}{c\sigma_i} = \frac{x_i - x_i^*}{\sigma_i}. \quad (34.4)$$

Consequently V and exact EBU are unchanged. The marginal becomes μ_i/c , while q becomes cq ; their product is unchanged. This is a useful dimensional check on an implementation specification.

Changing σ while keeping stocks fixed is not a unit conversion. It changes the model's relative sensitivity. Treating those operations as the same would hide a substantive calibration decision inside a formatting step.

Worked checks

A rate of six units per hour over a quarter hour transfers $3/2$ units. Over two hours it transfers twelve. A routine that uses six in both cases has silently assumed a unit-duration interval.

In the chain matrix, take $a = 3$, $b = 2$ and predecessor $(5, 1, 0)$. The declared endpoint is $(2, 2, 2)$, with total six before and after. Whether the middle cell may send two while initially holding one depends on timing: one may allow a declared inflow-funded simultaneous process, or require outgoing commitments to be covered by frozen stock. The matrix alone cannot settle that physical convention.

34.5 Why the transpose appears in the field equation

The transpose is not decorative notation. It makes the dimensions of an operation visible. If there are n cells and m available action directions, S has n rows and m columns. A quantity vector q has m entries, so Sq has n entries: one stock change for each cell. The marginal μ has n entries. Its transpose is a row, and $\mu^T Sq$ is a single number. That number is the tangent potential change produced by the complete quantity vector.

Associativity lets us calculate the same scalar in two stages. We can first form the physical change Sq and then value its direction with μ^T . Or we can first form the row $\mu^T S$, one marginal change per action direction, and then multiply by the quantities. Since $\mu^T S = (S^T \mu)^T$, defining $f = -S^T \mu$ gives

$$-\mu^T Sq = f^T q. \quad (34.5)$$

This is the whole sign convention in one line. A positive entry of f means that a small positive quantity in that declared direction reduces the potential. It does not authorize an action or select its quantity.

For the chain matrix earlier in the chapter, let $\mu = (2, 0, -2)^T$. Then $S^T \mu = (-2, -2)^T$, so both edge forces are two. For $q = (3, 2)^T$, the tangent benefit is ten. Directly, $Sq = (-3, 1, 2)^T$, and $-\mu^T Sq = -(-6 - 4) = 10$. This checks the matrix arithmetic, not the exact finite value. The latter must still subtract the quadratic curvature contribution.

34.6 Addition of vectors does not establish simultaneous permission

Suppose a proposed group has three children. Writing their increments as columns makes their sum easy to inspect. A coordinate that is positive in one child and negative in another may cancel in the aggregate. Algebra records that cancellation exactly. Physical interpretation requires more. Did the middle tank hold the resource long enough to make the outgoing delivery? Was an intermediate conversion required? Could the two devices operate concurrently? These questions are not answered by a zero in the aggregate vector.

The distinction is similar to adding deposits and withdrawals in a stock record. A zero net daily change does not prove that the stock was never empty during the day. A constant-rate simultaneous model can make a specific path declaration; an endpoint-only model leaves intermediate behavior unspecified. A chronological model needs actual ordered states. All three can share endpoints while representing different physical claims.

34.7 Weighted length and the quadratic form

The expression $v^T H v$ looks more complicated than it is. With diagonal H , first multiply each entry v_i by $1/\sigma_i^2$, then multiply again by v_i and sum:

$$v^T H v = \sum_i \frac{v_i^2}{\sigma_i^2}. \quad (34.6)$$

It is a squared length measured in the declared scales. Its value cannot be negative, and it vanishes only when every entry vanishes. Reversing a vector leaves this squared length unchanged, although it reverses the linear directional term $\mu^T v$.

This difference between the linear and quadratic terms explains a common finite-action surprise. A small movement toward the reference can be helpful, but extending the same movement far enough overshoots. The linear part grows with quantity; the curvature term grows with its square. A longer movement is therefore not just several copies of the original marginal improvement at an unchanged state.

For the chain example with equal scales two and increment $(-3, 1, 2)$, the squared weighted length is $(9 + 1 + 4)/4 = 7/2$. Its half is $7/4$. If the marginals above came from a compatible Gaussian declaration, the exact finite field benefit would be $10 - 7/4 = 33/4$. For instance, state $(18, 10, 2)$ and reference $(10, 10, 10)$ provide precisely those marginals. The endpoint is $(15, 11, 4)$, with potential $31/4$ rather than the starting sixteen. Their difference is again $33/4$.

34.8 Coordinate changes and model changes

Reordering the list of cells changes the written vector, but not the physical account if every associated object is reordered consistently. The state, reference, scales, incidence rows and ownership identifiers must all follow the same permutation. Reordering just the state can turn a formally well-shaped matrix multiplication into the wrong physical calculation without producing a syntax error.

Changing litres to millilitres is another consistent coordinate change, already checked above. Replacing a scale because one cell is judged more important is not. Nor is combining two distinct carriers into one scalar because they share a convenient numeric range. The first changes units; the second changes valuation; the third changes the represented world. Software needs identities that make these differences visible.

34.9 Practice: follow one symbol through every layer

Choose one edge quantity q_e . Identify its stock unit, its incidence column, the cells it changes, the constraints it consumes, and the corresponding force entry. Then identify the finite curvature term in which it appears alone and any cross terms in which it meets another quantity. Finally ask whether a named owner exists. The matrix will not supply that final answer.

This exercise is worth doing before proceeding to proofs. Many later errors are not failures of advanced calculus. They are failures to carry one symbol's meaning consistently through a chain of otherwise correct operations. With this dimensional foundation, the next chapter can introduce the potential without asking the reader to trust opaque notation.

Chapter recap

Vectors collect quantities; matrices express declared transformations. Units and column sums expose important errors, but they do not certify the whole physical process. Conservation, nonnegativity, timing and valuation each need their own check.

Gaussian reference geometry

The word Gaussian can suggest random noise, a bell-shaped histogram or a statistical fit. Here its primary role is geometric. We declare a reference and a positive scale at each cell, measure deviation in units of that scale, square it and add. No observations have to be normally distributed for that mathematical definition to make sense.

35.1 The potential and its matrix form

Define

$$V(x) = \frac{1}{2} \sum_{i=1}^n \frac{(x_i - x_i^*)^2}{\sigma_i^2} = \frac{1}{2} (x - x^*)^T H (x - x^*), \quad H = \text{diag}(\sigma_i^{-2}). \quad (35.1)$$

The factor $1/2$ simplifies derivatives. Each diagonal entry of H is positive. Thus for a nonzero vector v , $v^T H v = \sum_i v_i^2 / \sigma_i^2 > 0$. This is positive definiteness, and it gives $V \geq 0$ with equality exactly at $x = x^*$.

Because the reference is assumed compatible with the physical domain, it is also the unique minimizer on that domain. A minimum of a function is not yet an attracting state of a process. It is a geometric fact available before an actor law has been specified.

35.2 Deriving the marginal and curvature

Differentiate one summand. The derivative of $(x_i - x_i^*)^2$ is $2(x_i - x_i^*)$, cancelling the factor $1/2$. Hence

$$\mu_i(x_i) = \frac{x_i - x_i^*}{\sigma_i^2}, \quad \mu(x) = H(x - x^*), \quad \nabla^2 V = H. \quad (35.2)$$

The marginal is negative below reference and positive above it. There is no flat healthy band. A stock very close to its reference has a small, usually nonzero marginal. This is a substantive departure from the old hinge family, not a cosmetic rewriting of its coefficients.

Let $h_{\min} = \min_i \sigma_i^{-2}$ and $h_{\max} = \max_i \sigma_i^{-2}$. Then

$$\frac{h_{\min}}{2} \|x - x^*\|_2^2 \leq V(x) \leq \frac{h_{\max}}{2} \|x - x^*\|_2^2. \quad (35.3)$$

These inequalities follow by replacing each positive diagonal weight with its smallest or largest value. They relate potential to Euclidean distance under this declared calibration. They do not say that Euclidean distance is a complete measure of human or ecological function.

35.3 What changing a scale does

Take $(x_1, x_2) = (18, 2)$, reference $(10, 10)$ and scales $(2, 2)$. The standardized deviations are $(4, -4)$ and $V = 16$. If only the first scale becomes one, the potential becomes $\frac{1}{2}(8^2 + (-4)^2) = 40$. The physical state has not changed; the declared sensitivity has. That is why comparisons across parameter versions need explicit recalibration records.

The shape $\exp[-V(x)]$ resembles a Gaussian density, but it is not itself a normalized probability law. Restricting to fixed mass couples the coordinates, and the nonnegative boundary further changes the domain. A claim about the distribution of actual states needs a stochastic transition law and evidence or proof about that law. A bell-shaped reference geometry cannot supply those missing ingredients.

35.4 An incompatible-reference extension, not the active model

To see why mass compatibility is load-bearing, temporarily allow $\sum_i x_i^* \neq M$ and ignore the nonnegative boundary while minimizing on the affine mass plane. A Lagrange multiplier λ gives

$$H(x - x^*) = \lambda \mathbf{1}, \quad x_i = x_i^* + \lambda \sigma_i^2, \quad \lambda = \frac{M - \sum_i x_i^*}{\sum_i \sigma_i^2}. \quad (35.4)$$

This candidate solves the nonnegative problem only if all its entries are nonnegative; otherwise active-boundary conditions are required. Even in the interior, the constrained optimum need not have $\mu_i = 0$: its marginals are equal, so conservative transfers have no first-order benefit.

The active examples do not use this extension. It is included to show why “zero gradient,” “constrained minimum” and “declared reference” are not universally interchangeable phrases.

Exercises and answers

At $(14, 6)$ with reference $(10, 10)$ and scales $(2, 2)$, the marginals are $(1, -1)$ and $V = 4$. At $(10, 10)$ both vanish. At $(2, 18)$ the potential is again sixteen, but the marginals reverse sign. Equal potential does not mean equal state or equal consequences for a directed action menu.

Explain why removing the old Allee term does not prove regeneration safe. The new closed lossless domain contains no such regenerative dynamics. The old theorem has not been refuted; its plant is simply not the active one.

35.5 A constrained reference is not a Gaussian probability law

There are two different uses of the same exponential expression. One may define a geometric score $V(x)$ and note that $\exp[-V(x)]$ has a Gaussian shape. Or one may define a probability measure proportional to that expression on a specified domain, with a specified base measure and normalizing constant. The second is an additional model. It does not follow from the first and does not describe an actor process until a connection to that process is established.

Conservation makes the distinction concrete. In two cells with total mass twenty, choosing x_1 determines $x_2 = 20 - x_1$. They cannot be independent random coordinates on that mass line. A diagonal Hessian still gives a separable formula on the ambient space, but the feasible states are coupled by the constraint. A simulation histogram resembling a bell would also not prove independent Gaussian coordinates.

The same care applies to the scale σ_i . In this book it is a declared width in physical stock units. Calling it a standard deviation would require a specified probability distribution and calibration. Calling it a measurement error would require a sensor-error model. Neither role is obtained simply because statistical notation often uses the same letter.

35.6 The reference and the directions the world can use

In an unconstrained ambient space, a vanishing gradient characterizes the minimum of this strictly convex quadratic. A conserved world moves only in directions whose entries sum to zero. If every allowed directional derivative is zero, the state need not have zero ambient gradient in a more general incompatible-reference problem. Equal marginals can make all conservative first-order transfers neutral while the individual marginals are nonzero.

The active compatible-reference model avoids that particular ambiguity at its global minimum, but graph restrictions can create another. If the graph has disconnected components, no internal action can change a component's total mass. A reference compatible with global mass may still be incompatible with a component's available mass. Thus global compatibility is not a reachability theorem even for an otherwise continuous action set.

One can see this without running a process. Divide four cells into two disconnected pairs. If the first pair holds twelve units but its two references sum

to twenty, no transfer confined to that pair can meet both references. Movement elsewhere does not repair the missing route. A global potential may show a persistent deviation that is structurally impossible to remove under the declared topology.

35.7 Why a small potential is an incomplete functional statement

Because the potential sums standardized squared deviations, a small value does bound each represented standardized deviation. In fact,

$$\frac{|x_i - x_i^*|}{\sigma_i} \leq \sqrt{2V(x)} \quad \text{for every represented cell } i. \quad (35.5)$$

The proof uses nonnegativity: an individual summand cannot exceed the whole sum. This is a useful geometric guarantee. It concerns represented coordinates and declared scales only.

It does not establish the adequacy of the reference for health, the quality of the resource, access for an unrepresented household or the absence of damage outside the system boundary. Those concerns are not counterexamples to the inequality. They are reasons to avoid assigning the inequality a meaning it never had. Functional calibration and representational completeness require additional evidence and choices.

Conversely, a large potential does not prove that one actor caused the departure or owes compensation for it. Initial conditions and external forcing can create deviation before the actor chooses anything. The following chapters therefore bind value to a specific transition and baseline, rather than treating the state score as a debt automatically assigned to whoever happens to be present.

Chapter recap

Gaussian geometry supplies a smooth, strictly convex quadratic and constant curvature. It makes several derivations exact and simple. It does not supply calibration, a probability law, a controller or a theorem of recovery.

Graphs, supports and physically admissible groups

An edge is a possible physical transformation, not a promise that a helpful transfer can always occur. A route, a pump and a warehouse doorway may each impose limits independent of any potential. Begin with those limits, because an exact value for an impossible action is only a counterfactual calculation.

36.1 Three graphs with different jobs

The physical graph lists allowed transformations. The potential-factor graph lists which state coordinates a term in V depends on. The action-interaction graph records overlap or coupling between actions. In the diagonal Gaussian model every potential factor has one coordinate, yet two physical actions can interact because they both change the same source. A diagonal Hessian does not make all edge actions independent.

For action a , its write support is the set of coordinates with nonzero increment δ_a . Its constraint support includes capacities, reservations or aggregate budgets it uses. Two actions may have disjoint write supports but share a machine capacity. Conversely, reading the same immutable scale does not by itself create a shared physical constraint.

36.2 Joint feasibility is not singleton feasibility

Suppose Ridge contains five units. Two requests each withdraw three units from Ridge to different destinations. Each is individually nonnegative at its endpoint, but their group withdraws six and is invalid. Testing children one at a time against the same five units does not validate their sum.

With $q_a \geq 0$, a conservative frozen-stock rule might require

$$\sum_{a: \text{src}(a)=i} q_a \leq z_i. \quad (36.1)$$

This is an example of a declared rule, not a newly adopted programme policy. A plant that permits concurrent incoming flow to fund outgoing flow requires

a different timing and feasibility contract. Endpoint nonnegativity alone does not prove that an unmodelled intermediate mechanism can perform it.

Destination capacity is another independent constraint. If a tank can hold ten and begins at eight, two incoming deliveries of two each cannot both fit merely because their separate endpoints do. Physical capacity is not the Gaussian reference. A reference may be six even when the tank can hold ten; exceeding the reference can be feasible but increase potential.

36.3 The straight segment and its limited guarantee

The simplex is convex. If z and $z + \delta_G$ lie in it, then

$$z + s\delta_G = (1 - s)z + s(z + \delta_G), \quad 0 \leq s \leq 1, \quad (36.2)$$

also lies in it. This proves nonnegative conserved stock along the aggregate straight interpolation. It does not prove child-specific rate limits, mechanical synchronization, queue clearance or causal access to incoming stock. Those are additional constraints on a physical realization.

This distinction matters when a path integral is later described as physical. The same segment can be a valid mathematical interpolation even when the plant has not declared constant-rate joint execution. The integral identity survives; the physical-path interpretation does not follow automatically.

36.4 Grouping without losing atomic actions

A joint group retains child identities and increments. It is not an excuse to conceal a fuel-to-electricity-to-pump-to-water chain inside one undifferentiated operation. Distinct carriers and intermediate transformations must remain traceable when they belong to the represented scope.

The analytical bridge [5] groups coupled actions under a common accounting boundary and distinguishes physical grouping from receipt batching. Batching independent records can reduce administrative overhead, but it does not create a new physical synergy. Conversely, simultaneous actions sharing a source need joint evaluation even if their paperwork is submitted separately.

Exercises with reasoning

Take a source with six units and two proposed quantities (4, 4). A common cap of six with proportional scaling would yield (3, 3). The number three comes from that resolver, not from the value equation. If the resolver were priority-based, a different accepted vector could be physically admissible. Never price (4, 4) and execute (3, 3) under the old value record.

Now let two accepted actions use disjoint nodes and disjoint constraints. With a separable potential, their finite values add because the affected terms are disjoint. If a new pair factor couples one node from each action, the independence proof no longer applies even though the physical edges remain disjoint. The correct affected support follows the potential factors.

Chapter recap

The graph says what can connect, support says what changes, constraints say what is permitted, and the potential says how the represented change is valued. None can silently perform the job of another.

Why local evaluation can be exact

The project asks for local valuation without a global observer deciding the worth of every action. The mathematical possibility comes from cancellation, not from approximating a large world by a small one. A local calculation is exact when it includes every potential term that can change.

37.1 The separable theorem

Let $V(x) = \sum_i v_i(x_i)$ and let W contain all coordinates changed by δ . For $i \notin W$, $z_i + \delta_i = z_i$. Split both sums:

$$V(z) - V(z + \delta) = \sum_{i \in W} [v_i(z_i) - v_i(z_i + \delta_i)] \quad (37.1)$$

$$+ \sum_{i \notin W} [v_i(z_i) - v_i(z_i)] \quad (37.2)$$

$$= V_W(z) - V_W(z + \delta). \quad (37.3)$$

No information about an unchanged term is needed to calculate its zero difference. This identity holds for nonlinear v_i and does not require Gaussian geometry. The Gaussian model is one especially transparent specialization of the historical locality result [4].

37.2 A complete small example

Use references (10, 10, 10) and scales (2, 2, 2) at state (18, 2, 12). The first two potential terms are eight each; the third is one-half. A four-unit transfer from cell one to cell two gives (14, 6, 12), with potential terms two, two and one-half. The global difference is $16.5 - 4.5 = 12$. The two-endpoint local calculation is $16 - 4 = 12$. The remote cell's value was not assumed zero. It cancelled because it was unchanged.

If an action also changes cell three, the proof demands its inclusion. Calling the third effect small or inconvenient does not permit its omission while retaining the adjective exact. One may define an approximation, but then its error and claim must be stated separately.

37.3 The factor version

A more general potential has factors

$$V(x) = \sum_{\alpha \in \mathcal{F}} \phi_{\alpha}(x_{A_{\alpha}}), \quad (37.4)$$

where A_{α} is the set of arguments used by factor α . Let $\mathcal{F}_W = \{\alpha : A_{\alpha} \cap W \neq \emptyset\}$. Every factor outside $\mathcal{F}_W$ cancels, so the exact difference is the sum over $\mathcal{F}_W$. A factor in that set may require reading a coordinate outside W . Write support and read support need not coincide.

For example a declared term $k(x_1 - x_3)^2$ changes when an action changes cell one. Cell three may remain physically untouched, yet its current value is required to evaluate the factor. A two-endpoint view that omits it cannot be called complete for that expanded potential. This is an illustrative extension, not an adopted nonseparable Gaussian model.

37.4 Three different locality claims

Valuation locality means the finite value needs only complete affected factors. *Information locality* means a particular actor or module may access only specified records. *Coordination locality* concerns how candidate groups are assembled and selected. A proof of the first does not establish the second or third.

A central program could enumerate every group in the world and evaluate each group locally. Its values would satisfy cancellation, but the selector would still be global. Similarly a class named `LocalView` is not a complete security proof if a helper can read unrestricted state elsewhere. The implementation chapters later in this volume turn these distinctions into review questions without asserting that the future runtime already enforces them.

37.5 Frozen state and version completeness

Locality also requires agreement about time and parameters. Combining the source's present stock with the destination's stale stock does not evaluate one declared state. Changing a scale between the before and after terms does not compute one potential difference. Exact arithmetic cannot repair either mismatch.

Thus a reconstructible local quote binds the state version, affected factor identities, reference/scale version, action increment and process-burden declaration. Hashes can identify those inputs. They cannot certify that a sensor observed reality correctly; that is a measurement question.

Exercise and answer

Suppose a quote reports a difference of twelve while a global audit reports eleven. Name three possible causes before blaming numerical precision. An affected factor may be missing; the states may belong to different epochs; or one calculation may use a different parameter version. Each is a semantic discrepancy. The exact identity tells the reviewer where to look because, under its actual assumptions, such a difference is impossible.

Chapter recap

Local exactness is a cancellation theorem over complete affected factors. It is powerful precisely because its boundary is precise. It does not certify sensors, ownership, or decentralized selection.

Force, mobility, flux and finite action value

The potential describes a state. Its derivative describes how that potential changes in a direction. A constitutive response law describes how strongly a controller moves in response. A finite EBU quote describes the complete change caused by a specified action. These four statements fit together, but none should be substituted for another.

38.1 The force along a physical edge

For a lossless edge $i \rightarrow j$, the unit action column is $S_e = -e_i + e_j$, where e_i is the coordinate vector with one at position i . The directional derivative is

$$\mu^T S_e = -\mu_i + \mu_j. \quad (38.1)$$

Define the field force with the sign that makes a small improvement positive:

$$f_e = -S_e^T \mu = \mu_i - \mu_j. \quad (38.2)$$

For the Gaussian potential this is

$$f_e = \frac{x_i - x_i^*}{\sigma_i^2} - \frac{x_j - x_j^*}{\sigma_j^2}. \quad (38.3)$$

It compares standardized marginal deviations, not stocks alone. Two cells holding equal quantities can have different marginals if their references or scales differ. Conversely, unequal stocks can have equal marginals.

In the historical loss-aware edge model $S_e = -e_i + \eta_e e_j$, the same chain rule gives $f_e = \mu_i - \eta_e \mu_j$. The efficiency belongs inside the derivative because it changes the physical transformation. This general directional identity does not by itself adopt a loss-aware runtime or resolve its separate conservation and cost declarations.

38.2 Mobility and the Onsager-type response

The historical reference is Onsager's 1931 work on coupled irreversible processes [1, 2]. It connects linear force–rate relations and reciprocal coefficients

under microscopic-reversibility assumptions. EBU borrows a modelling structure, not an entitlement to those physical assumptions. A chosen deviation potential need not be free energy, and a declared mobility need not be an experimentally measured transport coefficient. This is why the qualifier “Onsager-type” matters.

The reason for introducing the structure is constructive: it separates the state’s driving slope from a mechanism’s responsiveness, and it permits an explicit conditional dissipation calculation. It does not follow from the finite accounting identity. We can change the response law while retaining that identity, but we cannot then retain the old response law’s dynamical proof.

The original EBU books studied the directed threshold-mobility law

$$J_e = M_e[f_e - \theta_e]_+, \quad M_e > 0, \quad \theta_e \geq 0, \quad (38.4)$$

where $[a]_+ = \max(a, 0)$. J_e is a rate. M_e is mobility: the strength of the declared response to force above threshold. In the present dimensionless potential normalization, M_e has units X^2/T and θ_e has units $1/X$. They are not the physical stock capacity or the persistent EBU capacity balance introduced later.

This is the project’s *Onsager-type* constitutive controller. The unthresholded signed relation $J = Mf$ is linear; the positive-part directed version is piecewise linear and blocks reverse flow on that edge. Neither form is derived merely by defining V . Choosing it is a separate modelling decision. The name does not establish universal thermodynamic validity for an economic system, nor does it imply a general coupled reciprocal-mobility matrix. The historical proofs concern their explicitly stated diagonal, directed controller.

If $M = 2$, $f = 1.4$ and $\theta = 0.5$, the requested rate is 1.8 in the corresponding units. Doubling mobility doubles this unconstrained request. It does not double the stock available at the source or authorize a physical cap to be exceeded. Permission may produce a different accepted rate, in which case a proof about the unconstrained law needs reinspection.

38.3 The exact finite quadratic

Let a declared action have finite increment δ . Expanding the square,

$$V(z + \delta) = \frac{1}{2}(z - x^* + \delta)^T H(z - x^* + \delta) \quad (38.5)$$

$$= V(z) + \mu(z)^T \delta + \frac{1}{2} \delta^T H \delta. \quad (38.6)$$

The two cross terms are equal because H is symmetric. Therefore

$$E(\delta) = V(z) - V(z + \delta) - C = -\mu(z)^T \delta - \frac{1}{2} \delta^T H \delta - C. \quad (38.7)$$

No small-action approximation was made. The first term is the tangent benefit; the second is the exact curvature correction. For $\delta = qS_e$ on a lossless edge,

$$E(q) = qf_e(z) - \frac{q^2}{2} (\sigma_i^{-2} + \sigma_j^{-2}) - C(q). \quad (38.8)$$

If $C = 0$ and $f_e > 0$, positive quantities give nonnegative value only up to $2f_e/(\sigma_i^{-2} + \sigma_j^{-2})$. Physical constraints can be tighter. The mathematical maximizer is half that upper zero, but the active random-affordable programme does not automatically choose the maximizer.

38.4 A force can be positive while a large action is harmful

At $z = (18, 2)$, $x^* = (10, 10)$ and $\sigma = (2, 2)$, the marginals are $(2, -2)$, so $f = 4$ and

$$E(q) = 4q - q^2/4. \quad (38.9)$$

At $q = 4$ the value is twelve. At $q = 8$ it is sixteen. At $q = 16$ it is zero, and at $q = 17$ it is $-17/4$. The initial force stays four in the frozen tangent calculation, but the field changes along the finite action. Quoting qf would ignore that change and overstate benefit.

The Onsager controller would supply a rate based on the force and its own mobility/threshold. A duration turns that rate into a quantity. Exact EBU then values the resulting accepted quantity. Keeping these steps explicit prevents the words force, flux and value from collapsing into one number.

Chapter recap

Force is a directional derivative; mobility is a constitutive response parameter; flux is the controller's rate; finite EBU is an endpoint difference. Chapter 44 derives the dissipation result for the Onsager-type law and explains exactly why it is not a recovery proof for a different actor.

From path integrals to committed quotes

A derivative tells us the effect of an infinitesimal displacement. A finite action travels through changing derivatives. Integrating them recovers the endpoint difference. This is the mathematical connection between force and exact EBU, and it remains valid whether or not a controller uses the force to choose the action.

39.1 One action, one chain rule

Let $\gamma(s) = z + s\delta$ for $0 \leq s \leq 1$, with V continuously differentiable along this path. The chain rule gives

$$\frac{d}{ds}V(\gamma(s)) = \nabla V(\gamma(s))^T \delta. \quad (39.1)$$

Integrate and change the sign:

$$V(z) - V(z + \delta) = - \int_0^1 \nabla V(z + s\delta)^T \delta ds. \quad (39.2)$$

For $\delta = qS_e$ with $q \geq 0$, substitution $r = sq$ yields $\int_0^q f_e(z + rS_e) dr$. The changing force, not just its initial value, is integrated. A declared process burden is then subtracted separately.

For the quadratic, $\mu(z + s\delta) = \mu(z) + sH\delta$. Integrating one and s over $[0, 1]$ immediately gives the finite expansion of Chapter 38. The average endpoint gradient is also exact:

$$V(z) - V(z + \delta) = -\frac{1}{2}[\mu(z) + \mu(z + \delta)]^T \delta. \quad (39.3)$$

This is not a general exact quadrature claim for arbitrary potentials. It works here because the gradient along the segment is affine.

39.2 Path arithmetic and path meaning

If the physical model declares constant-rate joint execution, s can parameterize its realized path after normalizing duration. If only endpoints are declared,

the same segment is an accounting interpolation. The endpoint identity is exact in both uses. Calling it an observed physical history requires the first additional declaration and, outside a simulation, appropriate measurement.

More generally, a smooth potential integrates to the same total along any admissible sufficiently regular path between the same endpoints. This does not imply equal physical cost for every route: process burdens and other state coordinates may differ. It also does not imply equal child allocations when a group path is decomposed in different ways.

39.3 A schedule is a function, not a promise about one quantity

A pre-action quote can commit $q \mapsto E(q)$ on a physically accepted domain. For the Ridge–Vale example the schedule is $4q - q^2/4$. If the accepted upper quantity is four and the measured quantity is three, the committed schedule gives $E(3) = 39/4$, not twelve. The measurement selects a point of an existing function; it does not authorize a new potential chosen after the outcome.

Quantity shortfall does not always mean a smaller signed value. On a decreasing part of a concave schedule, less movement can have greater value. The schedule, not a verbal monotonicity assumption, governs the account. If quantity exceeds the committed domain, the old positive quote cannot simply be extended. The historical foundation retains an explicit overexecution boundary rather than inventing punitive physics.

39.4 Convexity and activation burden

For a convex differentiable V , $V(z + \delta) \geq V(z) + \nabla V(z)^T \delta$. Hence exact value is no greater than its tangent value when both subtract the same C . For the positive definite quadratic the inequality is strict for nonzero δ .

Along a fixed edge, the field-benefit schedule is concave. The complete schedule remains concave when the declared $C(q)$ is convex. Nonnegative cost alone is not sufficient. A fixed activation cost with $C(0) = 0$ and $C(q) = c_0 > 0$ for $q > 0$ creates a downward jump at zero; interpolation across it would invent an unregistered discount. The first prospective lossless domain avoids this complication by declaring $C = 0$, not by claiming real operations are universally costless.

39.5 What a quote does not guarantee

A valid schedule does not establish permission, ownership, sensor integrity, or execution. A positive quote does not prove the action will occur. A negative

quote does not prove that essential service should be denied. The mathematics describes represented field effect. Institutions and the experimental capacity policy are additional objects with additional claims.

Chapter recap

The exact finite difference and the integrated force are two expressions for one field change. A committed schedule binds that expression to a baseline, rule and domain. Physical-path meaning and institutional consequences do not follow from integration alone.

Groups, cross terms and honest comparisons

Two actors can share a source without either actor's request being wrong on its own. The problem is that both requests refer to one source, not two copies of it. Nonlinear valuation must therefore evaluate their joint change before asking how any resulting value might be divided.

40.1 One group, one endpoint

For fixed accepted increments δ_a , define $\delta_G = \sum_{a \in G} \delta_a$. In the cost-free Gaussian domain,

$$E_G = -\mu(z)^T \delta_G - \frac{1}{2} \delta_G^T H \delta_G. \quad (40.1)$$

Expand the quadratic sum. Symmetry of H gives

$$E_G = \sum_{a \in G} E_a^{(0)} - \sum_{a < b} \delta_a^T H \delta_b, \quad (40.2)$$

where $E_a^{(0)} = V(z) - V(z + \delta_a)$ is the hypothetical value of child a alone at the same baseline. These singleton values are not sequential receipts. The cross terms are exactly what their naive sum leaves out.

40.2 A shared-source calculation

At $(18, 2)$ with the usual reference and scales, let two children each send two units along the same edge. Each alone has value seven. Together they send four and have value twelve, not fourteen. Their increments are both $(-2, 2)$, so $\delta_A^T H \delta_B = 2$ and the correction is -2 .

The same issue occurs when the destinations differ. If two children share only a source with quantities $a, b > 0$, their cross term contains $ab/\sigma_i^2 > 0$. The joint field value is smaller than the sum of their separate source-relief claims by that amount. Disjoint changed coordinates have zero cross term for the diagonal model.

Opposing increments can instead produce a positive correction. Take $\delta_A = (-2, 2)$ and $\delta_B = (2, -2)$ at the same baseline. Their standalone values are seven and minus nine. The net increment is zero, so the group value is zero

and the correction is $+2$. Positive nonadditivity here means cancellation of represented displacement. It is not proof that real physical circulation is free or socially beneficial.

40.3 Same-baseline nonadditivity is not a parallel advantage

Define $N_G = E_G - \sum_a E_a^{(0)}$. Now choose an admissible sequential schedule π , with complete endpoint x_π . Its field total is $V(z) - V(x_\pi)$. The comparator-relative difference is

$$I_{G|\pi} = E_G - E_\pi = V(x_\pi) - V(z + \delta_G). \quad (40.3)$$

When fixed additive increments can be serialized feasibly, the final endpoint is the same and $I_{G|\pi} = 0$. Yet N_G may be nonzero. The shared two-unit example has $N_G = -2$ and $I = 0$: it contains nonlinear same-baseline double counting but no endpoint advantage of simultaneity.

A resolver replay can change this comparison. If it recomputes accepted quantities after the first action, it may produce another endpoint. That is a different, explicitly named comparator, not evidence that the same accepted vector has two contradictory values. The bridge source [5] distinguishes quantity-fixed from rule-replayed comparisons.

40.4 The conditional Boolean-lattice calculation

For a set A of fixed children, let $F(A) = V(z) - V(z + \sum_{a \in A} \delta_a)$ and $F(\emptyset) = 0$. The pair inclusion–exclusion coefficient is

$$m_{ab} = F(\{a, b\}) - F(\{a\}) - F(\{b\}) = -\delta_a^T H \delta_b. \quad (40.4)$$

Because the indicator expansion of F has degree at most two, all coefficients of order three and higher vanish. This statement needs the same baseline, potential and fixed increments in every subset.

If a subset reruns a resolver, its increments can differ. The resulting set function need not be quadratic in membership indicators. Furthermore, physically inadmissible subsets cannot be turned into valid interventions by algebraic convenience. Every relevant subset must be admitted under the declared protocol before a physical Boolean-lattice interpretation is made. Möbius analysis remains an optional diagnostic, not extra issuance.

Questions to ask of a group claim

Were the children valued at one baseline or consecutive states? Were quantities held fixed or re-resolved? Is the comparator physically admissible? Does

it include the same costs and complete state? Is a correction being mistaken for a causal contribution? These questions determine the meaning of an interaction number before its sign is interpreted.

Chapter recap

The quadratic cross term explains why independent same-baseline values fail to close on a shared group. A named sequential comparator answers another question. Neither calculation supplies an ownership rule.

A path attribution is not an entitlement

Once the group value is known, one may want a number beside each child action. A decomposition can be mathematically exact without being a uniquely identified causal contribution. The difference must be kept visible before any such number is connected to a persistent account.

41.1 The declared common-path decomposition

Choose the straight group path $\gamma(s) = z + s\delta_G$. Define

$$R_a = - \int_0^1 \mu(z + s\delta_G)^T \delta_a ds = -\mu(z)^T \delta_a - \frac{1}{2} \delta_a^T H \delta_G. \quad (41.1)$$

The last equality uses the affine Gaussian gradient. Sum over children:

$$\sum_a R_a = - \int_0^1 \mu(z + s\delta_G)^T \sum_a \delta_a ds \quad (41.2)$$

$$= - \int_0^1 \mu(z + s\delta_G)^T \delta_G ds = E_G. \quad (41.3)$$

This proves closure for the cost-free domain. If $C_G \neq 0$, the path terms still sum to the field component; allocating or accounting for cost requires its own declaration. It must not disappear in the notation.

41.2 What the formula assigns in a small group

Let the children send one and three units from Ridge to Vale at $(18, 2)$. The group increment is $(-4, 4)$ and its value is twelve. Initial tangent benefits are four and twelve. Curvature corrections are one and three, so $R_A = 3$, $R_B = 9$. The total closes exactly.

Their standalone values are $15/4$ and $39/4$, summing to $27/2$. Their sequential live values depend on order. Sending one first gives $15/4$ followed by $33/4$; sending three first gives $39/4$ followed by $9/4$. Both sequential sums equal twelve. Neither ordered split is the same as $(3, 9)$, because it describes another allocation convention.

The sum identity cannot tell us which of these child descriptions is fair, deserved or causally identified. It tells us which declared arithmetic is being used and whether that arithmetic closes on the group.

41.3 Physical path versus interpolation

Under a declared constant-rate physical model, each child's normalized rate contributes δ_a along the common path. The integral then decomposes that model path. Under endpoint interpolation it is an accounting construction. Identical arithmetic does not erase the distinction.

Historical sources use the name radial Aumann–Shapley for related path allocation [4]. Using or avoiding that name does not resolve the interpretation. The scientifically useful record specifies the actual path and assumptions, labels the result as a mathematical attribution, and states what it does not establish: ownership, fairness, causal entitlement, or permission to spend.

41.4 Splitting and cancellation

Suppose one child is replaced by two proportional subchildren with increments $\lambda\delta_a$ and $(1 - \lambda)\delta_a$, while the group path is unchanged. The formula is linear in the child increment, so their attributed values add to the original R_a . This is a limited mathematical splitting property. A split that changes grouping, fees, ownership or feasibility is not covered by it.

If $\delta_G = 0$, the formula becomes $R_a = -\mu(z)^T \delta_a$. These child values can be nonzero and cancel. In the opposing two-unit example they are +8 and -8, while the group has zero value. This illustration makes the interpretive risk vivid: a cancelling endpoint does not establish a gift of eight units to one participant and a legitimate charge of eight to another.

41.5 From children to owners is another map

If a prospective policy assigns every child a unique owner, then finite-sum rearrangement gives

$$\Delta B_i = \sum_{a: \text{owner}(a)=i} R_a, \quad \sum_i \Delta B_i = \sum_a R_a. \quad (41.4)$$

The second equality is exact once the map exists. It does not select the map. A requester, provider, beneficiary and physical source can be different entities. Substituting one for another because a software field is convenient would change the policy while leaving the arithmetic looking correct.

Chapter recap

Common-path attribution provides an explicit, auditable decomposition. Its proof closes a sum. Ownership and spendability require separate prospective authority even for a group with only one child.

Sequences, cycles and external change

The simplest accounting theorem in EBU is also one of the most important. If every action begins where the previous one ended, the intermediate potentials cancel. A surprisingly large number of incorrect issuance rules fail because they stop using those live intermediate states.

42.1 Undriven telescoping

Let $x^0, x^1, \dots, x^N$ be a declared sequence under one fixed potential, with action values $E_k = V(x^k) - V(x^{k+1}) - C_k$. Then

$$\sum_{k=0}^{N-1} E_k = V(x^0) - V(x^N) - \sum_{k=0}^{N-1} C_k. \quad (42.1)$$

To prove it, write the first two differences: $V(x^0) - V(x^1)$ and $V(x^1) - V(x^2)$. Their shared term cancels. Repeating that cancellation leaves only the first and last endpoints. Neither convexity nor Gaussian geometry is required for this algebraic identity.

For a closed physical cycle $x^N = x^0$, the total is $-\sum C_k$. If all burdens are nonnegative, it cannot be positive. With $C = 0$ it is zero, not necessarily negative. A reversible movement can return every represented coordinate to its start without making the account grow.

42.2 A live-state calculation

Use (18, 2) with reference (10, 10) and scales (2, 2). Transfer two units, then three, then one. The successive potentials are

$$16, \quad 9, \quad 9/4, \quad 1. \quad (42.2)$$

The action values are 7, $27/4$ and $5/4$, whose sum is fifteen. That equals $16 - 1$. Quoting all three against the first state instead would replace the last two live values with unrelated counterfactuals.

Splitting a transfer preserves the field total when the resulting sequence has the same endpoint. It need not preserve separate activation costs, physical delays or intervening disturbances. A theorem about endpoint differences is not a theorem that every way of organizing work costs the same.

42.3 Forcing before the action epoch

The new programme proposes an external change from x_t to z_t before actors evaluate their candidates. Define

$$D_{\text{ext},t} = V(z_t) - V(x_t), \quad E_t = V(z_t) - V(x_{t+1}) - C_t. \quad (42.3)$$

Add and subtract $V(x_t)$ in the second expression and sum:

$$\sum_{t=0}^{N-1} E_t = V(x_0) - V(x_N) + \sum_{t=0}^{N-1} D_{\text{ext},t} - \sum_{t=0}^{N-1} C_t. \quad (42.4)$$

An actor who does nothing has zero field value at the shared post-forcing baseline. Nature's own change appears in D_{ext} , not in an idle actor's income. The external term can be negative: forcing may move the system closer to the reference.

The old D0 foundation uses a frozen-drive no-action successor. Its immediate cancellation reasoning is related, but its event order is not identical to this proposed forcing-first profile. An implementation needs an explicit profile contract rather than renaming one variable and claiming equivalence.

42.4 A forcing example with exact signs

Begin at reference (10, 10), where $V = 0$. A conservative external shift produces (14, 6), where $V = 4$. An actor then transfers two units back, ending at (12, 8) with $V = 1$. The external audit is +4 and the actor's field value is +3. The total physical potential change is only +1; confusing it with actor value would get both the baseline and the sign wrong.

If a later external shift takes (12, 8) to (10, 10), its audit is -1 . No actor performed that improvement. Crediting it to the next idle actor would violate the declared separation even if a global sum happened to close.

42.5 Changing the potential is another accounted event

If parameters vary, simple telescoping under one V no longer applies. Writing V_t for the version at event t , parameter changes create terms such as $V_{t+1}(x) - V_t(x)$ even at unchanged physical state. A recalibration can therefore create apparent value unless its effect is separated from physical actions. The first model freezes references and scales precisely to avoid silently mixing those effects.

The historical no-overissue theorem retains its original one-action and audit assumptions. This chapter derives a general endpoint identity; it does not

retroactively broaden that source theorem's settlement scope. For simultaneous groups, the joint transition must be valued once and any division needs the distinct treatment of Chapters 40 and 41.

Chapter recap

Live endpoints, fixed parameters and complete external accounting make the sum reconstructible. Closed-cycle closure rules out one accounting exploit. It does not prove that actors prefer recovery over cycling or refusal.

Capacity accounts and random affordable action

The new research question is not whether a controller can be written to move downhill. The Onsager-type example already shows how a particular response law can have a descent property. The proposed question is what happens when actors choose without ranking EBU, but their choices are restricted by a separately declared capacity account.

43.1 The policy, stated conditionally

Let $B_i \geq 0$ be persistent experimental capacity for owner i , initially zero. It is not physical stock, installed pumping capacity or a real-world monetary entitlement. Suppose an explicitly adopted owner/path policy would assign group changes ΔB_i satisfying

$$\sum_i \Delta B_i = E_G. \quad (43.1)$$

The proposed affordability condition is

$$B_i + \Delta B_i \geq 0 \quad \text{for every affected owner } i. \quad (43.2)$$

The intended policy permits same-event signed netting within one owner, not across all owners. It introduces no loan, refill or global pool. Its adoption is a modelling choice still requiring the programme's declared authority, not a consequence of Chapter 41.

43.2 Why positive group value is insufficient

Consider a candidate whose owner changes are $(3, -1)$ and whose aggregate value is two. With zero balances, it is unaffordable for the second owner. Positive total value does not cover that owner's debit unless a different policy explicitly permits pooling. Conversely, if one owner holds both child actions, same-owner netting would produce $+2$ and pass. Ownership can therefore change the feasible stochastic process without changing the physical endpoint or group field value.

With positive existing capacity, a negative-valued action may be affordable. That is not an accounting error. It is one intended reason this policy is not a monotone potential-descent controller. Whether spending later enables useful action or merely produces cycles is a research question.

43.3 The external-audit invariant

Let J accumulate $D_{\text{ext},t} = V(z_t) - V(x_t)$ and suppose the adopted cost-free updates are exact. Then

$$J_{t+1} = J_t + D_{\text{ext},t}, \quad (43.3)$$

$$\sum_i B_{i,t+1} = \sum_i B_{i,t} + V(z_t) - V(x_{t+1}). \quad (43.4)$$

Adding terms yields

$$V(x_{t+1}) + \sum_i B_{i,t+1} - J_{t+1} = V(x_t) + \sum_i B_{i,t} - J_t = K. \quad (43.5)$$

At reference with zero genesis and zero audit, $K = 0$. Forcing does not directly fund an actor. Its signed potential change enters J so that the full account can be reconstructed.

After a single shock, suppose $J = 4$ and $K = 0$. If an actor reduces potential to one, total capacity becomes three. Later an affordable negative action could raise potential to two while lowering total capacity to two. The identity permits this. It gives a constraint, not a ranking of choices.

43.4 What can be proved about the reference

At $x = x^*$, any nonzero net increment has $E_G = -\frac{1}{2}\delta_G^T H \delta_G < 0$. At zero balances, owner-wise affordability would require every $\Delta B_i \geq 0$, making their sum nonnegative. Thus no nonzero-net action can be affordable there under the stated cost-free policy. A cancelling group may have zero aggregate change; its owner-level feasibility still depends on the declared attribution/map.

This is a conditional statement about zero genesis at the reference. After a shock and earned capacity, the account can permit uphill movement. Therefore the zero-genesis observation does not establish attraction from disturbed states or long-run recovery.

43.5 Random choice is not mathematical neutrality

“Choose uniformly” is incomplete until the sample space is fixed. Uniform choice among edges differs from uniform choice among quantities, groups or

owners. If one physical option appears twice under different identifiers, sampling entries gives it twice the probability. A finite menu is also not the same as the continuum of all physically possible quantities.

The programme intends EBU-blind candidate construction and unweighted choice among affordable physically feasible options, with a physical-random control that lacks the affordability filter. Exact menu, duplicate, no-action and group-assembly semantics remain load-bearing decisions. They must not be chosen after inspecting which version makes EBU look better.

Chapter recap

Capacity and external audit can satisfy an exact conditional invariant. That invariant does not make the policy fair, decentralized, monotone or successful. Those are separate statements about the declared process.

Onsager-type dissipation and finite-step limits

The force–mobility–flux structure deserves its own proof, both because it motivated earlier EBU models and because it shows how much a dynamical conclusion depends on the chosen response law. We derive the result without calling every Gaussian actor an Onsager controller.

44.1 The continuous-time calculation

Suppose, on a time interval where the declared physical model is valid,

$$\dot{x} = u(x) + SJ(x), \quad f = -S^T \mu, \quad J_e = M_e[f_e - \theta_e]_+. \quad (44.1)$$

The chain rule gives

$$\dot{V} = \mu^T u + \mu^T SJ = \mu^T u - \sum_e f_e J_e. \quad (44.2)$$

For an active edge, $J_e = M_e(f_e - \theta_e) > 0$, so $f_e = J_e/M_e + \theta_e$. For an inactive edge both J_e and its product with f_e vanish. Consequently

$$\dot{V} = \mu^T u - \sum_e \left(\frac{J_e^2}{M_e} + \theta_e J_e \right). \quad (44.3)$$

With $u = 0$, $M_e > 0$ and $\theta_e \geq 0$, every subtracted term is nonnegative and $\dot{V} \leq 0$. This is a theorem about this declared continuous-time law. It does not require the historical hinge potential; the chain rule works for the smooth Gaussian potential too.

44.2 What the proof did not check

Nothing in the sign calculation alone ensures nonnegative stock, physical capacity, or existence of a trajectory for all time. A source could be asked to send beyond its available stock. A later cap or projection may repair feasibility but changes the vector field. One must then verify the actual accepted flux’s sign/constraint properties, not cite the unmodified equality.

External drive contributes $\mu^T u$ and can raise the potential. Vanishing flux also need not mean $x = x^*$. Directed edges, positive activation thresholds, disconnected components and physical constraints can create other stationary configurations. A nonincreasing scalar is not by itself a proof of convergence to its unique unconstrained minimizer.

44.3 The finite Euler update

Freeze $w = u + SJ$ and take $x^+ = x + \Delta t w$. For the Gaussian model,

$$V(x^+) - V(x) = \Delta t \mu^T w + \frac{1}{2} \Delta t^2 w^T H w. \quad (44.4)$$

The second term is nonnegative and exact. Continuous-time dissipation can therefore coexist with an overshooting finite step. Put $D = -\mu^T w$. If $D > 0$ and $w \neq 0$, a sufficient and in this quadratic case exact condition for nonincrease along that frozen direction is

$$0 < \Delta t \leq \frac{2D}{w^T H w}. \quad (44.5)$$

This does not certify physical feasibility. If $D = 0$ and $w \neq 0$, every positive step raises V ; if $w = 0$, the state does not change.

44.4 A complete controller example

At $(18, 2)$, use one edge, $M_e = 1/2$, $\theta_e = 0$ and $u = 0$. Then $f = 4$, $J = 2$ and $w = (-2, 2)$. We have $\mu^T w = -8$ and $w^T H w = 2$, giving

$$V(x^+) - V(x) = -8\Delta t + \Delta t^2. \quad (44.6)$$

The nonincrease interval is $0 \leq \Delta t \leq 8$. At duration four, the quantity is eight and the reference is reached in this particular two-cell frozen step. At duration eight, the endpoint is $(2, 18)$ and potential returns to sixteen. At duration nine, the physical endpoint $(0, 20)$ is still nonnegative but has potential twenty-five. Continuous descent did not license an arbitrarily large Euler step.

These are analytic substitutions into one frozen update, not a simulated time series or parameter search. Their role is to expose the curvature term.

44.5 Network geometry and a general bound

For a full vector of edge rates, the relevant quadratic is $J^T S^T H S J$. Shared nodes produce off-diagonal terms in $S^T H S$ even when H is diagonal. Thus a certificate derived by treating every edge in isolation can miss joint curvature.

The Gaussian gradient is globally Lipschitz with constant $L = h_{\max}$. Indeed $\mu(x) - \mu(y) = H(x - y)$ and the diagonal operator norm of H is $h_{\max}$. Therefore $w^T H w \leq L \|w\|_2^2$, yielding the familiar sufficient descent-lemma bound. The original network and timestep proofs follow as a historical reading in this chapter, but their old numerical constants must not be copied over the new Hessian or a different actor law.

44.6 Why keep Onsager in the books?

It supplies a transparent benchmark of reasoning: declare a response, derive its sign structure, check feasibility, then inspect discretization. It also explains the origin of mobility and why force is not itself a flux. What it does not supply is a hidden restoring plant for the new programme. Random affordable actors can spend capacity on negative-valued actions; their increments are not generally $\Delta t S J(f)$. Applying the theorem to them would change the subject of the experiment.

44.7 Solving the two-cell response, not merely one Euler step

The earlier example froze the initial flux before taking a finite step. We can also solve its continuous response exactly, which helps explain why those are different objects. Keep total mass twenty, references $(10, 10)$, scales two, one forward edge, mobility $1/2$, zero threshold and no external drive. Write

$$x_1 = 10 + y, \quad x_2 = 10 - y, \quad y(0) = 8. \quad (44.7)$$

For $y \geq 0$, the force is $y/2$ and the flux is $y/4$. Conservation gives $\dot{y} = -y/4$. Separate variables, or differentiate the proposed solution, to obtain

$$y(t) = 8e^{-t/4}, \quad V(t) = y(t)^2/4 = 16e^{-t/2}. \quad (44.8)$$

For every finite nonnegative time the stocks remain nonnegative and the force stays in the chosen forward direction. The reference is approached asymptotically. It is not reached at the finite duration four simply because the frozen Euler step of that duration landed there.

The solution is a conditional analytic example, not an experimental run. It depends on the chosen continuous response law. Replacing that law by random finite actions does not preserve the differential equation, so the exponential expression cannot be advertised as the predicted recovery curve of the new programme.

44.8 A response threshold can leave residual deviation

Keep the same example but take a fixed threshold $0 < \theta < 4$. While $y > 2\theta$, the forward flux is $(y - 2\theta)/4$. Therefore

$$y(t) = 2\theta + (8 - 2\theta)e^{-t/4}. \quad (44.9)$$

This tends to 2θ , not zero. The limiting potential is θ^2 . Below the activation threshold the directed law has no forward flow. If the initial y were already at most 2θ and nonnegative, this edge alone would remain inactive.

This elementary calculation shows why nonincrease of a potential is not the same as attraction to its minimum. The threshold is a response parameter, not an Allee stock threshold. Removing the latter from the Gaussian geometry does not remove the former from an optional controller that explicitly declares it. The random-affordable proposal contains neither response threshold by default.

44.9 What positive clipping preserves and what it changes

Suppose an otherwise admissible historical flux is reduced edge by edge by factors $0 \leq \alpha_e \leq 1$, so the accepted rate is $\tilde{J}_e = \alpha_e J_e$. On every active edge the original law has $f_e \geq \theta_e \geq 0$. Thus $f_e \tilde{J}_e \geq 0$, and with no drive the chain rule still gives

$$\dot{V} = - \sum_e f_e \tilde{J}_e \leq 0. \quad (44.10)$$

This sign argument is weaker than the original equality expressed in $J_e^2/M_e + \theta_e J_e$. The accepted rate no longer necessarily satisfies the original constitutive equation with the old mobility. Preserving a sign does not preserve every quantitative statement.

Nor does this argument cover an arbitrary resolver. A resolver that changes directions, introduces compensating transformations or changes the state before evaluation needs its own analysis. In discrete time even positive clipping still leaves a curvature term to check for the accepted increment. A feasibility improvement is not automatically a numerical descent proof.

44.10 The three meanings of a safe step

A step can be safe with respect to nonnegative stock, safe with respect to a specified reserve certificate, or safe with respect to nonincrease of the chosen potential. These properties can overlap, but they are not synonyms. The two-cell duration-nine example earlier was physically nonnegative and nevertheless

increased potential. A tiny step with the wrong physical support can be impossible however small its numerical truncation error would be.

For this reason the detailed historical proofs that follow should be read as several conditional certificates, not one universal safety theorem. Their line-by-line derivations remain valuable precisely because the reader can see where each premise enters and decide whether it belongs to the current model.

Chapter recap

Onsager-type dissipation is retained as conditional mathematics and an explicit controller alternative. Exact finite EBU survives controller changes; the old controller's dynamical guarantees do not automatically do so.

Historical reading H44: From the chain rule to network timestep certificates

Historical model; not Gaussian evidence

The next reading preserves the original line-by-line dissipation, discrete-step and network-geometry arguments. Its potential constants and admissibility assumptions belong to the historical model. Read it after the Gaussian derivation above: identify which steps use only the chain rule, which require the flux law, and which use a model-specific curvature bound. Original Part II, PDF pages 65–84. Internal chapter, section, equation and page references retain their original meaning. The new running folio belongs to this integrated volume.

What to carry forward. The shared mathematical bridge is the chain rule, not a universal controller. For the active quadratic use its declared Hessian; for a random-affordable actor there is no reason to substitute the historical flux law. A certificate that controls numerical overshoot also does not prove that an action is physically permitted.

Continuous-time dissipation, derived line by line

So far we have valued finite actor actions. We now study a related physical controller: a continuous edge flow generated by the local force. The purpose of this chapter is narrow and important. It proves that, in the absence of external drive, this particular flow law cannot increase the declared potential.

The result is sometimes described in one line as “the field dissipates”. We will not leave it at that. We will derive every term, check its unit, and show why the theorem does not apply to arbitrary circulation or arbitrary actor behaviour.

38.1 Continuous time and the dot notation

Let $x(t)$ denote the state at continuous time t . A dot above a variable means derivative with respect to time:

$$\dot{x}(t) = \frac{dx}{dt}(t). \quad (38.1)$$

If x has unit X , then $\dot{x}$ has unit X/T . The continuous network law is

$$\dot{x} = u(x) + SJ(x). \quad (38.2)$$

The vector $u(x)$ contains natural or imposed rates at nodes. The vector $J(x)$ contains edge rates. The matrix S converts edge rates into node rates. Every term therefore has unit X/T .

The edge controller remains

$$J_e = M_e[f_e - \theta_e]_+, \quad f_e = \mu_i - \eta_e \mu_j. \quad (38.3)$$

All quantities are evaluated at the same instantaneous state $x(t)$.

38.2 The chain rule for the field

Because V depends on time through the changing state $x(t)$, the multivariable chain rule gives

$$\dot{V} = \nabla V(x)^\top \dot{x} = \mu(x)^\top \dot{x}. \quad (38.4)$$

Written coordinate by coordinate,

$$\dot{V} = \sum_{i=1}^m \frac{\partial V}{\partial x_i} \dot{x}_i = \sum_{i=1}^m \mu_i \dot{x}_i. \quad (38.5)$$

This form makes the units transparent. μ_i has unit B/X and $\dot{x}_i$ has unit X/T , so each product and the sum have unit B/T .

Substitute Equation (38.2):

$$\dot{V} = \mu^\top (u + SJ) \quad (38.6)$$

$$= \mu^\top u + \mu^\top SJ. \quad (38.7)$$

The first term is the contribution of natural or external drive. The second is the contribution of edge flow.

38.3 Turning the matrix term into edge forces

Matrix multiplication can be associated in either order:

$$\mu^\top SJ = (\mu^\top S)J. \quad (38.8)$$

Column e of S is S_e . From Equation (29.12),

$$\mu^\top S_e = -f_e. \quad (38.9)$$

Therefore the row vector $\mu^\top S$ is

$$\mu^\top S = (-f_1, -f_2, \dots, -f_{|E|}) = -f^\top. \quad (38.10)$$

Multiplying by J gives

$$\mu^\top S J = -f^\top J = -\sum_e f_e J_e. \quad (38.11)$$

The product $f_e J_e$ has unit

$$\frac{B}{X} \frac{X}{T} = \frac{B}{T}. \quad (38.12)$$

It is a burden-reduction rate along edge e , analogous in mathematical form to force times flux in irreversible thermodynamics.

38.4 Using the thresholded mobility law

There are two cases for each edge.

If $f_e \leq \theta_e$, then $[f_e - \theta_e]_+ = 0$ and $J_e = 0$. Consequently $f_e J_e = 0$.

If $f_e > \theta_e$, then

$$J_e = M_e(f_e - \theta_e). \quad (38.13)$$

Solve this equation for f_e :

$$f_e = \frac{J_e}{M_e} + \theta_e. \quad (38.14)$$

Multiply by J_e :

$$f_e J_e = \frac{J_e^2}{M_e} + \theta_e J_e. \quad (38.15)$$

Both terms are non-negative because $M_e > 0$, $\theta_e \geq 0$, and $J_e \geq 0$. The same equation also holds on an inactive edge because all terms containing J_e are zero.

38.5 The continuous identity

Insert Equations (38.11) and (38.15) into Equation (38.7):

$$\dot{V} = \mu^\top u - \sum_e \left(\frac{J_e^2}{M_e} + \theta_e J_e \right). \quad (38.16)$$

This is an identity for the declared dynamics, not an approximation. It separates two effects:

$\mu^\top u$ the field-change rate caused by natural or external drive;

$\sum_e (J_e^2/M_e + \theta_e J_e)$ the non-negative dissipation rate of the derived edge flow.

38.6 The undriven theorem

If $u = 0$, the identity becomes

$$\dot{V} = - \sum_e \left(\frac{J_e^2}{M_e} + \theta_e J_e \right) \leq 0. \quad (38.17)$$

Thus V is non-increasing along every trajectory of the derived undriven continuous law for which the calculation is well-defined.

The phrase *non-increasing* is exact. It allows $\dot{V} = 0$. Equality holds when every generated flux is zero. That can happen at many states because viable bands are flat and thresholds can deactivate edges. The theorem does not say that the state converges to one unique point.

38.7 Numerical two-cell example

Let the current marginals be

$$\mu_i = -1, \quad \mu_j = -5, \quad \eta = 0.8. \quad (38.18)$$

Then

$$f = (-1) - 0.8(-5) = 3. \quad (38.19)$$

Choose $M = 2$ and $\theta = 1$. The flux is

$$J = 2[3 - 1]_+ = 4. \quad (38.20)$$

The edge dissipation rate can be calculated in two equivalent ways:

$$fJ = 3 \times 4 = 12, \quad (38.21)$$

$$\frac{J^2}{M} + \theta J = \frac{16}{2} + 1 \times 4 = 8 + 4 = 12. \quad (38.22)$$

With no drive, $\dot{V} = -12$ burden-units per unit time at that instant. This does not mean V will become negative. As the state changes, the marginals, force, and flux change; flow stops when the local force no longer exceeds the threshold.

38.8 A driven example

Keep the same edge dissipation 12. Suppose external demand and regeneration together give $\mu^\top u = 7$. Then

$$\dot{V} = 7 - 12 = -5. \quad (38.23)$$

The edge flow more than compensates for the drive at that instant.

If instead $\mu^\top u = 15$, then

$$\dot{V} = 15 - 12 = 3 > 0. \quad (38.24)$$

The controller still dissipates 12 units per time, but the external drive creates burden faster. The theorem has not failed. The driven identity reports the competition honestly.

38.9 Counterexample: arbitrary circulation

It is false that every local flow decreases V . Consider a lossless directed cycle with an externally imposed circulation $J \neq 0$ such that $SJ = 0$. The

state does not change, so $\dot{V} = 0$, even though edge rates are nonzero. If an arbitrary imposed flow moves against the gradient, it can increase V .

Why does this not contradict Equation (38.17)? Because that equation used the thresholded mobility relation between J and f . Arbitrary circulation does not necessarily satisfy it. The theorem concerns a specific gradient-derived controller, not all possible local actions.

38.10 Potential descent and useful service

A decrease in V is a mathematical property of the represented field. It does not guarantee that all demand was served. If physical resources are insufficient, the lowest-burden feasible trajectory can still contain unmet need. Conversely, an emergency action may temporarily increase V while serving a critical need.

The model must therefore record both field burden and functional outcomes. Treating one scalar as a complete social objective would erase the distinction the project is designed to preserve.

38.11 Connection to Lyapunov reasoning

A Lyapunov function is a scalar function used to study stability. A non-negative V with $\dot{V} \leq 0$ is a candidate Lyapunov function for the undriven derived dynamics. To infer convergence to a target set, additional facts are usually needed: bounded trajectories, forward invariance, regularity, and a characterization of the largest invariant subset where $\dot{V} = 0$.

The continuous identity supplies a powerful ingredient. It is not, by itself, a complete infinite-horizon homeostasis theorem for driven, constrained, multi-actor worlds.

Scope statement

Continuous dissipation is proved for the declared differentiable potential, loss-aware edge matrix, and thresholded mobility law, with all quantities evaluated at the same instantaneous state. Undriven monotonicity does not cover arbitrary flows, external actions, discontinuous settlement rules, or unrepresented fields.

Chapter recap

The chain rule converts state dynamics into field dynamics. The transpose of the edge matrix turns marginals into negative edge forces. The mobility law turns force times flux into a sum of non-negative dissipation terms. In an undriven system the declared potential cannot increase; in a driven system the drive and dissipation compete.

Exercises

1. Starting from $\dot{V} = \mu^\top \dot{x}$, reproduce every line leading to Equation (38.16).
2. Check the units of J^2/M using the units of mobility given earlier.
3. With $f = 2.5$, $M = 3$, and $\theta = 1$, calculate J , fJ , and $J^2/M + \theta J$.
4. Give one example in which $\dot{V} > 0$ in a driven system even though the edge flow follows the theorem.
5. Explain why $\dot{V} = 0$ does not imply one unique equilibrium state.

Discrete time, curvature, and the safe timestep

Computer models and transactional systems update in finite steps. A continuous law that decreases V can overshoot when implemented with a timestep that is too large. The discrete theorem quantifies this risk. It keeps the favourable first-order dissipation term and adds a second-order allowance for curvature.

39.1 Forward Euler as a finite update

For tick n , freeze the state x^n , calculate $u^n = u(x^n)$ and $J^n = J(x^n)$, and define

$$y^n = u^n + SJ^n. \quad (39.1)$$

The synchronous forward-Euler update is

$$x^{n+1} = x^n + \Delta t y^n. \quad (39.2)$$

y^n is a rate vector with unit X/T . The finite change $\Delta x^n = \Delta t y^n$ has unit X .

Forward Euler follows the tangent direction at the beginning of the tick. If the tick is long, the true field slope can change materially before the finite step ends. Curvature controls the difference between the tangent prediction and the exact field value at the new state.

39.2 What a Lipschitz gradient means

Assume the gradient satisfies

$$\|\nabla V(x) - \nabla V(y)\|_2 \leq L_V \|x - y\|_2 \quad (39.3)$$

through the relevant region. The constant L_V has unit B/X^2 . It bounds how quickly the marginal vector can change as the state changes.

For the separable squared-hinge potential, a safe global bound within one homogeneous field is

$$L_V = \max_i 2(\alpha_i + \beta_i + \chi_i), \quad (39.4)$$

although a sharper bound may use only penalty branches active on the represented region. A valid upper bound may be conservative; an underestimated bound makes the certificate unsafe.

39.3 The descent lemma

For a function with L_V -Lipschitz gradient, the descent lemma states

$$V(x + \Delta x) \leq V(x) + \nabla V(x)^\top \Delta x + \frac{L_V}{2} \|\Delta x\|_2^2. \quad (39.5)$$

The first added term is the tangent prediction. The final term is a guaranteed curvature allowance. It is quadratic in the size of the step.

Substitute $\Delta x = \Delta t y$:

$$V(x^{n+1}) - V(x^n) \leq \Delta t \mu^\top y + \frac{L_V \Delta t^2}{2} \|y\|_2^2. \quad (39.6)$$

The factor Δt^2 is why decreasing the timestep eventually makes the first-order term dominate whenever there is strict instantaneous dissipation.

39.4 Insert drive and edge dissipation

Since $y = u + SJ$,

$$\mu^\top y = \mu^\top u + \mu^\top SJ \quad (39.7)$$

$$= \mu^\top u - \sum_e \left(\frac{J_e^2}{M_e} + \theta_e J_e \right). \quad (39.8)$$

Insert this into Equation (39.6):

$$V(x^{n+1}) - V(x^n) \leq \Delta t \mu^\top u \quad (39.9)$$

$$- \Delta t \sum_e \left(\frac{J_e^2}{M_e} + \theta_e J_e \right) \quad (39.10)$$

$$+ \frac{L_V \Delta t^2}{2} \|u + SJ\|_2^2. \quad (39.11)$$

The three lines mean:

1. external or natural drive can raise or lower burden;
2. the generated edge flow contributes first-order dissipation;
3. finite stepping adds a non-negative curvature allowance.

39.5 The undriven one-edge certificate

Consider one active edge, $u = 0$, and $J > 0$. The update direction is $S_e J$. Its squared norm is

$$\|S_e J\|_2^2 = J^2 \|S_e\|_2^2 = J^2 (1 + \eta_e^2). \quad (39.12)$$

The last equality follows because S_e has entries -1 and η_e on two different coordinates:

$$\|S_e\|_2^2 = (-1)^2 + \eta_e^2 = 1 + \eta_e^2. \quad (39.13)$$

Ignoring the additional favourable threshold term gives the sufficient bound

$$V(x^{n+1}) - V(x^n) \leq -\Delta t \frac{J^2}{M_e} + \frac{L_V \Delta t^2}{2} J^2 (1 + \eta_e^2). \quad (39.14)$$

Factor out the non-negative $\Delta t J^2$:

$$V(x^{n+1}) - V(x^n) \leq \Delta t J^2 \left[-\frac{1}{M_e} + \frac{L_V \Delta t}{2} (1 + \eta_e^2) \right]. \quad (39.15)$$

The bracket is non-positive when

$$\Delta t \leq \frac{2}{L_V M_e (1 + \eta_e^2)}. \quad (39.16)$$

Check the unit. $L_V M_e$ has unit

$$\frac{B}{X^2} \frac{X^2}{BT} = \frac{1}{T}, \quad (39.17)$$

so the right side has unit T , as a timestep must.

39.6 A numerical certificate

Let $L_V = 4$, $M = 0.5$, and $\eta = 0.8$. Then

$$\Delta t_{\max} = \frac{2}{4 \times 0.5 \times (1 + 0.8^2)} \quad (39.18)$$

$$= \frac{2}{2 \times 1.64} \quad (39.19)$$

$$\approx 0.6098. \quad (39.20)$$

A registered timestep of 0.3 lies below the sufficient certificate. A timestep of 0.8 lies above it. Being above the bound does not prove that this particular tick will fail; it means the theorem no longer guarantees descent. The distinction between sufficient and necessary conditions is important.

39.7 A pure quadratic overshoot

Take the simple potential

$$V(x) = \frac{1}{2}x^2 \quad (39.21)$$

and continuous descent law $\dot{x} = -kx$ with $k > 0$. Then

$$\dot{V} = x\dot{x} = -kx^2 \leq 0. \quad (39.22)$$

Forward Euler gives

$$x^{n+1} = (1 - k\Delta t)x^n. \quad (39.23)$$

Therefore

$$V(x^{n+1}) = \frac{1}{2}(1 - k\Delta t)^2(x^n)^2. \quad (39.24)$$

The potential does not increase exactly when

$$|1 - k\Delta t| \leq 1, \quad (39.25)$$

which is equivalent to $0 \leq k\Delta t \leq 2$. For a larger step, the discrete state jumps across the minimum with increased magnitude. The continuous law is stable, but its coarse discretisation is not.

This elementary example explains why a timestep certificate is a scientific part of the discrete model rather than a mere programming preference.

39.8 Exact remainder and upper bound

The descent lemma gives an upper bound on the finite remainder. In a test we can also calculate the exact remainder

$$R_{\text{exact}} = V(x + \Delta x) - V(x) - \mu(x)^\top \Delta x. \quad (39.26)$$

The theorem requires

$$R_{\text{exact}} \leq \frac{L_V}{2} \|\Delta x\|_2^2. \quad (39.27)$$

These two numbers should not be confused. Nor is either one the stock-ledger residual. The exact remainder measures nonlinearity of V over one state step; the ledger residual measures whether physical inputs, outputs, and losses add up.

39.9 The driven sufficient condition

In a driven system, Equation (39.11) guarantees non-increase if

$$\sum_e \left(\frac{J_e^2}{M_e} + \theta_e J_e \right) \geq \mu^\top u + \frac{L_V \Delta t}{2} \|u + SJ\|_2^2. \quad (39.28)$$

The edge dissipation must cover both the first-order drive contribution and the finite-step curvature allowance. If it does not, V may increase even though the code correctly implements the derived edge law.

39.10 What the theorem excludes

The proof assumes a frozen synchronous update. State projection, clipping, sequential edge updates, mid-tick reclassification, or replacing accepted flow with a different executed flow changes the mathematical map. Those operations may be useful, but their effects need separate analysis.

A projected non-negative state can produce a visually plausible trajectory while hiding requested service that was not physically available. Closing a post-hoc stock ledger is not enough; the service and acceptance ledgers must also distinguish request, feasible acceptance, delivery, unmet need, and loss.

Scope statement

The discrete inequality is a sufficient theorem for the declared synchronous Euler map and a valid curvature bound. A certificate below which descent is guaranteed need not be the largest stable timestep. A run above the certificate is outside the guarantee, not automatically a counterexample. A run below it still says nothing by itself about service or infinite-horizon stability.

Chapter recap

Finite stepping adds a curvature term of order Δt^2 . The one-edge certificate makes the favourable first-order dissipation dominate this remainder. Timestep bounds have physical units and mathematical meaning. Sufficient bounds certify a region; crossing them removes the guarantee but does not prove immediate failure.

Exercises

1. Verify the units of every term in Equation (39.11).
2. Recalculate the numerical one-edge certificate for $L_V = 3$, $M = 2$, and $\eta = 0.5$.
3. For the quadratic overshoot example with $k = 4$, compare the updates at $\Delta t = 0.2$, 0.5 , and 0.75 .
4. Explain the difference between an exact curvature remainder, a curvature upper bound, and a stock-ledger residual.
5. Why can projection change the theorem even if it prevents negative numerical states?

Network geometry and multi-edge timestep certificates

The one-edge calculation shows where efficiency and mobility enter a timestep bound. A network introduces overlap: several edges may share a source or destination, so their state-change vectors can reinforce or partially cancel. The matrix norm in the network certificate measures this geometry.

40.1 Mobility as a diagonal matrix

Collect edge mobilities into

$$D_M = \text{diag}(M_1, \dots, M_{|E|}). \quad (40.1)$$

The word “diag” means a square matrix whose listed values lie on the diagonal and whose off-diagonal entries are zero. Its square root is

$$D_M^{1/2} = \text{diag}(\sqrt{M_1}, \dots, \sqrt{M_{|E|}}). \quad (40.2)$$

Similarly,

$$D_M^{-1/2} = \text{diag}(1/\sqrt{M_1}, \dots, 1/\sqrt{M_{|E|}}). \quad (40.3)$$

These matrices let us compare the state-change size $\|SJ\|$ with the mobility-weighted dissipation $\sum_e J_e^2/M_e$.

40.2 The spectral norm in plain language

For a matrix A , the spectral norm is

$$\|A\|_2 = \max_{y \neq 0} \frac{\|Ay\|_2}{\|y\|_2}. \quad (40.4)$$

It is the largest factor by which the matrix can stretch a vector in Euclidean length. It does not mean that the system literally chooses the worst vector on every tick. It supplies a conservative bound valid for every possible vector.

Set

$$A = SD_M^{1/2}, \quad y = D_M^{-1/2} J. \quad (40.5)$$

Then

$$Ay = SD_M^{1/2} D_M^{-1/2} J = SJ. \quad (40.6)$$

By the definition of the spectral norm,

$$\|SJ\|_2^2 \leq \|SD_M^{1/2}\|_2^2 \|D_M^{-1/2} J\|_2^2. \quad (40.7)$$

The final norm expands to

$$\|D_M^{-1/2} J\|_2^2 = \sum_e \frac{J_e^2}{M_e}. \quad (40.8)$$

Thus the same quantity that appears in dissipation also controls the maximum network state-change norm.

40.3 The global spectral certificate

In the undriven discrete inequality, discard the favourable threshold terms and apply Equation (40.7):

$$V(x^{n+1}) - V(x^n) \leq -\Delta t \sum_e \frac{J_e^2}{M_e} \quad (40.9)$$

$$+ \frac{L_V \Delta t^2}{2} \|SD_M^{1/2}\|_2^2 \sum_e \frac{J_e^2}{M_e}. \quad (40.10)$$

Factor the weighted flux sum:

$$V(x^{n+1}) - V(x^n) \leq \Delta t \left(\sum_e \frac{J_e^2}{M_e} \right) \left[-1 + \frac{L_V \Delta t}{2} \|SD_M^{1/2}\|_2^2 \right]. \quad (40.11)$$

A sufficient network-wide certificate is therefore

$$\Delta t \leq \frac{2}{L_V \|SD_M^{1/2}\|_2^2}. \quad (40.12)$$

For a graph containing only one edge, the squared norm becomes $M_e(1 + \eta_e^2)$ and the result reduces to the one-edge certificate.

40.4 Why shared endpoints tighten the bound

Suppose one source has four outgoing edges. Each edge column contains -1 in the same source row. When all four rates are positive, their source withdrawals add. The matrix can therefore produce a large state-change norm from a given mobility-weighted rate vector. The spectral norm detects this overlap.

If the edges acted on disjoint pairs of nodes, their columns would be more orthogonal and the worst amplification could be smaller. Network topology matters not because a global controller reads every state, but because local edge effects can share coordinates.

40.5 Active-set certificates

The full matrix includes every permitted edge, even when many are inactive at the current state. If only a subset A_E has $J_e > 0$, form the restricted matrix S_{A_E} and restricted mobility matrix D_{M,A_E} . Then

$$\Delta t \leq \frac{2}{L_V \|S_{A_E} D_{M,A_E}^{1/2}\|_2^2} \quad (40.13)$$

can be less conservative.

The active set must be determined from the frozen state used for the update. Recomputing it halfway through a synchronous tick would change the map being certified.

40.6 A state-specific certificate

For a particular nonzero flux vector in the undriven case, define the exact instantaneous dissipation

$$D(J) = \sum_e \left(\frac{J_e^2}{M_e} + \theta_e J_e \right). \quad (40.14)$$

The discrete inequality gives descent whenever

$$-\Delta t D(J) + \frac{L_V \Delta t^2}{2} \|SJ\|_2^2 \leq 0. \quad (40.15)$$

For $SJ \neq 0$, divide by Δt and solve:

$$\Delta t \leq \frac{2D(J)}{L_V \|SJ\|_2^2}. \quad (40.16)$$

For a nonzero flux produced by the present controller, SJ cannot equal zero. To see why, suppose $SJ = 0$. Then

$$0 = \mu^\top SJ = -D(J). \quad (40.17)$$

Since every term in $D(J)$ is non-negative, $D(J) = 0$ forces every $J_e = 0$, contradicting the assumption of nonzero generated flux.

40.7 Gershgorin and degree-based alternatives

Computing a spectral norm is a global mathematical operation on the declared graph. It can be performed during model certification even when runtime decisions remain local. When a transparent conservative bound is preferred, row sums or Gershgorin-disc arguments can bound the largest eigenvalue of the relevant weighted matrix using node degrees and edge weights.

Such a bound is usually easier to audit but more conservative. The important scientific requirement is not that one method always be used; it is that the chosen bound be valid, reproducible, and locked before the result it is meant to certify.

40.8 A small fan-out calculation

Let one source feed two destinations with efficiencies 1 and equal mobilities $M = 1$. The edge matrix is

$$S = \begin{pmatrix} -1 & -1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (40.18)$$

Then

$$S^\top S = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}. \quad (40.19)$$

The eigenvalues are 3 and 1, so $\|S\|_2^2 = 3$. If $L_V = 2$, the network certificate is

$$\Delta t \leq \frac{2}{2 \times 3} = \frac{1}{3}. \quad (40.20)$$

Each edge alone would give $\Delta t \leq 1/2$. The shared source tightens the multi-edge guarantee from 0.5 to approximately 0.333.

40.9 Local decision versus global certification

The presence of a spectral certificate does not mean each actor reads the whole global state. The graph and parameter set can be certified at design time. Runtime action selection can still read only permitted local endpoints, the local source budget, and declared edge parameters.

This distinction appears throughout the project: global mathematics can prove a property of a local rule without becoming a global decision input. A local thermostat can be analysed using a global building model; the thermostat does not need to solve that model every second.

Scope statement

The spectral, active-set, degree-based, and state-specific certificates are sufficient bounds for the declared synchronous update. They rely on compatible coordinate scaling and a valid curvature bound. They do not establish physical feasibility, reserve preservation, service success, or optimality of the chosen graph.

Chapter recap

The matrix norm measures how overlapping edge columns can amplify a flux vector into node-state change. Mobility weighting links this geometry to dissipation. The spectral certificate is global in proof but need not be global in runtime information. Active-set and state-specific versions can be sharper; transparent degree bounds can be more conservative.

Exercises

1. Verify the eigenvalues of the two-edge fan-out matrix $S^\top S$.
2. Explain in words why two edges sharing a source are not geometrically independent.
3. Derive Equation (40.16) from the undriven discrete inequality.
4. Why can a design-time global certificate coexist with a local runtime decision rule?
5. Give one reason an active-set certificate can be larger than a full-graph certificate.

Feasibility, routes and the long-run boundary

An action account can be exact while a world is unable to meet its needs. A potential can have a unique minimum while a process never reaches it. These are not paradoxes. They follow from the distinction between a landscape, a physical domain and a transition rule.

45.1 The historical preservation certificate

P1C addresses a specific regenerative-source export question. In its historical applicable state and typing, a robust export-rate budget has the form

$$Q_{\max} = \frac{[(x - \varepsilon_x) + \Delta t(u - \varepsilon_u) - R_{\text{eff}}]_+}{\Delta t}. \quad (45.1)$$

R_{eff} is a stock quantity. It is not the old homeostatic reserve R , the Gaussian scale, or the new capacity balance B . Under the certificate's feasibility and uncertainty assumptions, capping aggregate export protects its stated successor boundary. It does not move material toward a destination that lacks it.

The source classifier matters as much as this arithmetic. The preserved P1C design does not grant preservation-safe export from every finite stock merely because the expression is positive. Its regenerative-source premise cannot be removed to create a permission law for the closed finite Gaussian world. The historical reading in this chapter retains the original proof at its own scope.

45.2 A route is a sequence of physical transformations

Take a three-cell teaching state $(18, 10, 2)$, reference $(10, 10, 10)$, and equal scales two. Its potential is sixteen. Move four units from Ridge to the hub: the endpoint is $(14, 14, 2)$ and the potential is twelve. Then move four from the hub to Vale: $(14, 10, 6)$ has potential four. The first edge contributes four and the second eight, totaling twelve.

The intermediate hub matters even though its final stock is unchanged. A direct transfer, if it existed, would have the same field endpoint value in this lossless example, but need not have the same physical costs, timing or permissions in a fuller world. One cannot erase an intermediate fuel, electricity or storage transformation simply because endpoint algebra is convenient.

45.3 Planning is not local valuation

A planner might ask which sequence of actions reaches a target at minimum cost. Bellman reasoning compares present choices and future continuations. That can be a legitimate separately declared problem, but it introduces state, horizon, objective and information requirements beyond a present local quote. A random-affordable actor must not secretly call that planner and still be described as choosing without target-directed optimization.

There is also a difference between a route's total favorable field value and its affordability at every intermediate point. A useful route can begin with a negative-valued action. End-to-end positivity does not fund that first step under zero balances without an explicitly different policy. This is a possible limitation to report, not a reason to add silent borrowing.

45.4 Boundedness is built into the closed domain

Since $0 \leq x_i \leq M$,

$$V(x) \leq \frac{1}{2} \sum_i \frac{\max\{(x_i^*)^2, (M - x_i^*)^2\}}{\sigma_i^2} < \infty. \quad (45.2)$$

Both an EBU-filtered arm and a physical-random control inherit this bound if they stay in the same simplex. Showing that one arm's potential remains finite cannot discriminate their usefulness. A long-run study must ask about quantities that can genuinely differ: distance occupation, recovery time, cycling, refusal, distribution or declared service.

45.5 Invariance, stability and attraction

An invariant domain contains every allowed successor. Stability of a reference means sufficiently small departures remain small under the specified dynamics. Attraction means departures approach the reference. Stationarity is a property of a probability law under a specified transition kernel. These statements answer different questions.

Strict convexity alone proves none of them for an arbitrary actor process. A finite menu can trap the system away from reference; a capacity policy can permit cycling; a directed graph can block useful routes. Even if the physical state has a stationary distribution under continuing forcing, the audit J and balances can drift. A stationary claim about the augmented process (x, B, J) needs its own argument.

An adverse case worth keeping

Consider two cells with odd integer stocks and only even transfer quantities. If the reference requires even stocks, every reachable stock remains odd: adding or subtracting an even number preserves parity. The reference can be mass-compatible but unreachable under that menu. This exact obstruction is not a Gaussian calibration result and should not be removed after a study reveals it. It is a reason to state the menu before testing recovery.

Chapter recap

Permission, route reachability and long-run dynamics are separate from exact valuation. Keeping them separate makes negative results intelligible rather than embarrassing exceptions to be hidden.

Historical reading H45: Preservation, routes and the distinction between invariance and attraction

Historical model; not Gaussian evidence

These original chapters retain the full preservation certificate, route and Bellman discussion, and long-run distinctions. Their reserve, uncertainty, regeneration and threshold symbols are historical assumptions, not additional fields silently added to the Gaussian world. The route material is a planning comparison, not the active actor's algorithm. Original Part II, PDF pages 85–113. Internal chapter, section, equation and page references retain their original meaning. The new running folio belongs to this integrated volume.

What to carry forward. Keep three conclusions separate: a certificate may protect one declared source boundary; a planner may find a beneficial route; and a process may or may not recover. None implies the others. The next laboratory returns to fixed Gaussian references and lossless finite quantities, without importing this reading's regenerative plant.

One-step preservation, feasibility, and genuine need

Potential descent and source preservation are related but not identical. A transfer can lower the total represented burden while withdrawing too much from one regenerative source in one coarse tick. Conversely, a preservation gate can keep a source above its reserve while leaving demand unmet. This chapter derives the local export budget that separates those questions.

41.1 The source-side update

Consider a protected source node i . Let

x_i be its frozen starting stock;

u_i be its frozen natural or exogenous rate before actor export;

R_i^{eff} be the effective protected reserve, including any declared safety offset;

Q_i be the aggregate accepted outgoing rate during the tick;

I_i be the aggregate usable incoming rate during the tick.

Its synchronous update is

$$x_i^{n+1} = x_i + \Delta t(u_i + I_i - Q_i). \quad (41.1)$$

Incoming flow satisfies $I_i \geq 0$ in the simple stock model. For a conservative export permission, ignore that incoming help when calculating how much the source may release. Require

$$x_i + \Delta t(u_i - Q_i) \geq R_i^{\text{eff}}. \quad (41.2)$$

This is a one-step boundary condition, not a potential argument.

41.2 Deriving the safe export budget

Begin with Equation (41.2):

$$x_i + \Delta t u_i - \Delta t Q_i \geq R_i^{\text{eff}}. \quad (41.3)$$

Move the export term to the right and the reserve to the left:

$$x_i + \Delta t u_i - R_i^{\text{eff}} \geq \Delta t Q_i. \quad (41.4)$$

Because $\Delta t > 0$, divide without reversing the inequality:

$$Q_i \leq \frac{x_i + \Delta t u_i - R_i^{\text{eff}}}{\Delta t}. \quad (41.5)$$

An export rate cannot be negative. Define

$$Q_{i,\max} = \frac{[x_i + \Delta t u_i - R_i^{\text{eff}}]_+}{\Delta t}. \quad (41.6)$$

The numerator is the amount by which the no-export successor lies above the effective reserve. Dividing by tick duration converts that finite stock margin into an allowable rate.

41.3 Why the positive part needs a feasibility flag

Suppose

$$x_i + \Delta t u_i < R_i^{\text{eff}}. \quad (41.7)$$

The source crosses the reserve even with $Q_i = 0$. The positive part in Equation (41.6) returns zero, correctly permitting no export, but zero export does not make the next state safe. The state is locally infeasible under the frozen no-export dynamics.

A robust implementation must therefore report both

1. the permitted export rate, which is zero; and
2. the no-export infeasibility, which remains a material domain outcome.

Silently treating every zero budget as successful preservation would hide natural collapse or excessive imposed demand.

41.4 A four-state classification

One useful source-side classification is:

Protected and feasible (P) current stock is at or above the effective reserve and the no-export successor remains at or above it. A regenerative source may use the robust aggregate export budget. A finite or irreversible source receives zero preservation-safe export; extraction would be depletion, not preservation.

Recovery (R) current stock is below the reserve. Protected export is zero while recovery is required.

Locally infeasible (I) current stock is not yet below the reserve, but the no-export successor crosses it. Export is zero and the failure is reported.

Declared flow source (F) the represented source is a certified flow rather than a protected stock. Its budget is the smaller of its declared flow cap and the non-negative available-stock rate, $\min\{\text{flow cap}, [x_i + \Delta t u_i]_+ / \Delta t\}$.

The labels are an implementation aid. The theorem depends on the inequalities, not on the letters.

41.5 A numerical feasible case

Let

$$x_i = 10, \quad R_i^{\text{eff}} = 8, \quad u_i = -0.5 \text{ units/hour}, \quad \Delta t = 2 \text{ hours}. \quad (41.8)$$

Without export, the successor is

$$x_i + \Delta t u_i = 10 + 2(-0.5) = 9, \quad (41.9)$$

which is one unit above reserve. The maximum export rate is

$$Q_{i,\max} = \frac{9-8}{2} = 0.5 \text{ units/hour.} \quad (41.10)$$

At the full budget, the finite exported quantity is one unit and the conservative next stock is exactly eight.

If an incoming delivery of 0.2 units/hour also occurs, the actual next stock is

$$10 + 2(-0.5 + 0.2 - 0.5) = 8.4. \quad (41.11)$$

The incoming flow creates extra margin because it was not relied on when the export budget was certified.

41.6 A numerical infeasible case

Keep $x_i = 10$, $R_i^{\text{eff}} = 8$, and $\Delta t = 2$, but let $u_i = -1.2$ units/hour. The no-export successor is

$$10 + 2(-1.2) = 7.6 < 8. \quad (41.12)$$

Equation (41.6) returns zero. Nevertheless the expected successor is below the boundary. The correct report is not “the gate preserved the source”. It is “no export was allowed, and the source was locally infeasible under the declared drive”.

This distinction matters in a hospital system. If essential local consumption alone exceeds replenishment, refusing external shipments does not solve the shortage. The model should expose the need for emergency supply, demand management, substitution, or structural change.

41.7 Measurement margins

Suppose measured stock may overstate true stock by at most ε_x and the estimated drive may overstate true drive by at most ε_u . Use the conservative values $x_i - \varepsilon_x$ and $u_i - \varepsilon_u$:

$$Q_{i,\max}^{\text{rob}} = \frac{[(x_i - \varepsilon_x) + \Delta t(u_i - \varepsilon_u) - R_i^{\text{eff}}]_+}{\Delta t}. \quad (41.13)$$

This is not the same as a floating-point tolerance. ε_x and ε_u are physical uncertainty margins with units. A numerical tolerance only decides whether two finite representations should be treated as equal for diagnostics. It must not create material export capacity.

For a robust preservation certificate, the conservative current stock and conservative no-export successor must themselves be feasible:

$$x_i - \varepsilon_x \geq R_i^{\text{eff}}, \quad (x_i - \varepsilon_x) + \Delta t(u_i - \varepsilon_u) \geq R_i^{\text{eff}}. \quad (41.14)$$

If either inequality fails, the positive part may return a zero export budget, but refusing export does not prove preservation under the declared uncertainty envelope. Nominal State P and robust theorem eligibility are therefore distinct when the physical margins are nonzero.

41.8 Several outgoing requests

Let $\mathcal{O}(i)$ be the set of edges leaving source i . Each edge requests q_e^{req} . The aggregate request is

$$Q_i^{\text{req}} = \sum_{e \in \mathcal{O}(i)} q_e^{\text{req}}. \quad (41.15)$$

Write $Q_{i,\text{max}}$ for the applicable budget returned after classification. It is $Q_{i,\text{max}}^{\text{rob}}$ for a regenerative class-P source, the flow-cap/available-stock minimum for class F, and zero for class R, class I, or a finite or irreversible class-P source.

The released controller uses the following fail-closed diagnostic and scaling convention:

$$\sigma_i = \begin{cases} \min \left(1, \frac{Q_{i,\text{max}}}{Q_i^{\text{req}}} \right), & Q_i^{\text{req}} > 0 \text{ and } Q_{i,\text{max}} > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (41.16)$$

Thus a zero aggregate request is accepted as zero and recorded with $\sigma_i = 0$; a positive request against a zero budget is also scaled to zero. The value of σ_i in the zero-request branch cannot change a physical quantity, but matching the released convention makes diagnostic traces unambiguous.

Every outgoing request is scaled from the same frozen factor:

$$q_e^{\text{acc}} = \sigma_i q_e^{\text{req}}. \quad (41.17)$$

Then

$$\sum_{e \in \mathcal{O}(i)} q_e^{\text{acc}} = \sigma_i Q_i^{\text{req}} \leq Q_{i,\max}. \quad (41.18)$$

Proportional scaling is a physically order-independent way to respect the aggregate budget. It is not asserted to be the best social priority rule.

41.9 The one-step preservation theorem

Theorem: aggregate reserve preservation. Assume

1. $x_i \geq R_i^{\text{eff}}$;
2. the no-export successor satisfies $x_i + \Delta t u_i \geq R_i^{\text{eff}}$;
3. the applicable protected-source budget obeys $0 \leq Q_{i,\max} \leq [x_i + \Delta t u_i - R_i^{\text{eff}}] / \Delta t$;
4. accepted aggregate export obeys $Q_i^{\text{acc}} \leq Q_{i,\max}$;
5. incoming usable flow is non-negative;
6. the synchronous update uses the same frozen quantities.

Then $x_i^{n+1} \geq R_i^{\text{eff}}$.

Proof. From Equation (41.1),

$$x_i^{n+1} = x_i + \Delta t (u_i + I_i - Q_i^{\text{acc}}) \quad (41.19)$$

$$\geq x_i + \Delta t (u_i - Q_i^{\text{acc}}) \quad \text{because } I_i \geq 0, \quad (41.20)$$

$$\geq x_i + \Delta t (u_i - Q_{i,\max}) \quad \text{because } Q_i^{\text{acc}} \leq Q_{i,\max}, \quad (41.21)$$

$$\geq R_i^{\text{eff}}. \quad (41.22)$$

The final inequality follows from the no-export feasibility and budget assumptions. $\square$

For the robust version, additionally assume Equation (41.14), $x_i^{\text{true}} \geq x_i - \varepsilon_x$, $u_i^{\text{true}} \geq u_i - \varepsilon_u$, and $Q_i^{\text{acc}} \leq Q_{i,\max}^{\text{rob}}$. Then

$$x_i^{n+1} \geq (x_i - \varepsilon_x) + \Delta t(u_i - \varepsilon_u - Q_{i,\max}^{\text{rob}}) \geq R_i^{\text{eff}}. \quad (41.23)$$

If conservative no-export feasibility fails, $Q_{i,\max}^{\text{rob}} = 0$ is a refusal result, not a robust preservation conclusion.

41.10 Why the theorem is an invariant implication

The theorem has the logical form

$$x^n \in \mathcal{K} \text{ and assumptions hold at tick } n \implies x^{n+1} \in \mathcal{K}, \quad (41.24)$$

where $\mathcal{K}$ contains states above the protected reserve. If the same conditions are re-established at every tick, induction can extend the boundary property over many ticks. But the no-export feasibility condition may cease to hold under a shock, changing demand, model error, or exhausted regeneration. The one-step theorem does not make that condition permanent.

41.11 Preservation is not service

Suppose one tick lasts $\Delta t = 1$ hour. Two hospitals each request an oxygen transport rate of 80 cylinders per hour, while the safe source budget is 100 cylinders per hour. Proportional scaling accepts 50 cylinders per hour on each edge. Let both edge efficiencies be $\eta_e = 0.9$. The transport ledger for this one-hour tick is therefore

$$\text{requested transport rate} = 160 \text{ cylinders/hour}, \quad (41.25)$$

$$\text{accepted transport rate} = 100 \text{ cylinders/hour}, \quad (41.26)$$

$$\text{sent amount} = \Delta t \sum_e q_e^{\text{acc}} = 100 \text{ cylinders}, \quad (41.27)$$

$$\text{transport-delivered amount} = \Delta t \sum_e \eta_e q_e^{\text{acc}} = 90 \text{ cylinders}, \quad (41.28)$$

$$\text{transport loss} = \Delta t \sum_e (1 - \eta_e) q_e^{\text{acc}} = 10 \text{ cylinders}. \quad (41.29)$$

Transport delivery is not yet the bounded-service quantity. Suppose each hospital already has $x_j = 5$ usable cylinders, has no other local drive during the tick ($u_j = 0$), and has demand rate $d_j = 80$ cylinders per hour. For each hospital the released bounded-service semantics give

$$\text{available}_j = x_j + \Delta t(u_j + \eta_j q_j^{\text{acc}}) = 5 + 1(0 + 0.9 \cdot 50) = 50, \quad (41.30)$$

$$\text{demand amount}_j = \Delta t d_j = 80, \quad (41.31)$$

$$\text{service}_j = \min(\text{demand amount}_j, \text{available}_j) = 50, \quad (41.32)$$

$$\text{unmet}_j = \text{demand amount}_j - \text{service}_j = 30. \quad (41.33)$$

Total service is 100 cylinders and total unmet demand is 60. Service exceeds the 90 cylinders delivered in transport because the hospitals began with ten cylinders of usable inventory. In a different state it could be capped below transport delivery by lower demand. This is why requested transport, accepted transport, delivered stock, available stock, service, and unmet need must remain distinct fields.

An honest model does not manufacture service by projecting negative stocks back to zero or by counting requested quantities as delivered quantities.

41.12 Need priority and emergency borrowing

The preservation theorem does not decide how scarce service is allocated. A medical priority rule may direct more of the accepted stock to the patient with the most urgent need. If serving that need requires a high-burden alternative source, the negative EBU should remain visible. A later EBU institution

could allow borrowing, insurance, mutual aid, or public assumption of the burden.

The account should never make an essential need appear physically cheap by deleting its cost. High EBU can be the signal that mobilises long-term redesign: more local production, more resilient inventory, lower-loss transport, or a different hospital network.

41.13 What descent does not add

Even if the global potential decreases during the tick, a source reserve can be violated if a larger improvement at the destination outweighs source damage in V . The direct export budget is therefore not redundant.

Conversely, preserving every source boundary does not guarantee that V decreases; external demand or excess at destinations may raise it. Preservation and descent are two compatible certificates aimed at different failure modes.

Scope statement

The theorem preserves one declared source reserve for one synchronous tick when no-export feasibility, measurement bounds, and aggregate accepted export conditions hold. It does not prove that demand is feasible, that the reserve certificate is ecologically correct, that proportional allocation is fair, or that the conditions hold forever.

Chapter recap

The safe export budget is the no-export stock margin above reserve divided by tick duration. A zero budget can mean either safe refusal or unavoidable local infeasibility, so the status must be reported. Multiple requests share one aggregate cap. Under explicit conditions the next source state remains above reserve, while service and priority remain separate questions.

Exercises

1. Derive Equation (41.6) from the reserve inequality without skipping algebraic steps.

2. For $x = 12$, $R^{\text{eff}} = 9$, $u = -0.4$, and $\Delta t = 2$, calculate the no-export successor and maximum export rate.
3. Repeat with $u = -1.8$. Classify the source and explain why zero export is not a successful preservation result.
4. Three requests are 4, 3, and 5 units/hour and the budget is 6. Calculate σ and all accepted rates.
5. Explain the difference between a physical uncertainty margin and a floating-point comparison tolerance.
6. Construct a case in which global V falls while one source reserve is crossed.

Long-range needs, paths, and Bellman reasoning

An actor may need medicine, a replacement pump, a specialised repair tool, or another essential item that exists only far away. The local-field theory does not require every transaction to occur between neighbouring people. It requires the physical route to be represented as a sequence of accountable transformations.

This chapter explains how EBU can be calculated for paths across a grid or network, why route discovery is not the same as settlement, and how Bellman's principle replaces the misleading idea that distant EBU should follow one universal inverse-square formula.

42.1 From geographic points to accounting nodes

To evaluate actions “at every point”, first decide the spatial resolution of the field. A planet can be partitioned into cells, facilities, transport hubs, or other measurable nodes. The choice is a modelling and measurement decision: finer grids reveal more local variation but require more data.

A geographic coordinate alone does not define an EBU state. The node must carry the relevant measured field values and certified parameters. Two hospitals at the same distance from a factory can have different needs, storage capacities, transport efficiencies, and regenerative contexts.

Edges represent physically available steps: road segments, rail links, sea routes, electricity transmission, warehouse transfers, or production transformations. A long-range transaction is then a path

$$P = (e_1, e_2, \dots, e_K) \quad (42.1)$$

through the graph.

42.2 Settlement along a realised path

First consider an ordered sequence of action-only path segments under one fixed field, with no natural drive applied between receipt lines. Let the live state before segment k be x_{k-1} and after it be x_k . The segment receipt is

$$E_k = V(x_{k-1}) - V(x_k) - C_k. \quad (42.2)$$

Summing along the path gives

$$E(P) = V(x_0) - V(x_K) - \sum_{k=1}^K C_k. \quad (42.3)$$

Every segment can use local data from its complete support. The path total closes exactly because intermediate field states cancel. This compact form is the path version of action-only receipt closure, not a license to omit natural change during a journey that spans time.

In this undriven action-only case, if two paths start and end at exactly the same complete states under the same field definition, their field contributions are identical. Their EBU difference is then determined by declared process burdens:

$$E(P) - E(Q) = -C(P) + C(Q). \quad (42.4)$$

For a route whose segments occur across driven ticks, let z_k be the no-action successor from x_{k-1} and let

$$\delta_k = V(z_k) - V(x_{k-1}), \quad E_k = V(z_k) - V(x_k) - C_k. \quad (42.5)$$

The correct route ledger is then

$$E(P) = V(x_0) - V(x_K) + \sum_{k=1}^K \delta_k - \sum_{k=1}^K C_k. \quad (42.6)$$

Two driven routes with equal endpoints can still have different totals because their elapsed times and states can produce different $\sum_k \delta_k$. That difference is physical information, not a failure of local accounting.

In practice different routes may produce different loss, delay, spoilage, intermediate stock effects, or infrastructure wear, so their complete endpoint states may also differ.

42.3 Planning costs versus settled EBU

Before acting, a route planner needs estimates. Let $c_e(x)$ be a current planning cost for using edge e from state x . It may be the negative of a candidate exact quote, a burden estimate, a travel-time term, or a declared multi-criteria vector. The planner can compare feasible paths using these current local edge data.

The settled EBU is different. It is calculated from the measured action and live state when each segment occurs. If congestion, weather, inventory, or field state changes, the realised settlement can differ from the earlier plan. The system should replan rather than pretend the initial forecast remained true.

42.4 Bellman's principle

Suppose the planning cost is additive along a path. Let $D(j)$ denote the lowest known cost of reaching node j from a specified origin. For every edge $e = (i, j)$ entering j , a candidate path reaches i and then uses e . Thus

$$D(j) = \min_{e=(i,j)} \{D(i) + c_e\}. \quad (42.7)$$

Read the equation from inside outward:

1. $D(i)$ is the best cost already found to reach predecessor i .
2. c_e is the current cost of taking edge $i \rightarrow j$.
3. $D(i) + c_e$ is the cost of the complete candidate route through i .
4. $\min$ selects the smallest candidate among all incoming edges.

This is the principle of optimality associated with dynamic programming. It does not say that a package or photon mentally tests every route. It says that an optimal path has optimal subpaths under the stated additive, state-fixed planning model.

42.5 A four-node medicine example

Let node A be a medicine factory, B a regional warehouse, C an airport, and D the clinic. Suppose the current planning burdens are

edge	planning cost
$A \rightarrow B$	4
$B \rightarrow D$	3
$A \rightarrow C$	2
$C \rightarrow D$	8
$B \rightarrow C$	1

Set $D(A) = 0$. Then

$$D(B) = 0 + 4 = 4, \quad (42.8)$$

$$D(C) = \min(0 + 2, 4 + 1) = 2, \quad (42.9)$$

$$D(D) = \min(4 + 3, 2 + 8) = 7. \quad (42.10)$$

The planned route is $A \rightarrow B \rightarrow D$ with cost 7. The direct airport leg is shorter in one geographical sense, but its cold-chain burden makes the complete route cost larger.

If a storm changes the $B \rightarrow D$ edge cost to 10, the planner should recompute. The route $A \rightarrow C \rightarrow D$ may then be preferable. The receipt still settles the segments actually performed.

42.6 Negative edge costs and cycles

Positive EBU actions can appear as negative planning burdens if a planner uses $c_e = -E_e$. Algorithms such as Dijkstra's method require non-negative edge costs and are not valid without modification. Bellman-Ford-style relaxation can handle negative edges provided there is no reachable negative-cost cycle that makes the planning objective unbounded.

The EBU closed-cycle theorem helps conceptually: exact live-state settlement with non-negative process burdens cannot produce a positive net EBU on a

closed undriven state cycle. But a frozen planning graph can still contain an apparent negative cycle if it reuses stale favourable quotes. That is a warning that the additive static planning approximation has exceeded its valid scope.

Planning must not be mistaken for permission to settle the same frozen field improvement repeatedly.

42.7 Fermat's principle and what it teaches

Fermat's principle says that light follows a path making optical travel time stationary under the physical conditions. It does not require each photon to run a conscious search. The global path statement follows from local wave dynamics and variational structure.

The analogy suggests two useful questions for EBU:

1. Can route-level costs be composed from local edge contributions under explicit assumptions?
2. Can a local adaptation rule make repeated traffic reshape the network toward lower burden?

The first question has the path-closure and Bellman structures above. The second is a future adaptive-network research problem. The present local quote does not prove that factories, hospitals, railways, or settlements relocate optimally by themselves.

42.8 Why gravity is not the route equation

It is worth comparing the models directly.

Feature	Inverse-square gravity	EBU action field
Interaction	every mass pair in the ideal law	declared feasible transformations
Distance	explicit $1/r^2$ factor	route-specific efficiency, time, capacity, energy, and intermediate states
Source quantity	gravitational mass	measured state and local marginal
Direction	spatial attraction	directed action transformation
Finite action value	mechanical work along a path	exact field difference minus process burden
Decision layer	not applicable to a passive mass	actor or planner selects among feasible needs and routes

The shared mathematical idea is that a local gradient can guide motion and a line integral can accumulate a finite effect. The physical laws and variables are different.

42.9 Several non-substitutable fields

A long route may affect water, carbon, soil, biodiversity, and critical materials. Adding all burdens into one scalar EBU requires a calibrated conversion law. Without that law, the honest route account is a vector

$$\mathbf{E}(P) = \begin{pmatrix} E_{\text{water}}(P) \\ E_{\text{carbon}}(P) \\ E_{\text{soil}}(P) \\ \vdots \end{pmatrix}. \tag{42.11}$$

Route comparison then becomes a multi-criteria decision problem with declared constraints or priorities. A mathematical vector does not automatically provide one total ordering. This limitation is preferable to inventing an unsupported universal exchange rate.

42.10 Repeated needs can reveal structural opportunities

Suppose a rare medicine must initially travel by aircraft because no other route meets its temperature and time constraints. The action may carry large negative EBU and still be necessary. If the need recurs, the receipt history can identify which contributions dominate:

- repeated aviation energy;
- spoilage from poor cold storage;
- emergency production with inefficient batches;
- excessive distance between inventory and clinical need;
- lack of a compatible rail or regional distribution link.

Candidate adaptations can then be evaluated as actions in their own right. A regional store has construction burden but may reduce thousands of later route burdens. A local plant may be worthwhile only above a certain demand frequency. A hospital relocation could reduce logistics burden while increasing other access costs. The field account supplies evidence; an adaptive planning model must compare the full transition and future scenarios without crediting unrealised savings.

42.11 A repair-part example

A remote water-treatment plant needs a specialised pump seal. Without it, the plant will fail and a community will lose safe water. The nearest compatible part is overseas.

The immediate route has high transport burden, but the no-action alternative has a large destination deficit and possibly cascading field damage. The exact quote compares these live alternatives. Later, repeated seal failures may justify local spare inventory, redesigned pumps, regional manufacture, or standardised components. EBU does not turn the first emergency shipment into a luxury. It records why the present system made the genuine need expensive and provides a basis for reducing that cost next time.

42.12 What route planning must not read

If an experiment claims a local information boundary, the runtime selector may read only declared local states, edge parameters, and allowed planning labels. Global outcome metrics such as total planetary V , future realised service, or an experiment's final classification must not enter the decision path unless the architecture explicitly changes.

Researchers may calculate those quantities afterward to evaluate the system. This preserves the distinction between a globally analysed local rule and a centrally omniscient controller.

Scope statement

The compact endpoint path closure holds for ordered, action-only live-state segments under one fixed field, with complete supports and no duplicated settlement. A route spanning driven ticks uses Equation (42.6) and retains its drive residual. Bellman recursion applies to an additive planning model whose edge costs are held fixed during the calculation. State-dependent, time-dependent, multi-resource, capacity-constrained routes require corresponding extensions. Planned cost is not final settlement, and no inverse-square distance law replaces physical route data.

Chapter recap

Long-range action is represented by a path of local accountable transformations. Segment receipts telescope to the path field difference minus process burdens. Bellman's equation compares additive current route estimates; live EBU settles the realised sequence. Essential distant needs remain possible, and repeated high burdens can motivate infrastructure redesign without pretending that the redesign has already occurred.

Exercises

1. Add an edge $C \rightarrow B$ of cost 0.5 to the four-node example. Recompute the best current route to D .
2. Explain why a planned path quote can differ from final settled EBU even

when the route itself does not change.

3. Give a stale-quote example that creates an apparent favourable cycle in a frozen planning graph.
4. List the action supports needed to account for an overseas pump-seal shipment in three distinct resource fields.
5. Compare building a regional medicine store with continuing emergency shipments. Which burdens occur now, and which are only forecast?
6. Explain why a vector of field accounts may be more honest than one universal scalar.

Invariance, stability, and the long-run boundary

The phrase “in the long run the EBU system restores homeostasis” is an important research ambition. It is not yet a consequence of one finite quote, one preservation inequality, or one successful simulation. This chapter shows what can already be proved, why it is useful, and what additional structure a genuine long-run theorem would need.

The careful boundary is not a retreat from the project. It is how a strong theory grows. Each proved component becomes a lemma in a later composite result, while the missing assumptions are made visible enough to test.

43.1 Five terms that answer different questions

Forward invariance. A set $\mathcal{K}$ is forward invariant if a trajectory that begins in $\mathcal{K}$ remains in $\mathcal{K}$ at every later time. For the reserve-safe set, this means protected stocks do not cross their declared boundaries.

Boundedness. A trajectory is bounded if it remains inside some finite region. A bounded trajectory can oscillate forever and need not approach homeostasis.

Equilibrium. A state x^* is an equilibrium if the update leaves it unchanged. In continuous time, $\dot{x} = 0$ at x^* . In discrete time, $F(x^*) = x^*$ for update map F .

Stability. Roughly, an equilibrium or target set is stable if starting sufficiently close keeps the trajectory close. Stability alone does not require the distance to shrink to zero.

Attractivity. A target is attractive if trajectories from a declared region approach it. Asymptotic stability combines stability and attractivity. Practical stability allows trajectories to approach and remain within a bounded neighbourhood rather than converge exactly.

These definitions matter because an open regenerative system under changing needs may never settle at one fixed point. A robust homeostatic tube can be a more realistic target than exact convergence.

43.2 Extending one-step preservation by induction

The preservation theorem has the form

$$x^n \in \mathcal{K} \text{ and } H_n \implies x^{n+1} \in \mathcal{K}, \quad (43.1)$$

where H_n denotes the tick-specific hypotheses: no-export feasibility, correct measurement margins, aggregate budget compliance, and the declared synchronous update.

Suppose $x^0 \in \mathcal{K}$ and H_n holds for every $n \geq 0$. Then forward invariance follows by mathematical induction.

1. Base case: $x^0 \in \mathcal{K}$ is assumed.
2. Inductive step: if $x^n \in \mathcal{K}$ and H_n holds, the one-step theorem gives $x^{n+1} \in \mathcal{K}$.
3. Therefore every finite tick reached by repeating the step lies in $\mathcal{K}$.

This is a genuine multi-tick theorem, but notice its conditional phrase: H_n must hold at every tick. The proof does not show that regeneration, demand, uncertainty bounds, or source feasibility will guarantee H_n forever.

43.3 What monotone potential proves

Suppose an undriven certified discrete trajectory satisfies

$$V(x^{n+1}) \leq V(x^n) \quad \text{for every } n \quad (43.2)$$

and $V \geq 0$. Then the scalar sequence

$$V(x^0), V(x^1), V(x^2), \dots \quad (43.3)$$

is non-increasing and bounded below. A basic convergence theorem for real sequences says it approaches a finite limit $V_\infty \geq 0$.

This is already useful: represented burden cannot oscillate upward without bound, and the total amount of decrease is finite. But convergence of the number $V(x^n)$ does not necessarily imply convergence of the vector x^n .

43.4 Counterexample: the state moves on a level set

Consider

$$V(x_1, x_2) = x_1^2 + x_2^2. \quad (43.4)$$

The states

$$x^n = (\cos n, \sin n) \quad (43.5)$$

all satisfy $V(x^n) = 1$, but the state does not converge. It moves around the unit circle.

The present derived undriven flux law excludes some arbitrary level-set motions through its dissipation identity, but the example establishes the logical point: scalar convergence alone is not vector convergence. One must use properties of the actual dynamics.

The flat homeostatic set creates another benign example. A state can move inside a region where $V = 0$ without changing the potential. Such motion may be acceptable because homeostasis is a range, not one point.

43.5 Distance to a target set

For a target set $\mathcal{H}$, define

$$\text{dist}(x, \mathcal{H}) = \inf_{h \in \mathcal{H}} \|x - h\|_2. \quad (43.6)$$

The symbol $\inf$ means the greatest lower bound. In ordinary closed geometric sets it corresponds to the distance to the nearest target point. If $x \in \mathcal{H}$, the distance is zero.

For a viable band, set distance is more meaningful than distance to its mid-point. A water stock at either side of a healthy range can be fully viable without being driven toward one arbitrary centre.

43.6 A conditional convergence lemma

The following lemma shows one possible bridge from finite-step descent to long-run attraction.

Lemma. Assume a trajectory remains in the model domain and there exists $c > 0$ such that

$$V(x^{n+1}) \leq V(x^n) - c \operatorname{dist}(x^n, \mathcal{H})^2 \quad (43.7)$$

for every n . If $V \geq 0$, then

$$\operatorname{dist}(x^n, \mathcal{H}) \longrightarrow 0. \quad (43.8)$$

Proof. Rearrange Equation (43.7):

$$c \operatorname{dist}(x^n, \mathcal{H})^2 \leq V(x^n) - V(x^{n+1}). \quad (43.9)$$

Sum from $n = 0$ to $N - 1$. The right side telescopes:

$$c \sum_{n=0}^{N-1} \operatorname{dist}(x^n, \mathcal{H})^2 \leq V(x^0) - V(x^N) \leq V(x^0). \quad (43.10)$$

Letting N grow shows that the infinite sum of non-negative squared distances is finite. A non-negative sequence with finite sum must approach zero. Therefore the distance to $\mathcal{H}$ approaches zero. $\square$

This lemma is proved mathematics. What is not yet proved is that the current driven, constrained, multi-actor EBU system satisfies Equation (43.7) for a scientifically adequate target set.

43.7 A practical-stability recurrence

Open systems experience continuing drive. Exact convergence may be neither possible nor desirable. A useful theorem might instead establish a recurrence

$$V_{n+1} \leq (1-a)V_n + b, \quad 0 < a < 1, \quad b \geq 0, \quad (43.11)$$

where b bounds persistent disturbance or modelling error.

Iterate the inequality. After one step,

$$V_1 \leq (1-a)V_0 + b. \quad (43.12)$$

After two,

$$V_2 \leq (1-a)V_1 + b \quad (43.13)$$

$$\leq (1-a)^2V_0 + (1-a)b + b. \quad (43.14)$$

Continuing gives

$$V_n \leq (1-a)^nV_0 + b \sum_{k=0}^{n-1} (1-a)^k. \quad (43.15)$$

The geometric sum is

$$\sum_{k=0}^{n-1} (1-a)^k = \frac{1 - (1-a)^n}{a}. \quad (43.16)$$

Therefore

$$V_n \leq (1-a)^nV_0 + \frac{b}{a} [1 - (1-a)^n]. \quad (43.17)$$

Since $(1-a)^n \rightarrow 0$,

$$\limsup_{n \rightarrow \infty} V_n \leq \frac{b}{a}. \quad (43.18)$$

This is a mathematical form of practical stability: the eventual burden is bounded by disturbance strength divided by corrective strength. It is a template for future analysis, not a claim that current EBU parameters already provide constants a and b globally.

43.8 Persistent capacity and viability

Long-run homeostasis needs more than a decreasing penalty. Sources must retain the capacity to recover and to serve admissible genuine needs. A future theorem may require assumptions such as

- regeneration is lower-bounded inside the safe set;
- recurring need is bounded by sustainable capacity or supported by external supply;
- no-export feasibility can recover after bounded shocks;
- edge capacities keep the target set reachable;
- measurement errors remain within robust margins;
- actor requests and accepted actions respect information and safety rules;
- adaptation does not change the field faster than it is re-certified.

These are substantive physical hypotheses. A mathematical proof cannot create renewable capacity where the world lacks it.

43.9 Infeasible need remains an outcome

Suppose a clinic permanently requires six usable units per tick, while every feasible route together can deliver at most 2.7. No accounting system can make the remaining 3.3 units appear. Long-run service is physically infeasible in that world.

The scientifically correct model should maintain state safety where possible, record unmet need, classify physical impossibility, and expose which structural change could expand capacity. Calling such a run a failure of the local field law would confuse controller performance with the feasible set.

43.10 Why a long simulation is not a forever theorem

A simulation of one million ticks is stronger empirical evidence than a run of ten ticks for the tested initial states and parameters. It still examines a finite trajectory in a finite family of worlds. It cannot establish behaviour for every disturbance, measurement error, network topology, actor policy, or future parameter change.

Long-horizon simulation is valuable for finding slow failure modes: buffer depletion, periodic oscillation, rare threshold crossings, cumulative round-off, delayed shocks, or adaptation traps. Its role is to challenge and inform theorem design, not to replace proof.

43.11 What “fixing the planet forever” would require

A defensible long-run statement would name its domain. A possible form is:

For every initial state in a certified safe region, every admissible bounded drive, every compliant actor action, and every measurement error within the declared margins, the closed-loop state remains in the preservation set and approaches a declared homeostatic neighbourhood, while feasible essential need is served within specified bounds.

To prove such a statement, one would need at least:

1. a complete state and uncertainty model;
2. an invariant safe set;
3. a feasible-need definition and capacity assumptions;
4. a controller or decision policy specified for all domain states;
5. a strict or practical Lyapunov/barrier argument;
6. treatment of multiple actors, simultaneity, and adaptation;
7. a link from mathematical variables to measured planetary fields;
8. adversarial tests and independent review of implementation and calibration.

The existing exact quote, receipt closure, discrete certificate, and one-step preservation theorem would become components of this proof architecture. They need not be withdrawn because a larger theorem has not yet been completed.

43.12 Collaboration as a physical hypothesis

EBU does not require people to become altruistic. It may make cooperation materially advantageous whenever shared action meets the same genuine needs with less verified physical burden. Shared cold storage, pooled repair tools, coordinated transport, standardised spare parts, and regional emergency inventory can reduce repeated action costs.

Whether that advantage creates stable, fair, and resilient collaboration must be demonstrated rather than assumed. A future multi-actor model should compare individual and cooperative strategies under identical needs, measure how group benefits are attributed, test free-riding and exclusion, and include controls where sharing is physically inefficient.

The hypothesis is behavioural, but it begins at the action level: each actor can see that a lower-burden transformation produces a better signed receipt. That can change the question from “how do I obtain more money?” to “how can this need be met with a better verified field effect?” The cultural outcome is not a theorem; the micro-level incentive is a testable mechanism.

43.13 Adaptive infrastructure as a separate dynamics

Repeated receipts can influence investment decisions. Let G^n denote the network at a slower planning time and let R^n summarise recurring needs and burdens. An adaptive law might propose

$$G^{n+1} = \mathcal{A}(G^n, R^n), \quad (43.19)$$

where $\mathcal{A}$ can build, remove, or modify edges and nodes. This changes the feasible action set itself.

Proving that such adaptation lowers long-run burden is substantially harder than proving one action quote. Construction has present burden, demand can move, new capacity can induce use, and local improvement can shift harm to

another field. The appropriate study needs locked objectives, counterfactuals, and falsifiers. It should not be smuggled into the current theorem by saying “the system will naturally optimise”.

43.14 A theorem-development sequence

A careful next programme would proceed in stages:

1. prove forward invariance for a closed deterministic network under persistent feasibility assumptions;
2. establish set convergence or a practical bound for the same model;
3. add bounded drive and shocks;
4. add feasible essential-need constraints;
5. add multiple actors and controlled simultaneity;
6. add uncertain measurements and robust margins;
7. add slow infrastructure adaptation;
8. calibrate to empirical fields and seek independent replication.

Each stage should be preregistered and carry a stop condition. Failure at one stage is information about the theorem boundary, not a reason to reinterpret a null result as success.

Scope statement

Forward invariance follows across many ticks only while the one-step hypotheses continue to hold. Monotone non-negative V converges as a scalar but need not force state convergence. The strict set-descent and practical-stability results in this chapter are conditional mathematical templates; the current project has not yet proved their premises for a complete driven actor economy.

Chapter recap

Invariance, boundedness, equilibrium, stability, and attraction are different properties. Existing one-step theorems can be extended conditionally by induction, and monotone potential has a finite limit. Long-run homeostasis needs strict or practical set stability, persistent physical feasibility, uncertainty control, and an explicit multi-actor controller. Collaboration and structural adaptation are promising testable effects, not assumptions.

Exercises

1. Give a bounded trajectory that does not converge.
2. Explain exactly where the phrase “the hypotheses hold at every tick” enters the induction proof for reserve invariance.
3. Reproduce the telescoping proof of the conditional set-convergence lemma.
4. For $a = 0.2$, $b = 3$, and $V_0 = 100$, calculate the bound in Equation (43.17) at $n = 1$, 5 , and in the limit.
5. Name two physical assumptions a theorem would need before claiming long-run service of genuine needs.
6. Design one positive control and one negative control for the hypothesis that shared transport reduces cumulative EBU.
7. Explain why a million-tick simulation and an infinite-horizon theorem answer different questions.

A complete Gaussian field laboratory

This laboratory is a sequence of exact worked calculations. It is not a trajectory produced by the research engine and not a proposed configuration for a scientific campaign. Its purpose is to let the reader reconstruct every important distinction with pencil-and-paper arithmetic.

46.1 Write the declaration before calculating

For the first cases, use two cells with

$$z = (18, 2), \quad x^* = (10, 10), \quad \sigma = (2, 2), \quad H = \text{diag}(1/4, 1/4). \quad (46.1)$$

Both state and reference sum to twenty. The action is a lossless transfer from the first cell to the second, with finite quantity q . The increment is $(-q, q)$. The illustrative physical domain permits $0 \leq q \leq 18$; there is no separate destination cap in this example. Process burden is zero.

The initial potential is

$$V(z) = \frac{1}{2}[(8/2)^2 + (-8/2)^2] = 16. \quad (46.2)$$

The marginals are $(2, -2)$, so the field force is four. Substitution into the exact finite expression gives $E(q) = 4q - q^2/4$. Every later number should be checked both from that polynomial and from the endpoint potential.

46.2 A menu with favorable and adverse quantities

q	Endpoint	V after	$E(q)$
0	(18, 2)	16	0
1	(17, 3)	49/4	15/4
2	(16, 4)	9	7
4	(14, 6)	4	12
8	(10, 10)	0	16
12	(6, 14)	4	12
16	(2, 18)	16	0
17	(1, 19)	81/4	-17/4
18	(0, 20)	25	-9

Each row conserves twenty and has nonnegative stocks. Therefore physical feasibility alone does not choose the helpful rows. Conversely, the two zero-valued quantities mean very different things: $q = 0$ does nothing, while $q = 16$ exchanges the deviations and returns to the same potential. Equal potential is not equal state.

The table is not a controller instruction. Selecting eight because it has the largest value would define an optimizer. Choosing uniformly from a declared affordable subset would define another process. Listing analytical options does not authorize either one as the scientific study.

46.3 Check the integral independently

Along the transfer, $f(r) = 4 - r/2$. For $q = 4$,

$$\int_0^4 (4 - r/2) dr = [4r - r^2/4]_0^4 = 12. \quad (46.3)$$

The initial-tangent estimate is sixteen, overvaluing the action by four. The average force along the segment is $(4 + 2)/2 = 3$, and multiplying by quantity four again gives twelve. The endpoint-force averaging is exact because this particular gradient is affine, not because trapezoidal quadrature is exact for all field models.

At $q = 12$, the final force is minus two. The average is one and the integral is twelve. Some portions of the path improve the potential and later portions worsen it, yet the total remains positive. A positive integral is not a statement that every infinitesimal part was beneficial.

46.4 Separate force from an Onsager response

Declare, only for this illustration, $M_e = 1/2$ and $\theta_e = 0$. At the original state the requested rate is two. Duration $1/2$ gives quantity one and exact field value $15/4$. Duration two gives quantity four and value twelve. Duration four gives quantity eight and value sixteen. These are three frozen-step calculations, not repeated steps in one run.

If instead the flux were recalculated continuously as the state moved, the force would fall and so would the flux. It would be a mistake to interpret the frozen duration-four jump as the solution of that differential equation. “Same starting force” does not mean “same time evolution.”

46.5 Local evaluation with a nonzero remote term

Append a third cell at stock twelve, reference ten and scale two, but do not change it. The initial global potential becomes $33/2$. After the four-unit

transfer it is $9/2$. The remote half-unit contribution is present in both and cancels. Local valuation still gives twelve.

Now imagine the transfer also consumes stock from that third cell. The former two-cell formula is no longer a complete local account. The example does not need a numerical error to demonstrate the problem: its support assumption has changed. A correct implementation should reject the incomplete declaration rather than silently use the old result.

46.6 Sequential receipts and reversed order

Divide a four-unit transfer into quantities one and three. If one occurs first, its value is $15/4$ and the subsequent three-unit value is $49/4 - 4 = 33/4$. If three occurs first, its value is $39/4$ and the last one-unit value is $25/4 - 4 = 9/4$. Both sums are twelve.

The intermediate state explains the different child values. There is no contradiction to reconcile by averaging the orders unless a separate allocation rule asks for such an average. The physical sequence already defines which live state each action used.

If an extra nonnegative activation burden is incurred for each separate action, splitting can change the net total. The field sum still telescopes; the process-cost sum changes. An implementation must not quote the one-action cost and then execute two independently activated operations under it.

46.7 A simultaneous group on the same source

Let both two-unit children be accepted on the same baseline. The net increment is $(-4, 4)$ and the group field value is twelve. Same-baseline singletons each show seven, yielding a misleading fourteen if added as though independent. The missing cross correction is minus two.

Under the declared straight common path, the attribution to each child is six. This is easy to check from the average group gradient, or by giving each child half of the identical net direction. Closure is $6 + 6 = 12$. It does not prove that each actor owns six units of spendable capacity.

For unequal children of one and three units, common-path values become three and nine. These differ from both ordered sequential splits above. The group field result is the same because the endpoint is the same; the child interpretations refer to different declared constructions.

46.8 Cancellation and a signed child account

Use the same baseline but take opposing children $(-2, 2)$ and $(2, -2)$. The net endpoint is unchanged. Standalone values are 7 and -9 , while common-path values are 8 and -8 . Both the group value and the sum of path values are zero. The singleton sum is minus two.

If each child belongs to a different zero-balance owner, the proposed owner-wise affordability rule refuses the owner charged eight. If both belong to one owner, same-event netting gives zero. This is an explicit demonstration that ownership policy can matter even when the physical net change is identical. It is not a recommendation to choose the ownership map that produces a desired outcome.

46.9 A different scale is a different model

Keep the original state and reference but change scales to $(2, 4)$. The initial potential is ten, marginals are $(2, -1/2)$, and field is $5/2$. For quantity four the curvature coefficient is $\frac{1}{2}(1/4 + 1/16) = 5/32$, so

$$E(4) = 4(5/2) - (5/32)16 = 15/2. \quad (46.4)$$

The endpoint potential is $5/2$, agreeing with $10 - 5/2 = 15/2$. Nothing physical changed between this declaration and the earlier example. The change in value comes from the declared scale at the second cell. Any empirical use must therefore justify and version that declaration.

Changing to smaller physical units would instead rescale stock, reference, quantity and scale together and leave EBU unchanged. A field account that changes when only the unit label changes has a dimensional bug.

46.10 A route whose first step is unfavorable

Declare a new three-cell example with state $(10, 10, 2)$, reference $(6, 6, 10)$ and equal scales two. Both totals are twenty-two. The initial potential is twelve. A four-unit transfer from the first cell to the second gives $(6, 14, 2)$ with potential sixteen, so its value is minus four. A subsequent eight-unit transfer from the second cell to the third reaches the reference $(6, 6, 10)$ and has value sixteen. The route total is twelve.

This is a physically feasible quantity-fixed sequence under simple stock non-negativity. It illustrates the difference between favorable endpoint value and intermediate affordability. A zero-balance owner cannot pay the first negative action under the proposed no-borrowing rule. The later positive action does not retroactively authorize it. A different grouping, timing or ownership policy would be a different declared experiment.

46.11 Forcing and the complete account

Begin instead at (10, 10) with $V = 0$, zero balances and zero external audit. A conservative shock gives (14, 6), with $D_{\text{ext}} = 4$. One two-unit restoring transfer ends at (12, 8), with potential one and field value three. If the candidate account policy credits that action to its declared owner, the invariant reads

$$1 + 3 - 4 = 0. \quad (46.5)$$

Now consider a possible reverse two-unit action from (12, 8) back to (14, 6). Its value is minus three. An owner with balance three can afford it, returning to potential four and balance zero, still with audit four. Both accounts close. Recovery has been undone. This is a counterexample to the claim that the accounting invariant alone forces monotone recovery.

No random process has been run to obtain this pair of possibilities. It is an algebraic construction showing what the proposed rules allow. A later study must determine how often such events arise under its actual menus and sampling, without changing those rules after seeing results.

46.12 A reproducible hand calculation

For every candidate, a reader should be able to list the baseline, references, scales, child increments, complete support, physical constraints, before/after potential, group value and optional path values. If capacity is discussed, the owner map and prior balances must also be listed. If forcing occurs, the before-forcing state and signed audit must be listed separately.

An unexplained number in any one of those positions should interrupt the claim. The appropriate response is not to fill it with a plausible value. It is to identify which declaration, measurement or authority is missing. This is how a careful laboratory connects to the fail-closed software architecture developed in the following chapters of this volume.

46.13 A three-cell laboratory: the hub participates even when it does not change

The two-cell examples isolate curvature cleanly. A hub introduces another question: can two actions interact even when the hub's final stock equals its starting stock? Use

$$z = (18, 10, 2), \quad x^* = (10, 10, 10), \quad \sigma = (2, 2, 2). \quad (46.6)$$

Initial potential is sixteen and the marginals are (2, 0, -2). Action A transfers four from Ridge to Hub; action B transfers four from Hub to Vale. Their

increments are

$$\delta_A = (-4, 4, 0), \quad \delta_B = (0, -4, 4). \quad (46.7)$$

Alone, A ends at (14, 14, 2) and B alone ends at (18, 6, 6). Each endpoint has potential twelve, so each singleton value is four. Together their increment is (-4, 0, 4); their endpoint is (14, 10, 6), with potential four. The group value is twelve, not eight.

Calculate the cross term explicitly:

$$\delta_A^T H \delta_B = \frac{1}{4}[(-4)0 + 4(-4) + 0(4)] = -4. \quad (46.8)$$

The correction to the singleton sum is therefore plus four. It arises because the hypothetical singleton endpoints separately disturb the hub in opposite directions, whereas the joint endpoint does not. No nonseparable potential was introduced. Shared-coordinate action structure is sufficient to create the interaction.

46.14 Three descriptions of the same aggregate endpoint

Serialize A first, then B, keeping these quantities fixed and assuming the simple stock constraints allow both. A contributes four and B then contributes eight. Reverse the order: B contributes four and A then eight. Both serialized field totals are twelve. Hence the positive correction against hypothetical same-baseline singletons is not a positive advantage against either of these actual sequential endpoints.

For the constant-rate common group path, the hub stays at ten throughout. The gradient averaged along the path is (3/2, 0, -3/2). Therefore

$$R_A = -(3/2, 0, -3/2) \cdot (-4, 4, 0) = 6, \quad R_B = 6. \quad (46.9)$$

The split (6, 6) closes on twelve, just as the two sequential splits (4, 8) and (8, 4) do. They refer to different descriptions of how children contribute along a declared path. The equality of their totals does not identify one split as causal, fair or institutionally required.

The reader should be able to explain all three descriptions without saying that the same physical experiment has three contradictory outcomes. Only one physical chronology or path is actually declared in any given model. The other calculations are named comparisons or conventions.

46.15 A shared-source laboratory with three children

Now use four cells, all with reference ten and scale two, at

$$z = (18, 6, 6, 10). \quad (46.10)$$

The total is forty and the initial potential is twelve. Three children each withdraw two from the first cell, delivering respectively to cells two, three and four. The final group state is $(12, 8, 8, 12)$ and its potential is two. Thus the group field value is ten.

The singleton values are five, five and three. All three share the source; each pair has $\delta_a^T H \delta_b = 1$. The singleton sum thirteen therefore needs three corrections of minus one, yielding ten. A program that only checks whether destination supports are disjoint will miss the common source and return the wrong total.

For the common path, $\mu(z) = (2, -1, -1, 0)$ and $H\delta_G = (-3/2, 1/2, 1/2, 1/2)$. The path-average gradient is

$$\bar{\mu} = (5/4, -3/4, -3/4, 1/4). \quad (46.11)$$

Dotting it negatively with each child gives $(4, 4, 2)$. The values add to ten. Notice that the child ending at an already-at-reference destination still has a positive net value here: its source relief outweighs the destination's added deviation. The account describes the combined physical change, not only a destination label.

46.16 Write the entire subset table before interpreting it

For those same fixed increments, evaluate every subset at the same frozen baseline:

Subset	Field value
$\emptyset$	0
A	5
B	5
C	3
AB	9
AC	7
BC	7
ABC	10

Each pair coefficient is minus one. The third-order coefficient is $10 - 9 - 7 - 7 + 5 + 5 + 3 = 0$. This cancellation illustrates the quadratic membership structure. It is not a measured absence of all three-way physical effects in nature.

Before calling the table a physical subset experiment, check that each subset is admissible under the same protocol. If a coupled machine can perform only all

three children together, the singleton rows remain algebraic counterfactuals, not admissible interventions. If a resolver changes quantities separately for each subset, the table must be recomputed with those quantities, and the fixed-increment degree argument may fail. Do not fill missing physical rows with convenient numbers merely to make an inversion formula available.

46.17 An independence check with disjoint actions

Let four cells start at $(14, 6, 14, 6)$, again with reference ten and scale two. One action sends two from cell one to cell two; another sends two from cell three to cell four. Initial potential is eight. Each action reduces its pair's potential from four to one, so each field value is three. The group endpoint $(12, 8, 12, 8)$ has potential two, giving six.

The increments have disjoint support, and the diagonal Hessian gives zero cross term. Here independent singleton addition is valid. The lesson is not that singleton addition is always wrong; it is that its validity has a precise condition that can be checked. If both actions share a physical machine limit, valuation may still add while joint permission fails. Independence of field terms is not independence of every constraint.

46.18 A positive group can have an unaffordable child owner

Return to three cells, at $(14, 6, 10)$ with the same references and scales. Let A send two from cell one to cell two and B send two from cell two to cell three. The group endpoint is $(12, 6, 12)$. Potential changes from four to three, so the group value is one.

The initial gradient is $(1, -1, 0)$ and the net increment is $(-2, 0, 2)$. The average gradient along the group path is $(3/4, -1, 1/4)$. Hence

$$R_A = 7/2, \quad R_B = -5/2, \quad R_A + R_B = 1. \quad (46.12)$$

This is an exact example of a positive group total with a negative child attribution. If two different owners start with zero balances, the proposed owner-wise rule refuses B's debit. If one owner holds both children, same-event netting gives plus one and would pass that account condition. Physical feasibility is the same in both descriptions.

The example is not an argument for selecting whichever ownership map admits the group. It proves that the ownership map is a load-bearing model choice. Scientific comparison requires that choice to be declared before outcomes are inspected. It cannot be inferred from the favorable sign of the group total.

46.19 A missed factor can mimic an attractive improvement

Suppose the two-cell lossless example is expanded to represent a third physical resource consumed by pumping. The expanded potential might contain a factor of that resource, and the accepted action might change it. A quote that still uses only the water endpoints would omit a changed term. Even if its arithmetic for water is perfect, it would no longer be the complete exact local quote for the expanded declaration.

There are two legitimate ways to keep a teaching model simple. One is to leave pumping energy outside its scope and state the limitation. The other is to represent it and include the corresponding affected factors or a separately justified residual burden. What is not legitimate is to claim complete multi-carrier valuation while counting the same physical effect in both places, or in neither place, without disclosure.

This is also why a fuel-to-electricity-to-water chain should remain a linked sequence of atomic physical transformations when it is in scope. The water endpoint does not tell the reader which upstream transformations occurred or which loss boundary they crossed. A convenient aggregate action label cannot repair missing physical representation.

46.20 Recalibration is not physical work

Take the original two-cell state (18, 2) and leave it unchanged. With reference (10, 10) and scales two, potential is sixteen. If both scales are revised to four, potential becomes four. The twelve-unit difference is not an actor's physical improvement. No resource moved. It is a change in the declared comparison geometry.

The active first programme freezes scales and references, so this event is outside its action loop. A later adaptive-calibration model would need an explicit parameter-change account. Otherwise an actor could appear to earn value by changing the ruler rather than the world. Version commitments protect the distinction between these operations; numerical precision alone cannot.

46.21 Build a receipt that another reader can reconstruct

For the three-child source example, a minimal mathematical audit records the ordered state (18, 6, 6, 10), the ordered reference and scales, the three named increments, their sum, the before and after potentials, the group value ten, and the chosen path values (4, 4, 2). It also records that the quantities are source-side quantities, not rates.

An implementation-grade record needs more: state and parameter identities, the accepted physical constraints, path semantics, numerical policy, and evidence that the executed increment matched the valued increment. A capacity record additionally needs a declared owner map, prior balances and a separately authorized update rule. Each layer answers a question that the preceding arithmetic cannot answer by itself.

If a reader cannot determine whether the state list is ordered Ridge, Hub, Vale or another way, even the first calculation is not reproducible. If the owner map is missing, the field value can still be correct while the capacity account remains undefined. An honest report can say both things at once rather than treating the missing layer as permission to guess.

46.22 A final diagnostic exercise

Imagine a report for the three-child source example gives the group value thirteen, says every owner starts at zero, and claims that a closing receipt proves fair compensation. Identify three separate issues. First, thirteen is the sum of same-baseline singletons; the actual joint field value is ten. Second, balances alone do not identify owners or the rule mapping path values to accounts. Third, no corrected sum would prove fairness. Those defects require different repairs.

Now imagine the report instead gives ten and correct path values but executes only two of the three children. The formula has not failed. The event identity has: a different increment was performed. A receipt must correspond to that actual authorized event or the system must refuse under its declared deviation procedure. Carry this distinction into the software chapters. Their central job is to prevent a correct calculation from becoming a misleading record of a different occurrence.

46.23 A route-and-repair reading: the remote pump seal

The original laboratory also used a remote pump seal to explain genuine need, route choice and live-state replanning. That lesson deserves to survive the change in potential. A water-treatment plant needs a working seal before a deadline; rail and air offer different delays and physical burdens. A route is not valuable or worthless merely because it is long. The represented endpoint, timing, feasibility and actual process matter.

This repair is not an action inside the first homogeneous-scalar Gaussian world. A seal is a discrete functional item, not divisible water, and a plant's working condition is not automatically a scalar stock coordinate. A broader domain must declare the transformation and its functional representation. Keeping

that boundary visible preserves the example's meaning without pretending the first experiment already models maintenance.

The original example assigns illustrative planning burdens of nine to rail and seventeen to air before additional packaging. It also records usable factors $0.99 \cdot 0.99 = 0.9801$ and $0.99 \cdot 0.98 = 0.9702$. A fraction of a seal may be useless, so these products cannot simply be treated as divisible functional delivery. Packaging, reliability and item-count assumptions must be explicit. The historical exercise later assumes successful functional delivery; that is a teaching assumption, not a measured guarantee that probability one can be certified in a real transport system.

Suppose a flood closes the planned rail connection after the first segment. The planned endpoint is no longer the live physical situation. Replanning must start from the actual hub state and the new route constraints. An old positive forecast is not a receipt for an unperformed route. If the flood also changes represented physical state or parameters, that external change needs its own account; it cannot be hidden inside an actor's next value.

The original's separate fixed-state accounting exercise lists potentials 80, 76, 74, 68, 5 and burdens 3, 1, 8, 4. Under one fixed potential and with every intervening change included in those declared actions, the signed lines are 1, 1, -2, 59, summing to fifty-nine. Endpoint closure gives the same $80 - 5 - 16 = 59$. If an unrecorded external change occurs between those states, this simple actor-only telescoping would need the external correction derived earlier. The arithmetic cannot certify that all such events were recorded.

Repeated repairs can motivate separate studies of spare inventory, local manufacture, standardization or predictive maintenance. They do not justify crediting today's actor with all forecast future savings. Those interventions have their own physical costs, uncertainty and institutional questions. The future Part VI develops routes and transformations; Part VII develops explicit coordination policies. Neither is hidden inside the present random actor.

46.24 Proof workshop: rebuild the argument in eight passes

The goal of this workshop is not to memorize eight displayed equations. It is to learn how to reconstruct a result and state where it stops. For each pass, first write the objects and assumptions, then the algebra, then one conclusion that the algebra does not establish.

46.24.1 Pass 1: conservation from the transformation matrix

Begin with $x^+ = x + Sq$. Multiply by the row of ones. If every column of S sums to zero, the total before and after is equal for every quantity vector. This is a structural statement about the declared transformation matrix. To complete a physical validation, separately check nonnegative coordinates, allowed quantities and any joint constraint. An impossible negative source is not repaired by the conservation proof.

Now replace one destination coefficient by $\eta < 1$. The old column sum changes. State explicitly whether the missing carrier enters a loss coordinate or crosses an external boundary. Do not repair the mass equation by adding a burden number to it: burden and physical stock have different units and meanings.

46.24.2 Pass 2: differentiate the declared potential

Start with one term $\frac{1}{2}(x_i - x_i^*)^2/\sigma_i^2$. Keep the reference and scale fixed while differentiating. Recover $\mu_i = (x_i - x_i^*)/\sigma_i^2$ and curvature $1/\sigma_i^2$. Explain why differentiating a moving scale or reference would create additional terms. The fixed-parameter condition is part of the calculation, not a footnote to discard afterward.

Check the units: marginal has inverse-stock units and curvature has inverse-stock-squared units under dimensionless normalization. Finally explain why a derivative does not identify an actor's preferred action. It describes a directional property of a function before a choice rule has been defined.

46.24.3 Pass 3: expand a finite movement

Write $y = z - x^*$ and expand $\frac{1}{2}(y + \delta)^T H(y + \delta)$. The two cross terms are equal because H is symmetric. Subtract the initial potential and change the sign to obtain exact field benefit. The positive quadratic term in potential becomes a negative curvature correction in benefit. Check the result at zero increment and at the reference.

At the reference a nonzero net increment has strictly negative field value. A nonempty cancelling group has zero net increment instead. The proof distinguishes those cases and therefore cannot be paraphrased as "every nonempty group at equilibrium has negative value."

46.24.4 Pass 4: cancel unchanged factors

Partition the potential's factors into those touched by the action's write support and those untouched. Subtract before and after. Every untouched factor cancels exactly, leaving the complete affected-factor sum. For a factor involv-

ing several cells, include all arguments needed to evaluate it even when some are not changed.

The proof establishes the locality of this value calculation. It does not prove that a world-wide group enumerator is decentralized. Write that limitation beside the theorem so the word local does not quietly change meaning between a mathematical and an architectural claim.

46.24.5 Pass 5: recover the path integral

Let $\gamma(s) = z + s\delta$. Apply the chain rule to $V(\gamma(s))$ and integrate from zero to one. The endpoint difference follows from the fundamental theorem of calculus. In the quadratic case the gradient is affine, so integrating it gives the mean of its endpoint gradients exactly.

Then state which path meaning is in force. Is it a constant-rate model-realized trajectory or merely an endpoint interpolation? The integral identity does not choose. An observed real-world path requires more than an equality between two endpoint calculations.

46.24.6 Pass 6: compose children before dividing value

Expand the square of $\sum_a \delta_a$. Keep all pair cross terms. Their omission is the precise defect in summing independent same-baseline singletons. Next integrate each child against the same group gradient. Summing those child integrals recovers the integral of the net increment and hence group closure.

The closure proof has no requester, supplier, beneficiary or owner in it. That absence is informative. It is why choosing an owner map is another modelling step rather than a missing algebraic manipulation. Adding a label to a child does not add a theorem of entitlement.

46.24.7 Pass 7: telescope a sequence, then insert forcing

Write three consecutive field differences and cancel intermediate potentials visibly. Extend the pattern to any finite sequence. For a closed cycle the field sum vanishes; nonnegative separately declared burdens make net value nonpositive. Do not replace the live baseline of every line by the first state.

Now insert a forcing step before each actor epoch. Add and subtract the pre-forcing potential. The new external terms explain why an actor-only sum need not equal the overall endpoint difference. This is not a failure of telescoping; it is the account of an additional kind of change that must not be credited as actor work.

46.24.8 Pass 8: derive the account invariant conditionally

Assume an explicitly adopted owner rule whose signed changes sum to the field value, and a signed external ledger updated only by forcing. Add the potential change, balance change and negative audit change. All terms cancel under those definitions. Record the invariant and its initial constant.

Finally construct the restoring-and-reversing pair already shown in the laboratory. Both events satisfy the invariant, yet the second undoes recovery by spending prior capacity. This last step prevents the workshop from ending with an unjustified dynamical conclusion. The invariant constrains accounting; the declared stochastic process still determines which permitted events occur.

46.25 Ten mistakes a laboratory should expose

- 1. Omitting efficiency outside the lossless domain.** The destination increment changes, so both the physical balance and the directional derivative change. A lossless example cannot certify a loss-aware implementation.
- 2. Valuing from the pre-forcing state.** This folds nature's change into the actor's line. Keep the external record and the post-forcing baseline separate.
- 3. Ranking by value per unit without declaring an optimizer.** A ratio may be a legitimate diagnostic or policy in another model. Using it to select actions changes the current unweighted-random question, even if the finite quote itself is exact.
- 4. Settling the requested quantity instead of the accepted event.** The same action name can conceal a different increment. Bind quantity and baseline to the event identity, not just to a display label.
- 5. Counting one receipt twice.** Arithmetic correctness of each copy does not make the second copy a second physical occurrence. An account needs exactly-once identity and replay rules as well as a correct subtraction.
- 6. Pricing shared-source children independently.** The common source changes only once in the joint endpoint. Restore the cross terms or evaluate the complete endpoint directly.
- 7. Giving the full source budget to every edge.** Individually valid requests can be jointly impossible. A physical budget has one scope; duplicating its record does not duplicate stock.
- 8. Reading continuous dissipation as a finite-step guarantee.** The exact curvature term survives discretization. Check the accepted finite increment and physical feasibility independently.

9. Projecting an invalid endpoint while keeping its quote. A clipped successor is another physical map. It needs the value of that map and an explicit permission rule, not a repaired-looking state attached to the unrepaired record.

10. Inferring forever from a finite study. A completed horizon establishes observations on that registered horizon. Unlimited-time claims need their own theorem or evidential argument. Neither a large test count nor a smooth plot removes that boundary.

These mistakes are not ten ways the same equation fails. They are ten ways a reader or implementation can quietly substitute a different object for the one the equation describes. The following software chapters turn that diagnosis into inspectable responsibilities.

Final synthesis

The Gaussian model gives exact, local finite accounting and a transparent curvature correction. It supports conditional path and telescoping proofs, and it makes group interaction calculable. The Onsager-type controller gives an additional conditional dissipation law when it is explicitly chosen. Random-affordable action is a different prospective process. Its useful, adverse and inconclusive outcomes remain to be investigated, not written into the definition of success.

From a law to an executable claim

A mathematical equation can fit on one line while its faithful implementation requires many boundaries. The equation does not specify how state is frozen, how an action is identified, which module may read another cell, or what happens if a record is written twice. These are not decorative engineering details. They determine whether the program computes the object named in the theorem.

47.1 Four kinds of completion

A derived identity is mathematically complete when its statement and proof have the required assumptions. A component is implemented when its actual code performs its declared function. A test is completed when it has executed and produced inspectable output. A scientific study is completed only under its registered execution and finalization contract. A file named `runner.py` establishes none of the last three by its name alone.

The current Gaussian direction is prospective [3]. The repository contains historical executable engines, preserved candidate harness code and framework objects on source-locked lineages. Their presence does not establish an accepted Gaussian runtime. This chapter uses “would” and “must” for prospective responsibilities, not as euphemisms for completed implementation.

47.2 A module’s scientific contract

For each component, state its inputs, permitted reads, outputs, permitted writes, refusal conditions and version identity. A value evaluator should not choose the action. A physical checker should not improve an action’s score. A selector should not alter quantities after valuation. A logger should not repair a failed invariant by changing history.

For example, an evaluator might receive a frozen state projection, reference and scale declarations, and a finite accepted group. It would return the exact field value and declared attribution record without changing physical state. An execution adapter would later apply the exact selected group, under separate permission. Keeping those calls separate makes it possible to test arithmetic without accidentally running a model.

47.3 Why a generic framework is still useful

The source-inspected F2 decision [6] identifies reusable identity, canonical serialization, typed state/action/support records, information contracts, provenance and ledger structures. It does not say the whole framework already fits the Gaussian programme unchanged.

In particular, its potential declaration is metadata, not a numerical Gaussian evaluator. Its named scientific advancement routes are deliberately guarded. Its physical service-capacity records are not earned EBU balances. Reusing those names while changing their meanings would conceal a new scientific model inside an apparently routine software integration.

47.4 No framework replacement by default

The smallest honest path is to reuse inspected generic pieces and add separately versioned adapters where the active scientific meaning differs. Wholesale copying can import historical assumptions; wholesale replacement can discard proven provenance discipline. The planning decision is neither “everything works already” nor “nothing can be reused.”

The source locks matter. Some relevant framework files live on another lineage and are absent from the active checkout. An ignored bytecode directory is not an authoritative source package. A later implementation task must inspect the exact committed files it intends to reuse, rather than infer availability from import names or an old chat summary.

Worked review question

A function named `advance_epoch` exists but always refuses because its scientific execution permit is unavailable. Is epoch advancement implemented? The refusal guard is implemented. Successful scientific advancement is not established. Both facts belong in the status report; neither “there is no software” nor “the engine is ready” is accurate.

47.5 Follow one action through the proposed architecture

It helps to translate the mathematical laboratory into responsibilities without pretending to implement a runtime in the book. Begin with a declared post-forcing state and a fixed parameter version. A physical proposal identifies its child actions and finite quantities. A feasibility service checks the complete group and produces an accepted vector or a named refusal. It does not produce a reward merely because the proposal looks helpful.

An evaluator reads the frozen baseline, accepted vector and complete affected factors. It returns a field difference and, only if declared, path attributions. An account service checks the adopted owner map and the corresponding balance changes. A selector chooses under the registered sampling rule. An execution service would perform precisely that chosen vector, and a recorder would bind the physical and account effects to one event identity.

These names denote logical responsibilities, not necessarily separate processes or computers. A small implementation may place several in one module. What matters is that their inputs, outputs and authority remain distinguishable. Conversely, splitting one function across many files does not create those boundaries if every helper can read global state and mutate the ledger without a declared contract.

47.6 The difference between a refusal and a zero

Suppose a quote lacks the destination's scale. Returning zero would be dangerous: zero is also a legitimate field value for an idle action or a cancelling endpoint. A missing input should be a distinct refusal, not an invented neutral valuation. The caller must be able to tell whether the calculation succeeded and returned zero or never had sufficient information to calculate at all.

The same principle applies to an unavailable scientific execution route. A module can expose a named entry point whose implementation deliberately refuses. That is an implemented guard, not an implemented scientific engine. Describing it as either wholly absent or fully runnable would misstate the software's actual status.

47.7 An interface should carry assumptions, not just numbers

A function that accepts two arbitrary arrays and returns a floating-point number can be convenient for a classroom formula. A scientific boundary needs to know which arrays represent the same domain and epoch. The state dimension, coordinate identities, units, scale positivity, reference compatibility and potential version must be checked somewhere explicit. Otherwise a numerically valid answer can combine incompatible objects.

An action record likewise needs its support and quantity convention. If a rate is supplied where a quantity is expected, a unit-duration fixture may conceal the mistake. If a helper changes the accepted quantity but preserves the action identifier, a replay can appear to match while valuing different physics. Identities should bind semantic content, not merely a convenient display name.

47.8 Shared code and independent checks

Reusing a verified evaluator avoids inconsistent copies. But a test of that evaluator needs an expected result that does not simply call the same implementation twice. The laboratory's endpoint subtraction, quadratic expansion and path calculation provide independent routes through the mathematical specification. A deliberate wrong-sign or missing-curvature version should fail an appropriate check.

The physical layer needs independent checks too. A potential residual can close when the same wrong increment is used in both valuation and execution. Comparing only those two numbers would miss the fact that the accepted physical action was different. The action commitment, physical successor and account equation should cross-check distinct facts rather than echo one shared error.

47.9 Why the next chapters are ordered this way

The next chapter declares the smallest honest world. Physical permission then explains what that world permits. Only after those two objects are clear do we place them in an event chronology. Menus and selection follow because their meaning depends on both physical opportunities and the state at which they are assembled. Evaluation and accounting follow because the exact vector and baseline must already be identified.

Randomness, tests and provenance come afterward, not as optional polish, but as the conditions under which another reader could reconstruct the specified process and interpret an eventual result. This sequence joins the former mathematics and implementation books into one argument.

Chapter recap

The bridge from mathematics to software is a set of typed responsibilities and verifiable effects. The bridge from software to evidence requires an additional registered study, not a stronger adjective in the implementation description.

The smallest honest Gaussian world

The first active world should be small enough that every represented quantity has an unambiguous meaning. Small does not mean vague. A two-cell model with mixed units and hidden replenishment is less interpretable than a larger model with explicit boundaries.

48.1 Physical state and reference are different records

The stock vector x changes when physical actions or forcing occur. The reference x^* and scales σ are fixed configuration in the first domain. Both state and reference are nonnegative and share total mass M ; every scale is finite and strictly positive. NaN, infinity, a zero scale or a negative stock is not a small deviation to smooth away.

The potential is

$$V(x) = \frac{1}{2} \sum_i ((x_i - x_i^*)/\sigma_i)^2. \quad (48.1)$$

It is a derived value, not another physical carrier. Storing a cached value may be convenient, but replay must be able to recompute it from the exact declared state and configuration. Updating the cache without the state, or the state without invalidating the cache, risks valuing a nonexistent world.

48.2 Accounts belong beside, not inside, the stock vector

The proposed augmented record includes physical state x , persistent capacity B , signed external audit J , and random-stream replay coordinates. They have linked versions but distinct meanings. Adding an EBU balance to a water total is dimensionally invalid. Treating a physical capacity constraint as a spendable balance is semantically invalid even if both are numbers.

The old candidate C3 export budget also used a letter resembling B . That does not make it the new persistent account. A safe specification uses explicit type names and conversions where necessary, not a shared generic field called budget for every purpose.

48.3 Atomic action declarations

An action records its identity, source, destination, finite quantity, carrier, write support and relevant constraints. For a lossless transfer, $\delta_a = q_a(-e_i + e_j)$. A group retains every child and its accepted increment; it also records the aggregate δ_G .

The source-side quantity convention must remain explicit when importing historical rate-based examples. A duration belongs in the conversion $q = \Delta t j$, not in both the conversion and the finite evaluator. Non-unit-duration fixtures expose errors that unit-duration examples hide.

48.4 What is deliberately absent

The active first domain contains no Allee regeneration, lower/upper homeostatic band, hidden demand-serving controller or autonomous gradient flow. Its reference is not an empirically fitted stationary mean by default. Its scales are not automatically measurement-error bars. It contains one homogeneous conserved scalar, not an uncalibrated mixture of energy, water, medicine and welfare.

This is a scope statement, not a claim those phenomena are unimportant. Adding them later changes state, dynamics or valuation and must receive its own derivation and registration. An honest small world provides a controlled question; an underspecified broad world provides ambiguous answers.

A configuration audit

Suppose a proposed file gives $x = (18, 2)$, reference $(10, 10)$ and scales $(2, 2)$. It passes the elementary domain checks. Replace the reference with $(12, 12)$ and closed mass compatibility fails. Replace a scale with zero and valuation is undefined. Add a regenerative source label without a regeneration law and the label has supplied no physical mechanism.

Each failure should identify the exact predicate and field. “Invalid configuration” alone is difficult to audit. Silently normalizing the reference or clipping a stock would make the accepted model differ from the author’s declaration.

Chapter recap

Typed state, fixed geometry and complete action declarations make later checks meaningful. The first world is intentionally limited; its limitations must remain readable rather than filled with inherited historical defaults.

Permission before execution

The schema in the preceding chapter tells us what the world contains. It does not tell us which proposed movement the world permits. This missing step is easy to overlook when a quoted action has an attractive value. An evaluator can calculate a field difference for a negative-stock endpoint just as readily as for a valid one. Software must therefore distinguish the ability to calculate a number from authority to perform its movement.

49.1 Three questions, three records

First ask whether the declared physical mechanism can perform the action. Second ask what the action would do to the represented potential. Third, if the prospective account policy has been adopted, ask whether each owner can afford its assigned change. An action can pass one question and fail another. Keeping separate records lets a reviewer understand why it was refused without pretending that every refusal expresses one universal notion of safety.

Physical permission includes the source and destination constraints, allowed topology, carrier identity and timing. Field valuation depends on the frozen state and potential. Account permission depends additionally on the owner map and prior balances. The physical-random comparator omits the third filter; it does not get a different tank capacity or a privileged right to create resource.

49.2 Endpoint checks are necessary, not a complete plant

For a closed lossless action, a simple check verifies

$$z + \delta_G \geq 0, \quad \mathbf{1}^T \delta_G = 0. \quad (49.1)$$

It is tempting to call this the whole permission rule. Yet a pipe can have a finite delivery rate, a machine can serve only one request at a time, and a destination can have a storage ceiling. The first domain may intentionally omit these complications. It must say so. An omitted constraint is not a constraint that the software has proved satisfied.

For a declared constant-rate aggregate path, convexity of the nonnegative simplex keeps intermediate stocks nonnegative when both endpoints are valid. That argument concerns aggregate coordinates. It does not authorize an individual machine's rate or prove the causal availability of incoming material to an outgoing child action. The path and plant declarations must agree about what simultaneity means.

49.3 A supplied budget is not a permission theorem

Suppose a mechanism is given two requests of four units and a joint budget of six. Proportional scaling gives three units to each. This arithmetic can be reused wherever proportional scaling is the declared resolver. It says nothing about why the budget is six. A source-preservation certificate, a physical throughput limit and a deliberate experimental restriction can all supply the same numerical budget while making different claims.

The distinction is especially important for historical P1C. Its robust budget routine assumes an eligible regenerative source. Applying the formula to a finite store without that classification does not inherit the certificate. Equally, applying the old classifier's finite-extraction prohibition to every internal relocation would prevent the new closed world from expressing its intended question. Neither change should be smuggled through a helper function with a familiar name.

49.4 Export protection is not destination recovery

The historical quantity R_{eff} belongs to a source export certificate. A destination's old homeostatic reserve R belongs to the potential and condition of that destination. Even when a source budget protects its own boundary perfectly, it does not choose where to send stock or guarantee that another node's reserve is restored.

In the active Gaussian domain, the reference x^* is neither of these objects. It defines a comparison geometry, not a prohibition against physically crossing a reference value. Otherwise the physical permission layer would become a reference-seeking controller before the actor made a choice, and the proposed comparison would test a different mechanism.

49.5 Resolve first, then bind the exact vector

If a declared resolver modifies proposals, record both the requests and the accepted quantities. The evaluator receives the accepted vector and its com-

plete support. The eventual execution must apply that same vector, not a newly clipped version with the old value attached.

If a late physical condition invalidates execution, a prospective system needs an explicit refusal or re-epoching procedure. It may not retain the old positive attribution while silently reducing deliveries. Whether to retry is also a policy decision: repeated attempts can change selection probabilities and consume different random draws.

49.6 A permission review in plain language

For each refusal, a reader should be able to finish one sentence: “This proposal could not proceed because this named requirement failed against this version of the state.” A missing owner, a negative stock and an unavailable execution permit are different failures. Mixing them into one generic false value makes debugging easier only by making the science harder to audit.

The historical source reading that follows shows why the premises of an actual permission mechanism must remain beside its formula. After it, the event-order chapter will place these records into one explicit chronology.

Historical reading H49: The original P1C permission layer

Historical model; not Gaussian evidence

This source reading explains the original classifier and source-budget mechanism in full. It is retained because reusing its arithmetic without its source-typing premises previously obscured the difference between preservation and internal transfer. Its prohibition for finite extraction is not adopted here as a general prohibition on moving the conserved scalar. Original Part III, PDF pages 37–44. Internal chapter, section, equation and page references retain their original meaning. The new running folio belongs to this integrated volume.

What to carry forward. Reuse of proportional scaling does not adopt the permission policy that supplies its budget. An implementation must choose and authorize the latter explicitly. We can now discuss event order without allowing a convenient helper to define the plant in secret.

P1C: how a source says "not safely available"

The D0 force answers a directional question. It can point from a relatively comfortable source toward an urgent deficit. Direction alone does not answer how much a regenerative source may release during the next tick. If the source is close to a biological threshold, even a burden-reducing transfer can remove the capacity on which future regeneration depends.

P1C adds a source-local preservation layer. The name identifies a particular candidate controller in the V2.9 research sequence; it is not a general name for all safe control. Its core idea is familiar from a household emergency reserve: first calculate what must remain, then divide only the verified surplus among current requests.

49.1 Threshold, field reference, and certified reserve

Three quantities are easy to confuse.

- The viable-band floor L_i describes a desired functional condition in the burden V .
- An Allee threshold A_i belongs to the regenerative dynamics. Below it, the toy source can enter a self-reinforcing decline.
- The effective reserve R_i^{eff} is the operational boundary used by the preservation controller.

The controller's reserve may sit above the dangerous ecological threshold. That margin can represent uncertainty, discretisation, or a deliberately conservative operating rule. It must be certified outside the transaction itself. P1C uses a declared reserve; it does not discover the correct ecological threshold from price or from a convenient simulation result.

49.2 The no-export successor

Freeze a source with present stock x_i and local natural drive u_i . If it exports nothing, its one-tick successor is

$$z_i = x_i + \Delta t u_i. \quad (49.1)$$

This number is a first diagnostic. If $z_i < R_i^{\text{eff}}$, then even zero export cannot keep the source above the reserve for this tick. Rejecting all requests is still the least damaging action, but it is not a preservation success. The controller reports local infeasibility instead of pretending that inaction solved the problem.

49.3 Four source states

P1C classifies each exporting source from frozen local data.

State P – preservable. The source is at or above the reserve and its no-export successor also remains at or above it. A regenerative source may have a positive safe-export budget.

State R – recovery required. Present stock is already below the reserve. The export budget is zero even if natural drive might lift it during this tick. Recovery is not treated as fresh surplus.

State I – locally infeasible. Present stock is not below the reserve, but the no-export successor is. Export is zero, and the controller explicitly marks that zero export is insufficient.

State F – flow source. The source represents a declared external flow rather than a preserved regenerative stock. Its budget is limited by the flow cap and physically available stock.

Finite and irreversible stocks receive zero *preservation-safe extraction* budget. This does not prohibit all use in every future institution. It states a physical fact: positive extraction depletes such a stock and cannot be described as preserving it. A future depletion or restoration account must name that fact instead of laundering it through a preservation label.

49.4 Deriving the robust budget

Let Q_i be the aggregate accepted export rate from source i during the tick. Ignoring incoming flow when granting permission, the source successor is

$$x_i^{n+1} = x_i + \Delta t(u_i - Q_i). \quad (49.2)$$

To require $x_i^{n+1} \geq R_i^{\text{eff}}$, rearrange:

$$x_i + \Delta t(u_i - Q_i) \geq R_i^{\text{eff}}, \quad (49.3)$$

$$\Delta t Q_i \leq x_i + \Delta t u_i - R_i^{\text{eff}}, \quad (49.4)$$

$$Q_i \leq \frac{x_i + \Delta t u_i - R_i^{\text{eff}}}{\Delta t}. \quad (49.5)$$

Measurement and drive uncertainty are represented by physical margins ε_x and ε_u . The implemented robust budget is

$$Q_{i,\max}^{\text{rob}} = \frac{[(x_i - \varepsilon_x) + \Delta t(u_i - \varepsilon_u) - R_i^{\text{eff}}]_+}{\Delta t}. \quad (49.6)$$

The positive part prevents a negative budget. The uncertainty margins are not floating-point tolerances. They have physical units and must come from a measurement model. A separate numerical tolerance is used only when comparing finite-precision values.

49.5 A worked budget

Suppose a source contains $x = 13$, its certified reserve is 11, its local drive is $u = 0.5$ stock units per hour, and the tick lasts $\Delta t = 0.2$ hours. With no uncertainty margins,

$$Q_{\max} = \frac{13 + 0.2(0.5) - 11}{0.2} = 10.5 \quad (49.7)$$

stock units per hour. The maximum stock withdrawal during the tick is $0.2(10.5) = 2.1$. Starting from 13 and adding 0.1 through natural drive leaves exactly 11.

Now introduce $\varepsilon_x = 0.1$ stock units and $\varepsilon_u = 0.2$ stock units per hour:

$$Q_{\max}^{\text{rob}} = \frac{12.9 + 0.2(0.3) - 11}{0.2} = 9.8. \quad (49.8)$$

The smaller budget is not a fine or a price. It is the consequence of promising the reserve under the declared uncertainty envelope.

49.6 Why incoming flow is excluded from permission

An outgoing source may also be due to receive material through another edge. P1C records accepted incoming delivery in the final update, but it does not spend that uncommitted arrival when calculating the source's export budget. Otherwise two sources could each promise exports based on material expected from the other, creating circular permission with no independently available stock.

This is analogous to a hospital not releasing its last emergency medicine merely because a delivery truck is expected later. When the truck arrives, the new stock can support a new action in the next accounting epoch. The rule is conservative, local, and causally clear.

49.7 One source, many outgoing requests

Suppose three destinations request rates 4, 3, and 5 from the same source. The aggregate request is

$$Q_i^{\text{req}} = 4 + 3 + 5 = 12. \quad (49.9)$$

If the safe budget is 6, P1C computes

$$\sigma_i = \min \left(1, \frac{Q_{i,\text{max}}^{\text{rob}}}{Q_i^{\text{req}}} \right) = \frac{1}{2} \quad (49.10)$$

and accepts rates 2, 1.5, and 2.5. Their sum is exactly 6.

The important word is *aggregate*. Capping each edge separately at 6 would permit a total withdrawal of 18. This is the multi-out-edge duplication trap. The budget belongs to the source, not to each request line.

Proportional scaling answers how the current P1C controller shares a binding physical allowance. It does not prove that proportional allocation is the best social priority rule. Urgency, essential needs, and institutional guarantees may require a different layer, provided that the total physical withdrawal still respects the same source budget.

49.8 A complete P1C tick with two destinations

Let a regenerative source begin at $x_0 = 10.6$ with reserve 10. During a 0.2-hour tick its local drive is $u_0 = 0.5$, and uncertainty margins are zero. The

no-export successor is 10.7, so the source is State P. Its safe export rate is

$$Q_{\max} = \frac{10.6 + 0.2(0.5) - 10}{0.2} = 3.5. \quad (49.11)$$

Suppose edge requests are 4 and 3. The aggregate request is 7, hence $\sigma = 3.5/7 = 0.5$, and accepted rates are 2 and 1.5. With efficiencies 0.9 and 0.8, the destinations receive rates 1.8 and 1.2. The physical update of the source is

$$x_0^{n+1} = 10.6 + 0.2(0.5 - 2 - 1.5) = 10. \quad (49.12)$$

The source reaches but does not cross its reserve. Destination stock increments from transport are 0.36 and 0.24. The tick records requested and accepted rates, loss rates 0.2 and 0.3 per unit time, $\sigma = 0.5$, state P, the budget 3.5, and the boundary result.

If each edge had incorrectly received the full source budget, accepted rates could sum to 7 and the source successor would be

$$10.6 + 0.2(0.5 - 7) = 9.3, \quad (49.13)$$

well below reserve. This is why aggregate grouping is a theorem-relevant operation, not an implementation optimisation.

49.9 P1C and bounded service solve different boundary problems

P1C protects exporting sources. It does not by itself stop a consuming destination from subtracting more demand than physically available. The later bounded service wrapper protects that second boundary by saturating service at available stock.

In the example above, if destination 1 has stock 0.1 before delivery and tick demand 0.6, then available stock after delivery is 0.46. Bounded service delivers 0.46, records unmet demand 0.14, and leaves destination stock zero. Source preservation and destination non-negativity are both satisfied, but the need is not fully met.

This layered result is more informative than one “safe” flag. It says exactly which boundary held and where scarcity appeared.

49.10 Why State R does not immediately export re-covered surplus

Suppose a source begins at 9.9 below reserve 10 but has enough positive drive to reach 10.2 during the tick. P1C classifies it State R from its present stock and sets export to zero. It does not spend the forecast 0.2 surplus before recovery has actually occurred.

On the next tick, the newly observed state can be classified again. This one-epoch delay is conservative, but it avoids circular claims on unobserved recovery and keeps action permission tied to completed physical history.

49.11 The one-step preservation theorem

For a nominal State-P regenerative source, assume the frozen stock and drive are the quantities used by the controller, the uncertainty margins bound their errors as declared, and the conservative current stock and conservative no-export successor satisfy

$$x_i - \varepsilon_x \geq R_i^{\text{eff}}, \quad (x_i - \varepsilon_x) + \Delta t(u_i - \varepsilon_u) \geq R_i^{\text{eff}}. \quad (49.14)$$

If accepted outgoing flow does not exceed the robust budget and the update applies those accepted flows synchronously, then the no-incoming part of the next source state satisfies

$$x_i^{n+1} \geq R_i^{\text{eff}}. \quad (49.15)$$

The proof is the budget derivation read backward. Incoming delivery, if any, is non-negative and can only increase the successor. This is a controlled one-step invariance result. It is not a theorem that the entire coupled system converges, that all future demand is feasible, or that the reserve itself was ecologically certified correctly.

The robust proof written line by line

Let the true stock and drive be no worse than the conservative values used by the controller:

$$x_i^{\text{true}} \geq x_i - \varepsilon_x, \quad u_i^{\text{true}} \geq u_i - \varepsilon_u. \quad (49.16)$$

Let accepted aggregate export obey $Q_i^{\text{acc}} \leq Q_{i,\text{max}}^{\text{rob}}$. Then, excluding incoming delivery,

$$x_i^{n+1} = x_i^{\text{true}} + \Delta t(u_i^{\text{true}} - Q_i^{\text{acc}}) \quad (49.17)$$

$$\geq (x_i - \varepsilon_x) + \Delta t(u_i - \varepsilon_u - Q_{i,\text{max}}^{\text{rob}}) \quad (49.18)$$

$$\geq R_i^{\text{eff}}. \quad (49.19)$$

The last line follows because conservative no-export feasibility makes the budget numerator non-negative. If that feasibility condition fails, the positive part returns zero but the last inequality does not follow: zero export is then a refusal, not a robust preservation certificate. State I records the corresponding nominal failure. With nonzero margins, the conservative failure can also occur while the measured source is nominally State P.

Nominal State P is not sufficient for the robust theorem

The released conformance summary currently derives theorem eligibility from the nominal State-P classifier. It does not separately test Equation (49.14). With zero margins the two tests coincide, as in the registered D9/D10 conformance worlds. With nonzero margins, the eligibility diagnostic must be strengthened before it can be read as a robust theorem certificate. This is a bounded proof-and-diagnostic obligation, not evidence that a registered trajectory crossed its reserve.

If accepted incoming delivery is $I_i \geq 0$, the full successor is the expression above plus $\Delta t I_i$, so the inequality still holds. Incoming flow is useful in the realised state but unnecessary for the permission proof.

From one step to many steps

If the assumptions are verified again at every tick and the source remains eligible, the one-step implication can be applied inductively: the reserve holds after each completed tick. This is not yet a global stability theorem. The eligibility and uncertainty assumptions may fail, demand may become infeasible, and other state coordinates may deteriorate. Induction preserves the named boundary; it does not create regeneration, service, or convergence.

49.12 Permission is prior to EBU

The quote layer receives q^{acc} from P1C. It may evaluate any measured quantity between zero and that accepted maximum. It cannot enlarge the domain because a quote is attractive. In plain language: an excellent use of an unsafe quantity does not make the quantity safe.

This separation protects essential needs as well. If medicine is needed urgently but the local source cannot safely provide it, the system has discovered a real constraint. The response may be borrowing, another source, emergency reserves, or future infrastructure. Relabelling an unsafe local withdrawal as cheap would hide rather than solve the need.

Chapter recap

P1C classifies the source, calculates one source-level robust export budget, aggregates all outgoing requests, and scales accepted flow so that an eligible source remains above its certified reserve for the next tick. It reports recovery and infeasibility honestly. The theorem is local and one-step; EBU accounting comes only after physical permission.

Exercises

1. Calculate the robust budget for $x = 8.4$, $R^{\text{eff}} = 8$, $u = 0.1$, $\Delta t = 0.25$, $\varepsilon_x = 0.05$, and $\varepsilon_u = 0.04$.
2. A source receives requests $(2, 5, 1)$ and has budget 6. Calculate the proportional accepted rates.
3. Explain why State I is not counted as a failure of the one-step preservation theorem.

An event begins with its baseline

The same physical increment can receive different values at different states. It follows that the event boundary is part of the scientific definition, not merely a scheduler detail. A record must identify when forcing occurs, when the state is frozen and which version each candidate uses.

50.1 The intended forcing-first order

The new programme proposes the following logical sequence:

Phase	Responsibility
1	Apply the declared external forcing and record its signed audit.
2	Freeze one post-forcing physical baseline and its configuration.
3	Generate EBU-blind candidates/groups and check joint physical feasibility.
4	Evaluate exact group value and declared path attribution.
5	Aggregate by declared owner and screen affordability.
6	Make an unweighted random choice from the declared eligible menu.
7	Execute exactly that selected accepted vector, then record accounting effects once.

These are explanatory phases, not a new accepted enumeration of framework phase constants. The inspected framework has its own fixed ten-phase chronology, including natural drive in a different position. A recognized Gaussian event profile must resolve that difference before runtime adoption [6].

50.2 Why forcing is separate

Write x_t for the pre-forcing state and z_t for the post-forcing baseline. The external record is $D_{\text{ext},t} = V(z_t) - V(x_t)$. All candidates in that epoch compare

z_t with their own declared endpoints. An idle actor therefore has zero action value even if forcing just improved the field.

If one candidate uses x_t and another uses z_t , their comparison includes different external effects. If forcing changes again between quote and execution, the committed conditions need explicit treatment; the software cannot quietly call the new state the old baseline.

50.3 Epoch identity binds meaning

A useful epoch commitment identifies the state version, physical profile, potential and parameter version, accepted child vector, complete support, path semantics, owner-rule version and numerical policy. Not every module needs to read all of these values, but the composite record must connect them unambiguously.

Changing an accepted quantity changes the endpoint and generally every shared-path attribution. Removing one child can change the value of the others. Such changes require a new affected valuation and authorization record, not reuse of an old number whose identity happens to look familiar.

50.4 A worked baseline error

At reference (10, 10) the potential is zero. Forcing produces (14, 6) with potential four. A restoring action ends at (12, 8) with potential one. The actor's value is $4 - 1 = 3$. Comparing the original reference directly with the endpoint gives $0 - 1 = -1$, which includes the shock and answers a different question. Both subtractions are arithmetically correct; only one is the declared actor account.

This is why a test of subtraction alone is insufficient. The implementation must also test which state commitment supplies each operand.

50.5 Failure and publication order

A prospective runtime must specify what is durable before state advancement becomes possible, what constitutes one completed event and how partial publication is detected. Book prose cannot substitute for that execution contract. Historical Gate 1D-C has a particular durability protocol; it must not be copied as though every new study shares its exact permissions.

Chapter recap

The baseline, chronology and event identity are part of the model. A new diagram explains the intended process but does not amend the framework's existing scientific order by itself.

Historical reading H50: One historical synchronous tick, slowly

Historical model; not Gaussian evidence

The original walkthrough makes frozen evaluation and synchronous execution concrete. Its natural-drive and P1C chronology are not the proposed forcing-first Gaussian chronology. Compare the order of operations, not just the spelling of variables. Original Part III, PDF pages 31–36. Internal chapter, section, equation and page references retain their original meaning. The new running folio belongs to this integrated volume.

What to carry forward. Two event profiles can contain the same-looking operations and still implement different models. The prospective profile must specify when forcing is applied, when state is frozen and which accepted increment is valued. Reordering an implementation is not an editorial amendment.

One synchronous D0 tick, slowly

A *tick* is one discrete update from x^n to x^{n+1} . The word does not mean that nature changes in jumps. It means the program observes a frozen state, calculates rates, and advances its numerical model by the declared duration Δt .

48.1 The natural drive

Each cell has an on-site rate $u_i(x_i)$. It can include external supply, regeneration, demand in the unconstrained D0 model, constant leakage, and proportional leakage. For example, a logistic regenerative source uses

$$g_i(x_i) = \rho_i x_i \left(1 - \frac{x_i}{K_i}\right). \quad (48.1)$$

The natural drive is local: it depends on the cell's own state and declared parameters, not on the complete world.

An Allee-like regenerative source adds a threshold factor,

$$g_i(x_i) = \rho_i x_i \left(1 - \frac{x_i}{K_i}\right) \left(\frac{x_i}{A_i} - 1\right). \quad (48.2)$$

Above A_i and below capacity, regeneration can be positive. Below A_i , the last factor is negative, so the isolated model moves further downward. The equation is a controlled representation of threshold behaviour, not a claim that all ecosystems share one mechanism.

48.2 Marginal field signal

The local marginal is the derivative of the cell penalty:

$$\mu_i(x_i) = -2\alpha_i[L_i - x_i]_+ + 2\beta_i[x_i - U_i]_+ - 2\chi_i[R_i - x_i]_+. \quad (48.3)$$

It measures the local sensitivity of the burden to a small increase in stock. A large negative value at a deficit cell means that adding stock would strongly reduce its penalty. A positive value above the upper band means that removing some stock may reduce excess burden.

For edge $e : i \rightarrow j$, the loss-aware force is

$$f_e = \mu_i - \eta_e \mu_j. \quad (48.4)$$

The destination marginal is multiplied by efficiency because one unit leaving the source produces only η_e units at the destination. Omitting this factor pretends that lost material arrived.

The raw one-way flux is

$$J_e = M_e [f_e - \theta_e]_+. \quad (48.5)$$

Mobility M_e scales the response. Threshold θ_e creates a dead band: small forces do not activate the route.

48.3 Freeze first, then calculate every edge

At the start of a tick, the program copies the complete state into an immutable tuple. Every natural drive, local view, force, and flux reads this same x^n . No edge sees the partial result of an earlier edge.

This is called a *synchronous* or *frozen-state* update. It has two important consequences. First, permuting the stored edge order does not change the mathematical result. Second, information travels at most one graph edge per tick. In a chain $0 \rightarrow 1 \rightarrow 2$, cell 2 cannot react during the same tick to material that cell 1 has only just received from cell 0.

48.4 The incidence action without matrix anxiety

The compact equation is

$$x^{n+1} = x^n + \Delta t(u(x^n) + SJ(x^n)). \quad (48.6)$$

The matrix S is an efficient incidence operator. The notation can look abstract, but its job is simple. For every edge, it places $-J_e$ in the source cell and $+\eta_e J_e$ in the destination cell. Adding the contributions from all edges gives the transport rate of every cell.

For one edge $0 \rightarrow 1$ with flux J , the transport contribution is

$$SJ = (-J, \eta J). \quad (48.7)$$

For two outgoing edges $0 \rightarrow 1$ and $0 \rightarrow 2$, it is

$$SJ = (-J_1 - J_2, \eta_1 J_1, \eta_2 J_2). \quad (48.8)$$

No matrix multiplication is required to understand the bookkeeping.

48.5 A numerical tick

Let $x^n = (10, 2)$, $u = (1, -0.2)$, $J = 3$, $\eta = 0.8$, and $\Delta t = 0.1$. The net rates are

$$u + SJ = (1 - 3, -0.2 + 2.4) = (-2, 2.2). \quad (48.9)$$

The stock changes are $0.1(-2, 2.2) = (-0.2, 0.22)$, so

$$x^{n+1} = (9.8, 2.22). \quad (48.10)$$

The source loses 0.3 units to transport, the destination receives 0.24, and 0.06 is transport loss. Natural drive adds 0.1 to the source and removes 0.02 from the destination. Keeping these terms separate makes the result auditable.

48.6 The stock ledger

Summing all cell changes gives

$$\mathbf{1}^\top (x^{n+1} - x^n) = \Delta t \mathbf{1}^\top u - \Delta t \sum_e (1 - \eta_e) J_e. \quad (48.11)$$

Internal lossless transfers cancel. Only natural drive and declared transport loss change total stock. A small numerical residual is recorded because floating-point addition is not exact arithmetic.

Ledger closure does not mean that the action was beneficial. It means the stated physical bookkeeping is internally consistent. A perfectly balanced ledger can describe a harmful withdrawal; a falling burden can coexist with transport loss.

48.7 Why D0 does not clip

The unconstrained D0 engine allows an update to leave the declared domain and flags the event. It does not silently replace a negative stock with zero or an over-capacity stock with K_i . Silent clipping would create or destroy material without naming the correction and could make an unstable step appear safe.

A separate projected diagnostic wrapper exists, but it records the correction and sits outside the D0 theorem. This is another example of the evidence ladder: a convenient bounded wrapper is not automatically covered by a theorem proved for an unconstrained equation.

48.8 The timestep certificate in plain language

The explicit update uses the slope measured at the beginning of the tick. If the tick is too long, the state can pass through the low-burden region and land on the other side. The finite-step certificate bounds Δt using three kinds of information: the maximum curvature of the cell penalties, the edge mobilities and efficiencies, and the number of incident edges that can act on a cell.

The implementation computes a one-edge certificate and a degree-aware Gershgorin certificate. “Gershgorin” names a matrix-bound technique; the reader does not need its full proof to understand its role. A cell connected to several strong edges can change faster than a cell connected to one weak edge, so the safe explicit timestep must account for the combined local coupling.

The reported ratio

$$r_{\Delta t} = \frac{\Delta t}{\Delta t_{\text{certificate}}} \quad (48.12)$$

is a compact audit field. A value 0.5 means the study uses half the computed bound; 0.9 means it runs near the bound; 1.2 means it is deliberately outside. The ratio is not a probability of failure and not a physical uncertainty margin.

48.9 A multi-edge trace row by hand

Return to a source 0 with two accepted edge rates $q_1 = 2$ and $q_2 = 1.5$, efficiencies 0.9 and 0.8, and $\Delta t = 0.2$. Let the natural drive vector be $u = (0.4, -0.1, 0)$. The transport rate contributions are

$$SJ = (-3.5, 1.8, 1.2). \quad (48.13)$$

Adding drive gives

$$u + SJ = (-3.1, 1.7, 1.2). \quad (48.14)$$

If $x^n = (12, 2, 4)$, the successor is

$$x^{n+1} = (12, 2, 4) + 0.2(-3.1, 1.7, 1.2) = (11.38, 2.34, 4.24). \quad (48.15)$$

The total stock change is -0.04 . Natural drive contributes $0.2(0.4 - 0.1) = +0.06$. Transport loss is

$$0.2\{(1 - 0.9)2 + (1 - 0.8)1.5\} = 0.10. \quad (48.16)$$

Thus $+0.06 - 0.10 = -0.04$, and the ledger closes.

A trace row would keep more detail than the compact state equation: both edge rates, efficient deliveries, loss, source and destination identifiers, before and after states, and residual. A reviewer can therefore reconstruct the update without trusting a precomputed total.

48.10 Why summation order is recorded carefully

Floating-point addition is not exactly associative. Adding many contributions in different orders can produce tiny last-bit differences even when the algebraic sum is identical. The released engine accumulates edge contributions and uses a stable summation function. Permutation tests then check the declared invariance within the appropriate numerical standard.

This is not evidence that the physical world depends on Python's addition order. It is evidence that the implementation recognises a numerical representation issue and does not confuse it with a scientific effect.

Chapter recap

One D0 tick freezes the state, calculates local drives and loss-aware edge fluxes, sums all edge contributions, and applies one simultaneous update. The state vector notation compresses ordinary source withdrawal and efficient destination delivery. The stock ledger checks accounting, while domain flags expose rather than conceal impossible numerical states.

Exercises

1. Repeat the numerical tick with $\eta = 1$. Which term changes in the stock ledger?
2. In the chain $0 \rightarrow 1 \rightarrow 2$, explain why a sequential implementation can transmit a disturbance two edges in one nominal tick.
3. Why can a closed stock ledger not prove that the burden V decreased?

Candidates, physical permission and group selection

The candidate menu is where an apparently innocent implementation detail can become an undeclared controller. If the generator proposes only actions that point toward reference, the later selector is not testing unranked physical opportunities, however random its final choice may be.

51.1 EBU-blind does not mean physically ignorant

The proposed generator may use the physical information permitted by its contract: topology, available stock and declared capacity constraints, for example. It must not rank or shape opportunities using EBU, global potential, future recovery or an undeclared service objective. Exactly which local physical information it may read remains a declared design choice.

A finite menu restricts the experiment. If physical quantities are continuous but the menu contains only a few values, the study concerns that menu, not all possible divisible action. This is acceptable when explicit. It becomes misleading when a failure to reach reference is generalized without checking whether the menu ever made the necessary quantity available.

51.2 Joint constraints come before value

The physical layer validates a complete group. It checks nonnegative end-points, conservation, aggregate source and destination constraints, allowed edges and declared timing. It uses the same rules for the EBU arm and the physical-random control.

Two individually valid children may exceed a shared cap. If a resolver changes their quantities, the new accepted vector is the one to value. After selection, execution must not run another hidden resolver and keep the old quote. That would settle one vector while performing another.

The order “physical permission before value” does not mean physical permission chooses the most valuable allocation. A proportional allocator, priority allocator and refusal rule are different policies. A source-locked study must say which one it uses or avoid unresolved contention in its declared domain.

51.3 Uniform over what?

Suppose a menu contains physical options A and B, but A appears under two identifiers. Uniform sampling over entries gives A probability two-thirds; uniform sampling over unique physical options gives one-half. Neither interpretation can be inferred from the word random alone.

Group size matters too. Uniform sampling of groups can give an action that belongs to many groups a high chance of execution. Uniform sampling of owners first produces another distribution. A no-action option can affect refusal and time occupation even when it has zero field value. All of these choices change the transition kernel and must be fixed prospectively.

51.4 Local calculation and global assembly

An exact group value can be evaluated from its affected factors, while the group assembler still reads a world-wide menu. Thus factor-local valuation does not prove a decentralized actor process. The specification must name the chooser, its view, its communication and its authority over owners.

There is no need to settle every distributed-system research question in the first model. It is necessary to state what was actually chosen. A centralized experimental sampler can be studied honestly if it is not advertised as a proof of fully local institutional implementation.

A useful negative control

Consider two baseline states with identical permitted local observations but different hidden remote potential. The candidate generator should not change its output merely because the hidden diagnostic changed. Designing such an information-boundary fixture is useful; executing a full transition with it belongs to the separately authorized integration stage.

Chapter recap

The menu, duplicate policy, group assembly and sampling unit are part of the science. They should not be left to incidental list construction or silently optimized in the name of convenience.

An exact evaluator and a trustworthy quote

An evaluator’s task is narrow: given a declared baseline and accepted finite action, return the corresponding value without moving the model. A narrow task is easier to audit because unexpected side effects have no legitimate place to hide.

52.1 Independent arithmetic routes

For the Gaussian domain, two direct routes should agree:

$$E_G = V(z) - V(z + \delta_G), \quad (52.1)$$

$$E_G = -\mu(z)^T \delta_G - \frac{1}{2} \delta_G^T H \delta_G. \quad (52.2)$$

Local affected-factor evaluation provides a third comparison with the full diagnostic difference. These routes share a mathematical specification but can expose different implementation mistakes: wrong signs, omitted curvature, incomplete support or a stale baseline.

A test that compares a function with another call to the same function is not an independent check. A fixed rational example with an independently written expected value is often more revealing than thousands of repeated calls through one mistaken helper.

52.2 Exact algebra and numerical representation

The formula is exact mathematics. Floating-point evaluation is not exact real arithmetic. A future protocol must specify whether representation is rational, fixed-point or floating-point, and what refusal or tolerance rules apply near zero, affordability and conservation boundaries.

Reporting a residual of zero on selected fixtures does not prove all real inputs will produce exact machine zero. Conversely, choosing a convenient tolerance after observing a failure can weaken the scientific contract. The numerical policy must be fixed before outcomes can influence it.

52.3 Value, attribution and settlement have different outputs

The group value is a single field difference. A common-path evaluator can add R_a for each child and verify $\sum_a R_a = E_G$. A settlement service would additionally require an accepted owner/policy map and a valid event identity. Combining all three in one return value called reward invites the caller to assume an authority that the evaluator never established.

The preserved candidate code already distinguishes physical constant-rate path semantics from endpoint interpolation. That distinction is reusable as an idea. Its mere presence in a candidate module does not authorize spendable Gaussian capacity or prove every interpretation guard sufficient.

52.4 A quote must describe what was accepted

For a one-dimensional accepted schedule, the record can identify the full function on its accepted domain. A measured shortfall can select a point on that precommitted function. For a simultaneous group, changing one child may alter every path attribution. The applicable multidimensional or re-epoching contract must therefore be declared rather than extrapolated from the simplest single-action schedule.

An overshoot outside a committed domain is a protocol violation. It does not justify an improvised positive credit, nor an arbitrary punishment presented as physical EBU. Physical measurement, account calculation and sanctions remain separate questions.

Worked evaluator audit

At (18, 2), reference (10, 10), scales (2, 2) and quantity four, the endpoint is (14, 6). The potential route gives $16 - 4 = 12$. The tangent is sixteen and the curvature correction four, again giving twelve. A returned value of sixteen suggests the finite evaluator has silently become a first-order approximation. A returned value of fourteen for two two-unit children suggests independent singleton values were added.

Neither error is cured by increasing decimal precision. The wrong mathematical object was computed.

Chapter recap

Exact evaluation needs a complete baseline, support, increment and numerical policy. A successful calculation is not permission to execute, and a closing decomposition is not authorization to credit an owner.

Historical reading H52: The original exact-quote implementation

Historical model; not Gaussian evidence

This complete original account explains quote domains, exact local evaluation and the historical no-action baseline. The cancellation principle survives; the historical baseline and cost assumptions do not automatically become the Gaussian forcing-first contract. Original Part III, PDF pages 45–52. Internal chapter, section, equation and page references retain their original meaning. The new running folio belongs to this integrated volume.

What to carry forward. For the new quadratic, use the exact endpoint expansion and the post-forcing baseline established earlier. Keep schedule commitment and measurement separate. Updating a potential family is not permission to erase a quote’s original epoch, domain or overexecution boundary.

The exact local EBU quote

After P1C has defined the permitted quantity range, the accounting layer asks what the accepted action would do to the represented field. The quote is signed. A positive value means the action reduces the represented burden after process cost; a negative value means it increases that burden; zero means no net change under the declared equation.

It is important to say what the quote is not. It is not a market price, a moral judgement, a prediction of all future consequences, or permission to act. It is an exact finite difference on a declared local field model.

50.1 Why the baseline is the no-action successor

During the tick, both source and destination may regenerate, leak, receive external inflow, or experience other local drive even if the transfer does not occur. Crediting the actor for those background changes would be incorrect.

The quote therefore begins at

$$z = x^n + \Delta t u(x^n), \quad (50.1)$$

the frozen no-action successor. For one edge $e : i \rightarrow j$ and withdrawal rate q , the action changes only the affected components:

$$z_i \mapsto z_i - \Delta t q, \quad (50.2)$$

$$z_j \mapsto z_j + \Delta t \eta_e q. \quad (50.3)$$

Everything else is held at the same no-action baseline. This is a controlled counterfactual: compare the world with the action to the same tick without it.

50.2 The exact equation

Let $C_a(q)$ represent a declared action-process burden that is not already carried by the affected state terms. The exact signed quote schedule is

$$\Delta_e(q) = v_i(z_i) + v_j(z_j) - v_i(z_i - \Delta tq) - v_j(z_j + \Delta t\eta_e q) - C_a(q). \quad (50.4)$$

Because V is separable and only cells i and j change, this local expression is equal to the global field difference for the represented action support. The calculation does not approximate the rest of the world; unaffected terms cancel exactly.

50.3 The zero branch

The schedule defines $q = 0$ as an exact zero action. No stock changes and the process cost is zero:

$$\Delta_e(0) = 0. \quad (50.5)$$

This matters when a process model contains a fixed setup cost c_0 for any positive action. The cost function is defined piecewise:

$$C_a(0) = 0, \quad C_a(q) = c_0 + c_1 q + c_2 q^2 \quad \text{for } q > 0. \quad (50.6)$$

Resting does not incur the limit of an action that was never performed.

50.4 Exact finite difference versus linear diagnostic

The first-order expression is

$$\Delta_e^{\text{lin}}(q) = \Delta tq(\mu_i(z_i) - \eta_e \mu_j(z_j)) - C_a(q). \quad (50.7)$$

It uses the local slope at the baseline. It is valuable for interpretation and for deriving direction. It is not the settlement equation. The penalty is piecewise quadratic, so a finite action may cross a kink or overshoot a deficit. The initial slope cannot see the entire curved path.

Think of walking downhill. The slope under your first footstep tells you an initial direction. It does not prove that a ten-kilometre path ends lower, because the path may cross a valley and climb the other side. Exact settlement compares the two endpoints directly.

50.5 A worked overshoot

Suppose a destination is below its lower band and receiving some stock reduces its penalty. A small shipment can therefore have positive EBU. A much larger shipment may fill the deficit and push the destination above its upper band. The linear diagnostic continues the initial benefit along a straight tangent; the exact difference includes the new excess penalty.

Gate 1B, the exact-quote conformance gate examined later in this volume, contains a fixture where the exact result is -5.52 while the linear diagnostic is only -0.08 . The signs happen to agree there, but the magnitude error is large. Another deliberately wrong ranker can act in a constructed overshoot case where the exact schedule says to rest. These are not cosmetic disagreements: a linear settlement rule can reward an action whose exact finite effect is harmful.

50.6 Process burden and double counting

The allowed process-cost category is deliberately narrow: unrepresented action-process burden. The code rejects monetary cost, labour cost, fraud fines, and burden already represented in the state transition.

Route inefficiency first appears in the destination update: only $\Delta t \eta_e q$ arrives. Any consequence of that smaller delivery already represented by the destination state and its penalty—including a remaining shortfall—must not be charged again inside C_a . The undelivered carrier is a separate accounting question. If the amount $\Delta t(1 - \eta_e)q$ is physically destroyed and no state coordinate records that destruction, a calibrated category-2 process cost is permitted:

$$C_{\text{loss}}(q) = \lambda_L \Delta t(1 - \eta_e)q. \quad (50.8)$$

Here λ_L converts the measured destroyed stock into the declared burden unit and must have a documented calibration. If the same carrier loss is already represented by a loss, waste, or recovery state coordinate, adding C_{loss} would count it twice. Efficiency alone therefore neither requires nor forbids the term; the decisive question is whether that physical effect already has a state representation. Every process-cost term requires a named measurement interpretation.

Money is excluded because a high price can arise from scarcity, market power, fashion, or legal restriction without describing physical field change. Labour

is excluded because human effort is not automatically an ecological burden. A future social system can represent those matters in their own accounts without quietly changing the physical meaning of EBU.

50.7 The quote's information boundary

The quote input contains two endpoint views, their frozen local drives, the edge efficiency, timestep, requested and accepted rates, local identifiers, and the declared process cost. It does not accept a world object, the full state vector, global V , a wallet, health, price, or a future rollout.

This is how the software enforces the project's central local claim. The quote can be exact relative to the global separable field because the action support is local, not because it secretly reads a global outcome.

50.8 A complete numerical quote

Take a source at $z_i = 12$ with upper bound 10 and excess weight $\beta_i = 0.5$. Take a destination at $z_j = 1$ with lower bound 5 and deficit weight $\alpha_j = 1$. Ignore other penalty terms in this example. Let $\eta = 0.9$, $\Delta t = 0.2$, and $q = 2$.

Before the action,

$$v_i(12) = 0.5(12 - 10)^2 = 2, \quad (50.9)$$

$$v_j(1) = (5 - 1)^2 = 16, \quad (50.10)$$

so the affected-cell burden is 18. The source loses $\Delta tq = 0.4$, reaching 11.6. The destination gains $\Delta t\eta q = 0.36$, reaching 1.36. After the action,

$$v_i(11.6) = 0.5(1.6)^2 = 1.28, \quad (50.11)$$

$$v_j(1.36) = (3.64)^2 = 13.2496. \quad (50.12)$$

The exact state benefit is therefore

$$18 - (1.28 + 13.2496) = 3.4704. \quad (50.13)$$

If the declared unrepresented process burden is 0.05, exact settlement is +3.4204 EBU.

Now calculate the linear diagnostic. The source marginal is 2 and the destination marginal is -8, so

$$\Delta^{\text{lin}} = 0.2(2)\{2 - 0.9(-8)\} - 0.05 = 3.63. \quad (50.14)$$

The diagnostic exceeds exact settlement by 0.2096. Nothing is wrong with the slope calculation; curvature explains the difference. The example shows why “exact” is an operational distinction, not a preference for more decimal places.

50.9 Why the same shipment can change sign tomorrow

Suppose repeated shipments lift the clinic from 1 toward its viable band. The first doses remove a large deficit penalty. Later doses remove less. Once the clinic is sufficiently supplied, the same quantity may have only process cost or may create excess burden. Meanwhile the source may move closer to its reserve, increasing the harm of withdrawal.

The quote is state-specific by design. A positive schedule from one epoch cannot be treated as a permanent product label. This is the field-theory idea in practical form: action value depends on where and when the action occurs, not only on the object moved.

50.10 Concavity does not mean monotone benefit

Within the registered linear-cost O14 setting—the multi-edge capability study examined later in this volume—the exact quote schedule is piecewise quadratic and concave over its domain. Concavity means chords lie below the graph and marginal benefit does not increase with quantity. The schedule can still cross zero. A concave curve may rise, reach a maximum, and then fall below zero as an action overshoots.

O14’s certified quantity menus happened to remain on an increasing segment in all registered worlds. That is a structural fact about those parameters and timesteps, not a universal reason to choose the largest permitted quantity.

50.11 Four sign patterns

Readers often ask whether sending a useful object must produce positive EBU. The answer depends on the complete affected field state.

Comfortable source, urgent deficit, efficient route. Withdrawal re-

moves source excess and delivery relieves a destination deficit. Exact EBU can be strongly positive.

Source near reserve, urgent deficit. The destination benefit may be real, but source damage can dominate. P1C may restrict the quantity before the quote; the permitted remainder can still be positive or negative.

Comfortable source, already supplied destination. Delivery may overshoot the upper band. Exact EBU can be negative even though the transported object is normally useful.

Essential need on a costly route. The quote can be negative because loss and source burden exceed local field relief. Institutional need can still justify the action while its physical cost remains visible.

This state dependence is not inconsistency. It is the purpose of a field account. The same litre of water has a different effect when moved into a drought cell than when moved into a full reservoir.

50.12 A quote is not a forecast of adaptation

The exact schedule evaluates the immediate represented action. It does not add future benefits because the delivery might inspire a new factory, nor future harms because an actor might become dependent on the route. Those consequences require new state variables and future actions.

This present-action discipline keeps the quote auditable. Longer-horizon planning may compare sequences of future quotes, but historical settlement remains tied to what was measured in the committed epoch.

50.13 A needs-based example: emergency medicine

A remote clinic needs a medicine available only from a distant store. The route has low efficiency because refrigeration and transport consume capacity. The EBU quote may be strongly negative. That does not prohibit the shipment or say the patient is at fault. It records the physical field cost of the action that meets the need.

The high cost becomes information for collective adaptation: establish a regional store, improve refrigeration, coordinate shipments, or manufacture a

stable form closer to need. Until those alternatives exist, the necessary action remains necessary and its accounting remains honest. Hiding the cost would remove the signal that motivates better infrastructure.

50.14 Three essential transactions and their improvement paths

Remote medicine. Immediate service may require air transport with a large negative quote. Near-term support guarantees the shipment. Repeated receipts can justify regional stock, rail cold-chain investment, local licensing, or a more stable formulation.

Water-pump repair. A rare bearing may travel far because every local pump uses a proprietary design. The action remains necessary. Future burden can fall through standardised parts, repairable design, shared regional workshops, and predictive maintenance.

Disaster evacuation. Moving people rapidly can consume large energy and infrastructure capacity. The need is non-negotiable. Later adaptation may improve local shelters, warning systems, resilient transport, and settlement planning so fewer high-burden emergency movements are required.

In each case the accounting pattern is the same: meet the present need, record the true represented burden, protect access through institutions, and use repeated evidence to redesign the structure. A negative action line is not a command to abandon the person who needs help.

50.15 Why cheap money can coexist with expensive EBU

A subsidised flight may be monetarily cheap to the hospital while physically burdensome. A patented medicine may be monetarily expensive while its marginal physical delivery burden is small. Neither observation makes money false or EBU complete. They answer different questions.

When the two signals conflict, the conflict is informative. A monetary system may reward an action that externalises field damage, while an EBU receipt

reveals that damage. Conversely, society may choose to finance a physically costly essential action because justice and survival require it. Transparent parallel accounts make that choice visible rather than forcing one scalar to pretend it contains every value.

Chapter recap

The exact EBU quote compares an action with the same tick's no-action successor, uses only the affected local cells, subtracts declared unrepresented process burden, and evaluates finite quantity exactly. The linear expression is a slope diagnostic, never settlement. A signed cost describes field effect; it does not prohibit an essential action or blame its recipient.

Exercises

1. Explain why using x^n rather than $z = x^n + \Delta tu$ can credit an actor for natural regeneration.
2. Give an example of a process burden that might belong in C_a , and one that would double count a state-carried effect.
3. Why can two actions with the same transported quantity receive different EBU signs at different field states?

Ownership, capacity and atomic accounting

The most important unimplemented boundary in the new programme is not a missing sum. It is the authority to interpret a particular sum as an owner's capacity change. Software can enforce an adopted rule, but it cannot obtain that rule from the fact that several numbers add correctly.

53.1 A child action is not its owner

A request can be made by a clinic, performed by a carrier and supplied by a depot. The action record may identify all three. None of those labels automatically defines who owns its path attribution. The prospective programme needs an explicit, versioned $\text{owner}(a)$ map.

Once each child has one declared owner, aggregation is straightforward: $\Delta B_i = \sum_{a:\text{owner}(a)=i} R_a$. Closure follows by regrouping the finite sum. The difficult question is not whether the sum can be implemented, but which policy it represents and what claims may be made about it.

53.2 Same-event netting is a policy

Suppose one owner has child lines $+5$ and -3 and starts at zero. Owner-level same-event netting produces $+2$ and passes the proposed affordability rule. A policy requiring each negative child to be prefunded would refuse the negative child. A global-pooling policy could do something else again. These are distinct stochastic processes.

Likewise, a group with owner changes $(+5, -3)$ is not affordable from two zero balances merely because its total is positive. One owner's surplus does not become another's funding under the no-transfer proposal. The implementation must not substitute a single aggregate-balance check for owner-wise checks.

53.3 All-or-nothing effects

If a selected group is accepted under a joint contract, physical and account updates must refer to that same group. A partial credit followed by a failed debit can create spendable capacity without the corresponding authorized

event. A physical update followed by a lost receipt can make replay execute the same action twice.

A later execution design must therefore define commit boundaries, durable identities and recovery behavior across the linked records. “Atomic” here means an all-or-nothing committed event under that contract; it does not mean hiding a multi-carrier physical chain inside one mega-action.

53.4 Audit is not a spendable pool

The external audit J records signed forcing effects. It is not a wallet, loan facility or reward for the next actor. In the cost-free proposal, $V + \sum_i B_i - J = K$. A software routine must be able to check that identity without making J available to the selector as spending capacity.

The invariant alone cannot identify which record is wrong. An erroneous physical update and matching erroneous credit can cancel. Independent conservation, action-identity and owner-level checks are necessary in addition to the aggregate account residual.

53.5 Existing settlement machinery has a narrower meaning

The source-locked framework includes a repaired share-plus-residual closure validator on another lineage [6]. That is an implemented validation mechanism, not an adopted Gaussian owner rule. Its historical amendment’s authoring-stage label is not the whole later implementation history; neither should later implementation be mistaken for scientific approval of the new capacity mechanism.

Chapter recap

Capacity requires a declared ownership and update policy, exact owner-wise affordability, linked event identity and durable all-or-nothing effects. The book specifies the boundary; it does not claim that runtime exists.

Historical reading H53: From a historical quote to its receipt

Historical model; not Gaussian evidence

The original receipt and settlement discussion is preserved with its original scientific scope. Publication-era references to a receipt do not grant the new programme an owner map or a persistent spendable-capacity policy. Read the records as a sequence of distinct commitments, observations and checks. Original Part III, PDF pages 53–59. Internal chapter, section, equation and page references retain their original meaning. The new running folio belongs to this integrated volume.

What to carry forward. The Gaussian proposal adds a prospective persistent account, not merely a new name for an old receipt. Therefore its owner mapping, signed updates and atomic publication need their own authority. An exact field difference is still not a wage or an entitlement.

From quote to receipt and settlement

A quote becomes useful only if the system can prove what was quoted, what was permitted, what was performed, and whether the record has already been settled. This chapter follows one action from proposal to an auditable receipt.

51.1 Request, acceptance, measurement, and delivery

Four quantities must remain distinct.

q^{req} . The requested withdrawal rate derived from the local action rule or proposed by an actor.

q^{acc} . The maximum rate accepted by the physical permission layer. It satisfies $0 \leq q^{\text{acc}} \leq q^{\text{req}}$.

q^{meas} . The rate verified as actually performed.

q^{del} . The useful rate arriving at the destination, $q^{\text{del}} = \eta q^{\text{meas}}$.

If a source accepts 4 units per hour, a device performs 3.8, and route efficiency is 0.9, then the source withdrawal is based on 3.8 while delivery is 3.42. The receipt must not replace all four values with the vague word “quantity.”

51.2 A schedule, not one floating-point promise

The quote object commits the complete exact function

$$q \mapsto \Delta_e(q), \quad 0 \leq q \leq q^{\text{acc}}. \quad (51.1)$$

If the measured execution is smaller than accepted, settlement evaluates the same committed schedule at q^{meas} . It does not rebuild a new formula after seeing the result.

The family is piecewise quadratic and can be reconstructed exactly from its committed parameters. A sampled price table is unnecessary and could introduce interpolation error near penalty boundaries.

51.3 The quote epoch

An *epoch* identifies the precise state and rule under which a schedule was built. Its commitment binds at least:

- equation version and configuration identifier;
- frozen local state commitment;
- source, destination, and allocation pass;
- Δt , η , q^{req} , and q^{acc} ;
- process-cost identifier;
- tick, micro-step, and attribution identifier.

Canonical JSON places the fields in a deterministic byte representation. SHA-256 then produces the epoch identifier. On settlement the identifier is re-derived. A schedule from yesterday's state or a different accepted quantity cannot be substituted without detection.

51.4 What happens on overexecution

If $q^{\text{meas}} > q^{\text{acc}}$, the action lies outside the committed positive schedule. The current minimum envelope issues no positive EBU, records a protocol violation, and reports the physical overdraw. It does not invent a punitive number and call that number physical truth.

Full overexecution debit semantics remain open. This restraint is important. Security sanctions, legal responsibility, and physical field damage are different objects. A future system may need all three, but combining them in one hidden formula would make the receipt impossible to interpret.

51.5 Duplicate and stale settlement

An epoch registry prevents the same quote from being settled twice. It also rejects an expired epoch, a state mismatch, an equation-version mismatch, and an incorrect commitment. These controls are comparable to preventing the same warehouse receipt from being redeemed more than once, but EBU is not transferred from one actor to another. The registry protects uniqueness of an action account.

51.6 The receipt line

A reconstructible receipt contains the action identifier, contributor or actor identifier, affected resource account, source and destination, epoch, equation version, requested, accepted, measured, and delivered quantities, efficiency, loss, process cost, exact settlement, and violation status. It may additionally record measurement device, calibration version, and uncertainty.

The action is primary. Actor totals are derived by summing the receipt lines attributed to that actor. This is fundamentally different from buyer-to-seller payment. If two participants perform two verified burden-reducing components of one cooperative task, both may receive positive receipt lines. If both components increase burden, both may receive negative lines. There is no rule requiring one participant's sign to be the negative of the other's.

Part II used the simplest receipt granularity: every line had exactly one accountable actor. That remains true here. Part III merely allows one completed action to be refined into several measured contributor lines, each still assigned to one actor or attributable component. Those lines must add back to the action settlement. This refinement does not solve O3: when simultaneous actions share a nonlinear source state, the scientifically valid rule for allocating their joint effect remains an open aggregate-accounting problem.

51.7 Exact actor-receipt closure

Suppose a completed action a has an already defined exact settlement Δ_a . Its measured contributor lines are $r_{1a}, \dots, r_{ma}$, defined by an attribution rule that closes on the action:

$$\sum_{k=1}^m r_{ka} = \Delta_a. \quad (51.2)$$

Then actor k 's account over a set of actions $\mathcal{A}$ is

$$E_k = \sum_{a \in \mathcal{A}} r_{ka}, \quad (51.3)$$

and finite-sum rearrangement gives

$$\sum_k E_k = \sum_{a \in \mathcal{A}} \Delta_a. \quad (51.4)$$

This is an exact closure theorem once the receipt lines are defined from measured contributions. It does not by itself select a fair attribution rule for several simultaneous actions that affect the same nonlinear source penalty. That is the aggregate shared-state problem O3. Closure of recorded lines and scientific validity of the rule that produced those lines are separate questions.

51.8 Why EBU does not have to sum to zero like electric charge

Positive and negative EBU are useful analogies to signed charge because signs change how a field interaction is interpreted. The analogy stops before charge conservation. An action can reduce the represented burden, so the sum of action EBU over a period can be positive. Another period can be negative. The zero identity is receipt closure – actor lines equal action settlements – not a law that all EBU in the universe must cancel.

An action that moves a system from deficit toward its viable band can create positive EBU for all verified contributors. This is not free physical energy. It is an accounting statement that the represented field burden fell. The stock ledger still records all material and transport loss.

51.9 Observational quoting must be inert

An observational arm reads the accepted physical action and produces a quote, but cannot modify the action. The strongest implemented check compares an otherwise identical physical baseline with and without observational quoting. Every state, request, acceptance, delivery, service value, and ledger line must remain byte-identical; only the added EBU record may differ.

If merely observing a quote changes a state float, the architecture has mixed accounting with permission or selection. Later studies use this B-versus-C identity as a negative control.

51.10 Two actors do not imply opposite signs

Consider a repair action with two measured components. Actor A recovers and tests a reusable motor. Actor B installs it and restores a water pump. Suppose the exact action-group settlement is +12 and the already certified contribution rule assigns receipt lines +5 and +7. Both actors have positive entries because both verified components participate in the burden-reducing action.

Now consider an emergency air shipment with exact settlement −20. One actor prepares the medicine efficiently and earns a +2 component line; another action line representing the unavoidable high-burden flight is −22. The group closes at −20. The patient does not need an opposite +20 transfer, and the pilot's account is not defined by the hospital's money payment.

These examples show why EBU is action accounting rather than a conserved token passed between two parties. They also show why attribution must be measured and auditable: choosing 5/7 or 2/ − 22 arbitrarily would make the closure equation true but the receipt scientifically empty.

51.11 One actor can perform several accountable actions

A real transaction often contains a chain:

1. withdraw material from a source;
2. transport it;
3. install or consume it;
4. recover packaging;
5. restore or maintain capacity.

It can be clearer to record these as several action epochs than to force one giant number onto the entire story. Each epoch has defined support and measured quantity. The actor's total is the sum, while the physical state advances between epochs. This prevents an early positive quote from being reused after the state has changed.

For a long-range medicine route, the warehouse release, regional transport, local cold storage, and final delivery may therefore settle sequentially. A path planner can estimate future segments, but only performed segments become historical receipt lines.

51.12 Receipt closure versus monetary balance-sheet closure

Double-entry money accounting asks where a monetary claim came from and where it went. Receipt closure asks whether all actor-attributed EBU lines reconstruct the exact action settlement. The two ledgers can coexist, but their zero identities mean different things.

If a hospital pays a carrier 1,000 currency units, the hospital's cash decrease and carrier's cash increase can balance. The physical action may still have -20 EBU, and its contributor lines must sum to -20 . Converting the payment amount directly into EBU would confuse institutional claims with field effect.

51.13 A complete receipt story

A repair depot proposes to send a pump component to a flooded district. The source-local rule requests 2 units per hour. P1C accepts 1.5. The quote schedule is committed for $0 \leq q \leq 1.5$. The measured shipment is 1.4, route efficiency is 0.85, and the district receives 1.19 units per hour. Settlement evaluates the original schedule at 1.4, subtracts the declared unrepresented process burden, and records the result under the action epoch.

The recipient's need explains why the action was chosen institutionally. The exact EBU explains its represented physical effect. The receipt does not pretend that paying a negative number repaired the field. It creates traceable information for later restoration, infrastructure design, and actor accounting.

Chapter recap

Settlement binds an exact schedule to a frozen epoch and evaluates the quantity actually performed. Receipts distinguish request, permission, execution, delivery, and loss. Actor accounts sum action contributions rather than transfer EBU between buyer and seller. Receipt closure is exact when contribu-

tions are defined, while nonlinear shared-state attribution remains an open scientific problem.

Exercises

1. For $q^{\text{req}} = 5$, $q^{\text{acc}} = 3$, $q^{\text{meas}} = 2.6$, and $\eta = 0.75$, list withdrawal, delivery, and loss rates.
2. Why should an overexecution violation not automatically be converted into an arbitrary negative EBU punishment?
3. Construct two actor receipt lines that close on an action settlement of +7. Explain what the arithmetic proves and what it does not prove.

Randomness, paired controls and reproducibility

“Use the same seed” sounds like a complete reproducibility instruction. It is not. Two implementations can start from the same seed and consume draws in different orders, causing every later disturbance or choice to diverge. Scientific pairing requires a more explicit account of randomness.

54.1 Separate the sources of randomness

The programme distinguishes nature or forcing draws from actor-selection draws. If proposals are random, their stream also needs an explicit role. A counter-addressed design can associate each draw with coordinates such as stream, epoch, event and attempt, so an extra actor draw does not shift future nature draws. This is a candidate implementation technique, not a substitute for registering the actual stochastic law.

The generator, distribution transform, coordinate schema and retry behavior all belong to the reproducibility contract. Independent seeds alone do not prove independence of every downstream mechanism or prevent a helper from reading a future draw.

54.2 The physical-random comparison

The proposed comparison uses the same declared physical domain and menu construction but omits the EBU-capacity affordability filter. It is not an EBU maximizer, a service optimizer or the historical full multi-edge P1C arm. Those alternatives answer other questions.

Control parity means equal rules where equality was promised, not equal realized actions. Once states differ, physical feasibility can make their menus differ. A comparison must report that difference rather than force one arm to perform an invalid action to keep a superficial pairing.

54.3 Common raw draws versus common applied forcing

Imagine a raw forcing draw proposes moving three units out of a cell. One arm currently holds four there and another holds two. The same raw proposal is feasible for one and infeasible for the other. Rejection, clipping or resampling can yield different applied disturbances.

Thus shared raw randomness does not guarantee shared physical exposure. A future protocol must declare the handling rule, record raw and applied objects separately and determine how exposure differences affect the comparison. Silently promising both forms of equality would be false.

This is not a reason to abandon paired designs. It is a reason to specify what is paired. The scientific question may concern response under common raw environmental opportunities, or response to identical admissible physical inputs. They are different estimands when feasibility depends on the evolving state.

54.4 Restart without a new experiment

A reproducible checkpoint must restore stock, balances, audit, parameter version, pending event identity and random-stream coordinates together. Restoring only the stock can leave an apparently identical physical world with different future affordability or random choices.

Whether a restart is permitted after partial execution belongs to the study's operational contract. A book cannot grant an extra scientific attempt by calling it recovery. Historical Gate 1D-C illustrates why execution attempts and completion sentinels are separately recorded.

Practice question

Two arms have equal initial state and equal nature seed but use one shared random generator for nature and actors. The EBU arm rejects more candidates and consumes extra draws. Are future nature inputs paired? Not necessarily. The rejection path has changed draw consumption. An independently addressed nature stream would remove this particular coupling, while state-dependent forcing feasibility would remain a separate issue.

Chapter recap

Reproducibility concerns the exact stochastic process, not a seed label. Fair comparison requires explicit stream ownership, exposure accounting and restart semantics.

Tests that can genuinely detect being wrong

A useful test is designed so that a plausible defect makes it fail. A suite that merely confirms its own assumptions can stay green while the scientific meaning has changed. The goal is not a large number of checks; it is coverage of the ways the implementation could compute the wrong object.

55.1 Execution classes must be named honestly

Static inspection covers source, schema, hashes and declarations. Pure arithmetic evaluates a fixed mathematical expression without advancing model state. A transition test executes a physical update, even if there is only one synthetic tick. A scientific study executes its registered process and interprets outcomes under a frozen protocol.

These classes have different authorization boundaries. A test's filename or speed does not determine its class. Importing a module can itself have side effects, so calling a test static requires inspection of what it actually does. A source scan that sees no direct transition call cannot prove that an imported helper never transitions.

55.2 Arithmetic checks for the new family

The future Gaussian foundation should compare finite endpoint evaluation with the exact expansion, local with global differences, path sums with group value, and rate/quantity conversions at non-unit durations. It should include reference states, overshooting quantities, cancelling groups and unequal scales, not only beneficial transfers.

An especially useful defect injection is replacing exact finite value by the initial tangent. The quantity-four example changes from twelve to sixteen. Another is summing two same-baseline singleton values: the shared group changes from twelve to fourteen. A negative control should detect those substitutions, not merely run them and count a pass.

55.3 Physical checks are independent

Conservation, nonnegative stock, allowed support and joint capacity must be tested independently of the potential. A negative source and positive destination can preserve total mass. A changed sink can preserve one aggregate residual while violating its declared physical update. Passing one scalar identity is therefore not a complete physical validator.

At the lossless first-domain boundary, any loss coordinate should remain inert under its declared rule. A future loss-aware extension needs its own carrier-by-carrier checks; the success of the lossless suite cannot be reported as certification of that extension.

55.4 Information and account controls

Required prospective controls include forbidden remote reads, future draw access, stale epochs, duplicate settlement, post-selection quantity changes, missing owners, wrong parameter versions and cross-owner pooling. Each should specify the expected refusal and the state that must remain unchanged.

A refused campaign entry is not proven by any nonzero process exit. A syntax error also exits unsuccessfully. The check must verify the intended refusal class and message, and where relevant that no model state changed before it. Refusal is a behavior with a contract, not the absence of a successful printout.

55.5 What green checks do not establish

Even complete conformance to a specification does not prove the specification is scientifically useful. A correctly implemented process can cycle, ration, refuse, or fail to improve its declared metric. Those are possible scientific outcomes, not automatic reasons to weaken tests or alter the world.

This book's worked arithmetic is checked independently as teaching material. Those checks are not reported as implementation acceptance counts. Historical Q-test and D-test counts in the historical readings remain attached to their own code and configurations; none is inherited by an unbuilt Gaussian runtime.

Chapter recap

Test the equation, the physical transformation, information boundaries, accounting semantics and execution guards separately. Then keep conformance distinct from scientific success.

Historical reading H55: Original tests and their failure targets

Historical model; not Gaussian evidence

The original chapter gives concrete examples of what a test is meant to catch. Counts and pass statements refer only to that historical implementation and its recorded configuration. No test count is transferred to a Gaussian runtime. Original Part III, PDF pages 60–65. Internal chapter, section, equation and page references retain their original meaning. The new running folio belongs to this integrated volume.

What to carry forward. The enduring lesson is to make plausible mistakes detectable. The current book’s arithmetic checks test teaching examples, not runtime acceptance. A future transition test must be classified as a transition even if its fixture is small.

Tests that can detect being wrong

A test suite is informative only if it would fail when the implementation is wrong in a relevant way. Checking one easy positive example is insufficient. For instance, a function that always returns zero passes every identity evaluated only on zero actions. The EBU suites therefore combine conformance points, non-vacuity requirements, structural scans, runtime poison probes, and deliberately wrong controls.

52.1 What a unit test can establish

A unit test supplies inputs, executes a small code path, and asserts an expected property. It can show that the code returned an exact fixture value, rejected an invalid object, preserved a ledger within tolerance, or did not mutate its input. It does not prove the equation for all real numbers, certify ecological parameters, or demonstrate social usefulness.

The strongest suites connect tests to named scientific risks. Instead of “test helper 17,” a group asks whether a stale epoch can settle, whether independent fan-out duplicates a source budget, or whether a linear diagnostic can enter the settlement path.

52.2 Non-vacuity

A non-vacuous fixture makes the intended property capable of failing. Quote tests include positive, negative, and curved schedules; zero and positive quantities; penalty-boundary crossings; and cases where exact and linear rankings disagree. Preservation tests include a budget that genuinely binds. Service tests include an infeasible world with explicit unmet demand.

Non-vacuity is not the same as realism. A small synthetic fixture may be highly informative precisely because it isolates one mathematical mistake.

52.3 Negative controls

The test suite deliberately implements wrong alternatives and requires their detection:

- a loss-blind force omits η ;
- a sequential update reads a partially modified state;
- an oversized timestep violates the descent certificate;
- a linear quote replaces the exact finite difference;
- the pre-drive state replaces the no-action successor;
- the same epoch settles twice;
- an action beyond q^{acc} receives a positive quote;
- independent edge budgets duplicate a shared source allowance;
- a decision function reads forbidden global information;
- a serializer attempts to emit NaN or Infinity.

A passing negative control says, “this known wrong construction is visible to the current instrument.” It does not say that every unknown error is visible.

52.4 AST scans and runtime poison

Python source can be parsed into an abstract syntax tree, or AST. Unlike a text substring search, an AST distinguishes an actual import from the same words in a comment or string literal. Tests use AST inspection to reject forbidden imports and global identifiers on decision paths.

Runtime poison is complementary. A forbidden function such as global V_{total} is temporarily replaced with an object that raises an error if called. The local selector is then exercised. If selection completes, the test has evidence that this path did not consult the poisoned global function.

Static scans see source structure; poison probes see executed access. Neither is complete alone, so both are used.

52.5 The runner self-match failure

The first authorized attempt to execute O14 failed before any trajectory. Its randomness guard read the runner's own source and searched for the literal text `import random` or `import secrets`. Those phrases appeared in the guard's own string literals, so it rejected itself unconditionally.

The failure is instructive for three reasons. First, fail-closed ordering worked: the runner exited before the banner, before the study call, and before any scientific output. The redirected stdout file was exactly zero bytes. Second, the earlier tests had asserted only that some sentinel scenarios raised `SystemExit`. They did not require the intended reason, so several tests passed because the self-match failed first. Third, the correction replaced the substring scan with AST import inspection and strengthened every failure test to match its expected fatal message.

The defect and zero-byte artifact were documented before cleanup. The corrected runner was committed and tested under a new authorization. Only then was a fresh single-execution authorization issued. This is what an audit trail is for: not to claim that mistakes never happen, but to prevent them from becoming invisible.

52.6 Numerical tolerances and physical margins

Floating-point calculations can place a mathematically exact boundary one unit in the last place above or below its ideal value. Gate 1D initially labelled four such states as reserve crossings. Gate 1D-A changed only the diagnostic predicate to use a scale-aware tolerance. The 56 trajectories, states, actions, service, unmet demand, and ledgers remained byte-identical, and there were zero material crossings.

This numerical tolerance is not an ecological safety margin. It tells the diagnostic when two machine numbers should be treated as boundary-equal. Physical uncertainty belongs in ε_x , ε_u , and reserve certification. Mixing those roles would let a software convenience silently relax a physical promise.

52.7 Strict serialisation

JSON's informal ecosystem sometimes permits NaN and Infinity even though they are not valid strict JSON numbers. Scientific artifacts in the later gates

are written with non-finite values forbidden. A recursive finiteness walk runs before output. If any required number is non-finite, the runner stops rather than writing a misleading partial record.

Earlier D9/D10 history includes a narrow serialization repair: non-finite stability diagnostics were replaced by JSON null from already committed traces, without regenerating trajectories or outcome classes. The incident helped define the stricter later rule.

52.8 How to read a test count

At the relevant committed stages, Gate 1B reported 184 quote-conformance checks plus 19 audit checks; Gate 1C reported 150 adversarial checks plus 47 reproduction audit checks; Gate 1D reported 304 checks; and the corrected O14 pre-execution suite reported 299 checks in 17 groups.

These numbers are coverage records, not counts of independent laws. One theorem can require many implementation assertions; one hundred assertions can still miss one crucial path. The group names, fixtures, negative controls, and source boundaries carry more scientific meaning than the total alone.

52.9 A reader's test checklist

When a report says “all tests passed,” ask:

1. Which equation or invariant did each group target?
2. Was the fixture non-vacuous?
3. Was a known wrong alternative required to fail?
4. Did the test assert the reason for failure, not only the exception type?
5. Did it run a short conformance example or the registered scientific study?
6. Was any output written or any locked artifact modified?

This checklist turns test output from a badge into evidence that can be evaluated.

52.10 Following one assertion from theorem to failure mode

Take aggregate source preservation. The mathematical statement says accepted outgoing rate must not exceed one source budget. The implementation groups all outgoing requests by source and calculates one scaling factor. A direct test uses several edges whose sum exceeds the budget and checks the accepted sum. A negative control instead applies the full budget to each edge and must be detected. A short tick then verifies that the source successor remains above the reserve.

Four different facts appear in that chain:

1. algebra identifies the required inequality;
2. source inspection confirms one shared grouping path exists;
3. the fixture confirms numerical conformance at one state;
4. the negative control proves the fixture can see the duplication error.

This pattern is more informative than a single test named “source safe.” The reader can identify exactly which link would need new evidence under a changed allocation rule.

52.11 Why short test runs are not hidden scientific runs

Some suites execute one tick or a labelled ten- or twelve-tick horizon to validate record construction. They do not execute the registered 200-tick Cartesian product. Runner tests use synthetic results or a mock function to exercise output validation without calling production dynamics.

The distinction is enforced in source scans and reported explicitly. It permits the runner’s success path to be tested before authorization while preserving the claim that the official scientific sweep has not yet run.

52.12 Tests should survive refactoring but not semantic drift

A brittle test may assert an internal line order that has no scientific meaning. A weak test may assert only that the program returned a number. The desired test targets semantics: same frozen menu across arms, exact schedule domain, one settlement per epoch, or byte-identical physical trajectory under observation.

When code is reorganised, semantic tests should continue to pass. When an equation, parameter, or information boundary changes, they should fail until a new gate explicitly authorises the change.

Chapter recap

Good tests demonstrate both correct behaviour and the detectability of relevant wrong behaviour. The project uses numerical fixtures, non-vacuity, negative controls, source scans, poison probes, strict serialization, and reason-specific failures. The O14 self-match incident shows why preserved failures and stronger tests increase credibility rather than diminish it.

Exercises

1. Design a non-vacuous test for aggregate source budgeting with three outgoing edges.
2. Explain why asserting only `SystemExit` was insufficient in the O14 sentinel tests.
3. Give one example of a physical margin and one example of a numerical tolerance. Why must they remain separate?

Provenance, execution and immutable evidence

A result is not reproducible merely because its final number is saved. Another reader needs to know which model, protocol, implementation and environment produced it, whether the run completed, and whether the records were changed afterward. Provenance connects a claim to that chain.

56.1 The scientific packet precedes the run

A registered packet identifies hypotheses, domain, arms, parameters, randomness, metrics, falsifiers and interpretation rules. It also binds the implementation and execution environment. These decisions must precede outcome inspection; otherwise a successful-looking configuration can be selected from many attempts without saying so.

The present book does not fill those fields for the Gaussian programme. It describes what must be declared. No horizon, seed family, numerical tolerance, performance threshold or cloud resource is frozen by an example in these chapters.

56.2 Attempt, run and completion are different objects

An invocation can fail before a scientific state transition. A campaign can contain many registered runs within one authorized attempt. A runner can finish producing output before the full study is durably finalized. These distinctions prevent both inflated execution claims and unauthorized retry disguised as routine continuation.

Historical Gate 1D-C's manifest explicitly distinguishes the runner summary from the full-study completion sentinel [8]. That exact operational contract belongs to that study. A future Gaussian packet must define its own compatible mechanics and permissions, not inherit an old launch command.

56.3 Content identity and scientific meaning

Hashes identify bytes. A manifest should record the protocol, configuration, source versions and produced artifacts so their identities can be checked. Hashes do not certify a model's validity or a sensor's honesty. They make it harder to substitute one object for another unnoticed.

Canonical serialization matters when equivalent-looking objects can produce different byte sequences. Numeric representation, key ordering and schema version are part of the identity contract. A replay should not silently normalize a value in a way that changes the registered scientific object.

56.4 What to retain in a trace

For each event, retain enough information to reconstruct its physical baseline, forcing, candidate/selected identities, accepted increment, valuation inputs, owner changes, account audit and refusal state. The exact schema is prospective, but its purpose is fixed: an independent reader must be able to tell what happened without trusting a narrative summary alone.

Not every diagnostic belongs in the actor's view. A global evaluator may record quantities for analysis while the selector is forbidden to read them. Trace completeness and actor information locality are compatible when their data paths are separated.

56.5 Corrections should not erase history

If a result artifact is wrong, the correction stage should identify the original, the defect, the corrected interpretation and the authority for any replacement. Rewriting the original bytes until they fit a later story destroys the very evidence needed to understand the error.

The book revision follows the same principle. Original PDFs and historical result artifacts are preserved. New exposition explains which model-specific claims do not transfer, rather than editing old experimental outcomes to look like Gaussian evidence.

Chapter recap

Provenance makes a claim inspectable, not automatically true. Frozen scientific choices, exact identities, completion rules and preserved failures allow an independent review to determine what the evidence actually supports.

Historical reading H56: Reproducing evidence without rewriting it

Historical model; not Gaussian evidence

This full historical reproduction chapter preserves the original record structure and its caution about reinterpreting outcomes. Commands and paths inside the reading document that edition; they are not instructions to execute them now or launch a later study. Original Part III, PDF pages 119–132. Internal chapter, section, equation and page references retain their original meaning. The new running folio belongs to this integrated volume.

What to carry forward. Reproducibility requires the actual scientific packet and operational permission, not a copied command. Historical bytes remain inspectable while corrections and later findings are recorded separately. This distinction also explains why old experimental pages are retained rather than silently edited into new results.

How to reproduce the evidence without rewriting it

Reproducibility is often described as rerunning a script. In a preregistered project with protected result paths, that is only one possible activity and may be the wrong first action. A reviewer should first verify which plan, code, and artifacts produced the historical claim. A later computational reproduction should create a separately identified record rather than overwrite the original.

61.1 The four layers of a study record

Each later gate uses four kinds of artifact.

Protocol. Human-readable questions, hypotheses, falsifiers, equations, information boundaries, and stop conditions.

Machine plan. Exact parameters, worlds, arms, timesteps, run counts, thresholds, and schema in canonical JSON.

Implementation and runner. Production scientific functions plus the fail-closed orchestration that reconstructs the registered inventory.

Results. Stdout, summary, compressed trace, and manifest with provenance and hashes.

The protocol explains intent. The plan removes ambiguity. Code performs the declared calculation. Results preserve what happened. No single layer is a substitute for the others.

61.2 Commit and hash provenance

A Git commit identifies a snapshot of tracked files and their history. An annotated release tag identifies a named baseline. A file hash verifies one exact artifact. These mechanisms answer related but different questions.

For example, the O14 chain contains a preregistration commit, implementation commit, runner-preparation commit, fail-closed correction commit, execution SHA, and results commit. The manifest also records SHA-256 values of the plan, runner, summary, trace, and stdout. A reviewer can therefore establish that the plan predated the execution and that the committed result bytes have not changed.

61.3 Raw and canonical plan hashes

The raw hash fingerprints the exact file bytes, including whitespace and key order. The canonical hash fingerprints a normalised JSON representation with sorted keys and compact separators. Both are useful. The raw hash detects any byte change; the canonical hash identifies the same logical JSON object despite irrelevant formatting differences.

At preflight the runner recomputes both and compares them with locked values. A mismatch stops the study before the first trajectory. The runner does not ask the user whether the change is probably harmless.

61.4 Reading a summary

A summary is designed for analysis without scanning every tick. It may contain one record per run with:

- run identity, world, arm, and timestep;
- certificate and dimensionless ratio;
- service, unmet demand, and per-destination totals;
- EBU totals and sign counts;
- minimum and final stocks, viability, and burden;
- crossings, overuse, domain status, and ledger residual;
- action, rejection, rest, and quote-coverage counts;
- outcome class and paired comparisons.

The summary is derived from the trace. A trustworthy runner validates that the aggregates reconstruct before writing the completion sentinel.

61.5 What the comparison metrics mean

Several compact phrases in a summary carry methodological choices. A *burn-in* removes an initial interval from the primary comparison so the measurement is not dominated by the arbitrary starting transient. O14 uses 50 burn-in ticks in each 200-tick run. “Post-burn-in service” therefore means the sum from the registered comparison window, not the service delivered over the whole trajectory. Burn-in is defensible only when fixed before execution. Moving its boundary after seeing a favourable crossing would be outcome selection.

A *persistence window* asks whether an event or recovery remains true for a declared number of ticks. It prevents one noisy instant from becoming a recovery claim. The same idea appeared in D6 and D7: returning below a burden threshold once is weaker than staying below it while the world remains active.

A *paired comparison* matches two arms in the same world and at the same timestep. This removes world composition and numerical resolution as alternative explanations. A D/B service ratio, for example, divides post-burn-in service under the exact-quote selector D by the matched force selector B. The ratio is useful only when B’s denominator is positive and the absolute service difference is also reported. A ratio of one can mean identical action or merely identical saturated service; O14 demonstrated the second possibility.

Capability cost compares the full multi-edge action capability A with the restricted matched baseline B. It asks whether restricting simultaneous action changed service before quote ranking is interpreted. A nonzero cost would reveal a capability confounder. Zero cost does not prove the restriction harmless in all worlds; it records that the registered service instrument did not move.

Coverage is the fraction of eligible action opportunities for which the observational quote machinery produced its required diagnostic. It is an instrument-completeness measure, not a success score. A policy can have perfect quote coverage and poor service.

Domain failure means that the released numerical or physical domain conditions were violated. It is different from *physical impossibility*, where the model remains valid but declared demand exceeds safe capacity. W6 is valuable precisely because it reports unmet need without a domain failure.

Finally, a comparison tolerance states which floating-point differences count

as material. It must be tied to the quantity and frozen before results are read. Gate 1D-A showed why: four diagnostic labels changed at a one-unit-in-the-last-place boundary while every trajectory and scientific total remained identical. A tolerance is not permission to erase discrepancies; it is a declared rule for classifying their scale.

These definitions let a reader translate a summary sentence into a testable operation. “No alignment failure” then means that every registered matched pair passed one named predicate at one declared tolerance. It does not mean that every behavioural distinction vanished or that alignment has been proved for all future worlds.

61.6 Reading a JSONL trace

JSON Lines stores one JSON object per line. In O14 each line is one tick row, so a 60-run, 200-tick study contains 12,000 lines. The row repeats enough provenance to identify its plan and run, then records the per-tick physical and diagnostic fields.

To follow one run:

1. select rows with its exact run identifier;
2. verify ticks are ordered and unique from 0 through 199;
3. inspect x_{before} , selected candidate, accepted quantity, and x_{after} ;
4. reconstruct service and unmet demand;
5. sum per-tick totals and compare with the summary;
6. check the plan hash and finite-number rule on every row.

Compressed gzip changes storage size, not logical content. A gzip integrity check detects truncated compression, while the SHA-256 detects any byte change to the compressed artifact.

61.7 Walking through one committed O14 row

A trace becomes much less mysterious when one row is read as a physical receipt. Consider tick 1 of the conservative-timestep arm-A run in world W1.

The outer row says that the timestep is

$$\Delta t = 0.1722652885, \quad \Delta t_{\text{cert}} = 0.3445305771, \quad r_{\Delta t} = 0.5. \quad (61.1)$$

It also carries the world, arm, run identifier, certificate kind, and canonical plan hash. These are not physical variables. They are provenance fields that answer which registered calculation the row belongs to.

Inside the row's physical record, the three-cell state before the tick is

$$x^{\text{before}} = (15, 2, 2). \quad (61.2)$$

Cell 0 is the source and cells 1 and 2 are destinations. Both outgoing edges are active. Their accepted rates are 2.825 and 1.775. The source also receives a local drive of 2.25 during the tick. The source update can therefore be reconstructed without trusting the stored answer:

$$x_0^{\text{after}} = 15 + \Delta t(2.25 - 2.825 - 1.775) \quad (61.3)$$

$$= 14.5951765719. \quad (61.4)$$

That value matches the row's first `x_after` entry and its `min_source` diagnostic.

The first edge has efficiency 0.95, so its useful arrival rate is $0.95(2.825) = 2.68375$. The second has efficiency 0.60, so its useful arrival rate is $0.60(1.775) = 1.065$. Each destination consumes service at rate 1 over the tick. Their updates are thus

$$x_1^{\text{after}} = 2 + \Delta t(2.68375 - 1) = 2.2900516796, \quad (61.5)$$

$$x_2^{\text{after}} = 2 + \Delta t(1.065 - 1) = 2.0111972438. \quad (61.6)$$

The stored post-state is exactly these three numbers. This is a useful audit habit: read one vector entry as a sentence, then reconstruct it from named rates and the timestep.

The trace uses different units for several related fields. Accepted edge values are rates used by the physical step. The recorded service and demand entries are amounts accumulated during the tick. For each destination the row therefore stores

$$\text{service} = \text{demand amount} = 1 \times \Delta t = 0.1722652885, \quad (61.7)$$

and unmet demand is zero. Confusing a rate with a per-tick amount would introduce another factor of Δt and break reconstruction. Units are part of the schema, even when JSON itself cannot encode them.

The transport-loss field also has a direct check:

$$\ell = \Delta t [(1 - 0.95)(2.825) + (1 - 0.60)(1.775)] \quad (61.8)$$

$$= 0.1466408269. \quad (61.9)$$

The stored ledger residual is about -1.11×10^{-16} . That tiny value is floating-point roundoff around zero, not missing material. The test tolerance is defined in advance so a reviewer does not decide after seeing the value whether it is acceptable.

Arm A is the unrestricted multi-edge capability arm. It does not settle EBU, so the row stores zero for total, positive, and negative EBU. This does not mean the physical action had no field effect. The same row contains the settlement-free group diagnostic:

$$Q_{\text{group}} = 3.6877331183, \quad \sum_e Q_e^{\text{ind}} = 3.7595010483. \quad (61.10)$$

Their difference is 0.0717679300. The independent frozen edge quotes would over-credit the simultaneous source withdrawals by that amount. Keeping this number diagnostic prevents the known aggregate-allocation problem from being silently converted into actor balances.

Finally, the row records zero reserve crossings, zero Allee crossings, zero physical overuse, no domain failure, and no negative-state correction. Those zeros are not decorative. They distinguish four different claims: the source remained above a named reserve, did not enter the lower regenerative region, did not request more accepted material than the physical rule allowed, and remained inside the numerical domain.

One row cannot establish a trajectory result. It teaches how a result is made. Repeating the same reconstruction across 200 ordered rows checks one run; checking all 60 run identifiers checks the inventory; pairing arms under the registered rules produces the study comparisons. The hierarchy from arithmetic to claim is therefore visible rather than hidden inside a chart.

Two reconstruction mistakes deserve special attention. The first is to sum a state variable when the summary reports a flow. Source stock is sampled at each tick; adding those samples has no direct physical interpretation. Service and transport loss, by contrast, are per-tick amounts and may be summed over the declared window. Minimum stock is obtained with a minimum operation, final stock from the last ordered row, and a crossing from a comparison between consecutive states. The aggregation rule belongs to the meaning of the field.

The second mistake is to infer causality from two columns that move together. A fall in burden beside a positive EBU total does not by itself prove that the quote caused recovery. One must compare the registered selector with a matched baseline, verify that the action menus and physical capability were comparable, and inspect whether both trajectories were already on the same service plateau. This is why the project retains menus, selections, accepted quantities, and state series instead of publishing only a final score.

A useful independent audit therefore begins with a small table. For each summary metric, write its source fields, unit, aggregation operator, time window, pairing rule, and tolerance. Reconstruct three deliberately different runs: a normal feasible run, a binding-capability run, and an honestly infeasible run. Only then automate the same calculation across the full inventory. This sequence catches unit and interpretation errors before they are multiplied by thousands of rows. It also gives a reviewer a compact, repeatable route from stored numbers back to the physical story those numbers are meant to represent.

61.8 Why summary and trace are both needed

A summary can conceal timing. Two policies may deliver the same total while one allows a critical buffer to reach zero halfway through. A trace exposes the sequence. Conversely, a trace alone makes it easy to overlook registered paired comparisons or count one run twice. The summary provides the complete inventory and derived evidence.

O14 is a good example. The summary reports D/B service ratio 1, but the trace shows different destination selections in W2 and repeated switching in W5. The buffering diagnosis requires both levels.

61.9 The manifest as a human audit index

The manifest records the execution command, attempts, file hashes, plan hashes, code provenance, run counts, structural validation, headline outcomes, falsifiers, limitations, and final repository state. It is not another simulation result. It is a map that lets a reviewer connect claims to immutable sources.

A good manifest also records failures. The O14 manifest names the first preflight attempt and its zero-byte stdout. The D9/D10 record names the

serialization repair. The Gate 1D history preserves both the defective label rendering and corrected diagnostic attempt.

61.10 A reproduction ladder

Review can proceed in increasing order of cost and authority.

1. **Document review:** read the protocol, plan, manifest, and claim ledger.
2. **Hash verification:** recompute artifact and plan fingerprints.
3. **Static code review:** inspect module boundaries, equations, and forbidden imports.
4. **Artifact reconstruction:** recompute summary aggregates and figures from the committed trace without running dynamics.
5. **Test reproduction:** run conformance suites in a clean environment.
6. **Independent scientific reproduction:** execute the locked plan in a fresh, explicitly identified result location.
7. **Conceptual replication:** design a new study testing the same claim with different worlds or implementation.

The last two are different. Reproduction asks whether the same computation can be obtained again. Replication asks whether the scientific conclusion survives a meaningfully new test.

61.11 Why official runners resist a second execution

Several result directories use the summary as a completion sentinel. If it exists, preflight refuses to run. Orphan trace and manifest files also trigger a stop. Shell no-clobber protects the redirected stdout path.

This design means a casual “run everything” command is inappropriate. It prevents an accidental second execution from overwriting historical evidence or being mistaken for the original run. A reproduction belongs in a fresh directory with a new manifest and authorization.

61.12 Rebuilding figures from committed data

Figures in this volume are explanatory views of committed artifacts. The D9 trajectory plot reads source stocks from the compressed trace. The D10 map reads registered classifications from the summary. The O14 instrument plot reads paired service comparisons and S-arm unmet demand. None reruns a step function.

This distinction matters because a plot can be regenerated with different fonts or colours without changing scientific data. The caption should identify the source and explain whether the figure is a direct trace view, derived summary, or conceptual diagram.

61.13 How to read the code without being a Python expert

Begin from scientific nouns, not syntax.

1. Locate immutable data classes such as Cell, Edge, LocalView, and SourceConfig. Read their validation rules and units.
2. Locate pure scalar functions: penalty, marginal, regeneration, and robust budget. Compare them line by line with the equations.
3. Locate the tick function and write its update order in plain language.
4. Locate result data classes and identify which diagnostics are recorded.
5. Read tests for deliberately wrong alternatives.
6. Read the runner last, focusing on preflight, inventory, validation, and write order.

The reader does not need to understand every Python feature before checking that source loses q and destination receives ηq . Scientific review can proceed from the equation outward.

61.14 The technical stack and why it is deliberately small

Production scientific modules use the Python standard library. Data classes declare immutable records; `math.fsum` supports stable accumulation; `hashlib` supplies SHA-256; `json` supplies strict canonical records; `gzip` compresses traces; `ast` supports structural source inspection; and atomic file replacement protects completed output.

Plotting tools are used for documentation, not action selection. Tests and experiment runners are ordinary scripts rather than an opaque simulation framework. This small stack makes it easier to inspect which code owns the physics and reduces dependency-driven drift.

The choice does not prove correctness. It makes the executable surface small enough for independent review and long-term reproduction.

61.15 A map of the principal tests

Test file	Main responsibility
test_v29.py	D0 objects, penalties, drives, frozen flux, certificates, and ledgers
test_v29_behavior.py	registered deterministic wind-tunnel behaviour and negative controls
test_v29_p1c.py	source states, robust budget, aggregate fan-out, and one-step preservation conformance
test_v29_d9_d10.py	registered preservation and phase-map harness
test_v29_serialization.py	finite output and artifact discipline
test_v30_quote.py	Q1–Q21 exact quote conformance
test_v30_adversary.py	Gate 1C search harness without executing the official study
test_v30_service.py	bounded service, comparison predicates, and Gate 1D tolerance discipline
test_v30_o14.py	O14 worlds, menus, selectors, runner pre-flight, and output reconstruction

A reviewer should read the test that corresponds to the claim, not assume the largest test file covers every layer.

61.16 A code-to-claim example

Consider the claim “observational quoting is physically inert.”

1. Architecture: the quote function accepts immutable local input and returns a schedule without mutating the world.
2. Unit test: Q17 compares a small physical step with and without observation.
3. Trajectory control: Gate 1C A/B and later B/C comparisons require complete physical identity.

4. Result artifact: O14 records maximum state difference zero in all twelve pairs.
5. Limitation: the evidence covers these paths and fixtures, not every possible future integration.

This chain is stronger than citing one function or one result table alone.

61.17 Preserving failed attempts

Failure records answer questions that a polished final artifact cannot:

- Did any trajectory run before the correction?
- Which file was created?
- Was it empty, partial, or scientifically meaningful?
- Which line failed and why?
- What authorization allowed cleanup or repair?
- Was the corrected runner committed before execution?

The O14 first attempt can be proven pre-trajectory because the fatal guard ran before the banner and before the only function that calls production `run_arm`; stdout was zero bytes and no trace, summary, or manifest existed. This is stronger than a later assertion that “nothing important happened.”

61.18 A practical review session

A reader reviewing O14 can complete a useful session without executing dynamics:

1. verify the four result hashes against the manifest;
2. load the strict JSON summary and count 60 unique run identifiers;
3. verify the six-by-five-by-two Cartesian product and absence of arm E;
4. stream the gzip trace and count 12,000 rows;
5. verify 200 ordered ticks for each run;

6. reconstruct one B-versus-D service comparison;
7. inspect W2 selections to confirm behavioural divergence;
8. inspect W1 S stock to see the buffer reach zero;
9. reproduce the aggregate naive-versus-group quote total;
10. confirm F13 evidence is the empty discriminating-world list.

This procedure tests the report against the artifacts without creating a new scientific attempt.

61.19 Reproducibility is not truth

A perfectly reproducible model can be wrong about nature. Hashes can preserve a poor reserve. Deterministic code can execute an unrealistic demand model exactly. Reproducibility reduces uncertainty about what was done; it does not remove uncertainty about whether the model represents the world.

That is why external data, domain review, sensitivity analysis, and independent replication remain necessary. The audit architecture is a foundation for criticism, not a shield against it.

Chapter recap

Reproduction begins with provenance: protocol, machine plan, code, results, and manifest. Hashes identify exact artifacts; summaries provide inventory and aggregates; traces preserve timing; manifests connect claims to history. Figures can be rebuilt from committed data without rerunning dynamics. A fresh reproduction must create a new record, and reproducibility never substitutes for real-world validity.

Exercises and audit lab

1. Explain the difference between a raw plan hash and a canonical plan hash.
2. Why is a gzip integrity check not a substitute for a cryptographic hash?

3. Choose one claim in this book and list its equation, test, result artifact, and limitation.
4. Design a directory and naming rule for an independent O14 reproduction that cannot be mistaken for the official execution.
5. Explain why changing a chart colour does not create a new trajectory, while changing a service threshold can change scientific interpretation.

The historical engine and the Onsager connection

The older implementation is not simply the new model with a different potential expression. It has a particular natural drive, flux controller, permission layer and event order. Understanding those choices is necessary both to preserve its results and to avoid importing them accidentally.

57.1 Why force, mobility and flux entered EBU

Onsager's primary papers concern coupled irreversible processes and linear force–rate relations under microscopic-reversibility assumptions [1, 2]. The useful modelling pattern for EBU was to separate a directional driving force from a response coefficient. The earlier D0 design made that relationship explicit instead of treating the potential's derivative as though it were already a physical flow.

Its local marginal came from the historical hinge potential. For an edge with usable-delivery efficiency η_e , the directional force was $f_e = \mu_i - \eta_e \mu_j$. The controller requested

$$J_e = M_e[f_e - \theta_e]_+. \quad (57.1)$$

Mobility M_e controls responsiveness; θ_e is an edge activation threshold. It is distinct from Allee or homeostatic stock thresholds. Neither coefficient is an EBU balance or a source's physical stock.

This adaptation is called Onsager-type because the model declares its own potential and directed threshold response. It does not establish that the potential is thermodynamic free energy, or that a social network satisfies microscopic reversibility or reciprocal transport coefficients. The reason to use it is the transparent conditional force–flux calculation, not the prestige of an analogy.

57.2 D0's synchronous update

The historical map freezes state, computes the natural drive $u(x)$ and edge requests, then applies a synchronous Euler update of the form $x^+ = x + \Delta t(u +$

SJ). Frozen synchronous evaluation prevents a newly updated neighbor from influencing another edge within the same nominal step. Sequentially mutating an edge loop would implement a different propagation rule.

Chapter 44 derives the continuous-time dissipation structure and its finite-step curvature correction. The historical tests concern their registered hinge model and certificates. Replacing the evaluator with a Gaussian formula while retaining the old drive and selector would create a hybrid model, not automatically implement the new programme.

57.3 P1C is a permission layer with premises

The historical P1C classifier distinguishes regenerative and other source types and their admissibility states. The robust budget's arithmetic assumes the applicable source classification. For a finite or irreversible stock, the preservation policy does not become an internal-transfer permission merely because a helper can compute a positive number.

The distinction between R_{eff} and homeostatic R also survives. The former bounds export under a source certificate; it does not itself deliver resource to a destination below its own reference or reserve. A resolver's protection cannot be used to delete another controller's service logic while claiming the same function remains.

57.4 What is retained and what is not imported

Keep historical D0/P1C, quote code and evidence immutable. Reuse reviewed ideas such as complete support, frozen identities and exact finite evaluation. Do not import Allee drive, gradient-best selection, service-first comparators or finite-source export prohibitions into the Gaussian random actor process without a separate scientific declaration.

Likewise, retaining the Onsager explanation does not make it the active actor policy. It remains a mathematically meaningful controller alternative and an important part of the project's history. The new process asks a different question about random affordable actions.

Chapter recap

The historical engine supplies mechanisms, proofs and evidence at a known scope. Its force–mobility–flux architecture should be explained, not hidden; its controller should not be silently reused under a different study name.

Historical reading H57: The original executable world and its scientific safeguards

Historical model; not Gaussian evidence

The original implementation introduces its real state fields, response law and software boundaries. Thresholds, regenerative drive and old controller choices remain visible because they are needed to interpret the tests that follow. They are not requirements of the new first Gaussian domain. Original Part III, PDF pages 19–30. Internal chapter, section, equation and page references retain their original meaning. The new running folio belongs to this integrated volume.

What to carry forward. We now have the necessary historical vocabulary for the evidence sequence: the plant, controller, permission and measurement layers are identified. The next chapter follows experiments in their original order instead of presenting a collection of favorable outcomes detached from their design.

The smallest executable world

Before reading a controller, we need a precise picture of the world it controls. The project represents a spatial system as cells connected by directed edges. This is the same graph language used for roads, electrical networks, river reaches, supply routes, and communication systems. The graph is not meant to say that nature is literally made of squares and arrows. It is a measurement model: it states which local quantities are recorded and which transfers are possible.

46.1 A cell is a measured local account

A *cell* is one locally indexed part of the model. It may represent a reservoir, a district, a habitat patch, a hospital store, or a repair depot. A cell carries parameters describing the local physical account:

- x_i : the presently measured stock or functional capacity;
- L_i and U_i : the lower and upper edges of its viable band;
- R_i : a danger or reserve reference used in the field penalty;
- K_i : declared physical capacity;
- $\alpha_i, \beta_i, \chi_i$: weights attached to deficit, excess, and reserve danger;
- local supply, demand, leakage, and regeneration parameters.

The subscript i is only an address. If the world has four cells, i may be 0, 1, 2, 3. It does not mean that the quantity is imaginary or that cell zero is more important than cell three.

46.2 The state vector is a list of simultaneous measurements

The symbol

$$x = (x_0, x_1, \dots, x_{N-1}) \quad (46.1)$$

is called the *state vector*. A vector here is simply an ordered list of cell stocks measured at the same model time. If

$$x = (13, 2, 6), \quad (46.2)$$

then cell 0 contains 13 stock units, cell 1 contains 2, and cell 2 contains 6. The vector is not one mysterious global substance. It is a convenient way to write all local readings together.

The superscript n in x^n denotes a tick number, not a power. Thus x^4 means “the state at the beginning of tick 4.” A single component is written x_i^n : the state of cell i at that time.

46.3 An edge is a possible directed action

An *edge* $e : i \rightarrow j$ says that the model permits a transfer from source cell i to destination cell j . Direction matters. A water pipe with a one-way pump, a scheduled truck route, or an export connection may be represented by a directed edge. If return transport is possible, it is represented by a second edge $j \rightarrow i$ with its own parameters.

Each released V2.9 edge contains:

- $M_e > 0$, its mobility or response coefficient;
- $\theta_e \geq 0$, an activation threshold;
- $0 \leq \eta_e \leq 1$, its delivery efficiency.

If the source releases q units per unit time, the source loses the full rate q , while the destination receives $\eta_e q$. The difference $(1 - \eta_e)q$ is the transport loss rate. With $\eta_e = 0.9$ and $q = 2$, the source gives up 2 units per hour, the destination receives 1.8, and 0.2 is recorded as loss. The efficiency is inside the physics. It is not a later monetary discount.

46.4 The positive-part bracket

The notation

$$[y]_+ = \max(y, 0) \quad (46.3)$$

is read “the positive part of y .” If $y = 3$, then $[y]_+ = 3$; if $y = -3$, then $[y]_+ = 0$. It is used to express one-sided penalties and one-way activation without repeatedly writing case distinctions.

The local field penalty is

$$v_i(x_i) = \alpha_i[L_i - x_i]_+^2 + \beta_i[x_i - U_i]_+^2 + \chi_i[R_i - x_i]_+^2. \quad (46.4)$$

Below L_i , the first term grows quadratically. Above U_i , the second grows. Below R_i , the third records regenerative danger. Inside a region where none of those conditions is active, the corresponding terms are zero.

46.5 From local penalties to a global field burden

The global diagnostic is the sum

$$V(x) = \sum_i v_i(x_i). \quad (46.5)$$

This formula is the reason exact local accounting is possible. If an action changes only two cells, every other term in V cancels when we subtract the after-state from the before-state. The full-world difference is therefore equal to a calculation on the affected cells. The implementation exploits this fact; it does not inspect the whole vector when constructing a local quote.

46.6 Units and the meaning of time

Let stock be measured in units such as cubic metres of usable water or a normalised unit of functional capacity. A drive or flux is a rate, measured in stock units per unit time. The timestep Δt has units of time. Therefore

$$\Delta t \times \text{rate} = \text{stock change}. \quad (46.6)$$

This simple units check prevents a common error. If $q = 2$ units per hour and the tick lasts $\Delta t = 0.25$ hours, the source loses 0.5 stock units during the tick, not 2.

The symbol T sometimes denotes a final horizon or a number of ticks in other texts. In this implementation the individual time step is written Δt , and a study separately declares an integer tick count such as 200. The physical horizon is then $200\Delta t$.

46.7 A compact symbol-reading guide

Several typographic conventions repeat throughout the book:

Form	How to read it
x_i	stock or capacity at cell i
x^n	the complete state vector at tick n , not x raised to a power
$e : i \rightarrow j$	directed edge e from source i to destination j
q_e or J_e	rate on edge e ; multiply by Δt for stock moved
$\eta_e q_e$	useful rate arriving after route loss
$\sum_i$	add one term for every indexed cell
$\sum_{e \in \text{out}(i)}$	add only edges leaving source i
$[y]_+$	keep y when positive, otherwise use zero
μ_i	local derivative or marginal field signal
$V(x)$	global diagnostic formed by summing local penalties
$\Delta_e(q)$	exact signed EBU schedule for action quantity q

An index tells the reader what is being selected. A subscript i usually selects a cell; a subscript e selects an edge; superscript n selects time. A star, “max,” or “rob” modifies meaning and should be read from its definition rather than guessed.

46.8 Scalar, vector, and function

A *scalar* is one number, such as Δt or η_e . A *vector* is an ordered list, such as x . A *function* maps an input to an output, such as $v_i(x_i)$ or $g_i(x_i)$. The quote schedule is a function because the same committed object can be evaluated at several measured quantities.

This vocabulary is ordinary mathematical bookkeeping. It does not add a new physical assumption.

46.9 A complete three-cell example

Consider one medical-supply depot and two clinics. Cell 0 stores a temperature-sensitive medicine; cells 1 and 2 represent clinic stocks. The state is $x = (12, 1, 3)$. Two edges are available: $0 \rightarrow 1$ with efficiency 0.95 and $0 \rightarrow 2$ with efficiency 0.80. The first route is shorter and loses less medicine. The second clinic may nevertheless be in greater deficit.

This small world already contains the central scientific tension. Direction is not determined by distance alone. The local field includes the state of the source, the state and bounds of each destination, route efficiency, activation threshold, and the quantity under consideration. Preservation adds another question: can the depot provide either shipment while remaining above its own certified reserve?

46.10 From a parameter card to two edge requests

It is useful to calculate one complete graph by hand. Let the depot have $x_0 = 12$, $L_0 = 5$, $U_0 = 10$, $R_0 = 8$, weights $(\alpha_0, \beta_0, \chi_0) = (1, 0.5, 1)$, and zero natural drive for this example. Because the depot is above its upper band, its active penalty is

$$v_0(12) = 0.5(12 - 10)^2 = 2, \quad (46.7)$$

and its marginal is

$$\mu_0(12) = 2(0.5)(12 - 10) = 2. \quad (46.8)$$

Clinic 1 has $x_1 = 1$, lower bound 5, and deficit weight 1. Its penalty and marginal are

$$v_1(1) = (5 - 1)^2 = 16, \quad \mu_1(1) = -2(5 - 1) = -8. \quad (46.9)$$

For route $0 \rightarrow 1$ with $\eta_1 = 0.95$, $M_1 = 0.5$, and $\theta_1 = 0.05$, the force and raw flux are

$$f_1 = 2 - 0.95(-8) = 9.6, \quad (46.10)$$

$$J_1 = 0.5[9.6 - 0.05]_+ = 4.775. \quad (46.11)$$

Clinic 2 has $x_2 = 3$, so $v_2(3) = 4$ and $\mu_2(3) = -4$. Its less efficient route has $\eta_2 = 0.8$ with the same mobility and threshold:

$$f_2 = 2 - 0.8(-4) = 5.2, \quad (46.12)$$

$$J_2 = 0.5[5.2 - 0.05]_+ = 2.575. \quad (46.13)$$

The program stores these as two *edge results*, but their source is shared. Before any preservation cap, the source is being asked for the aggregate rate $4.775 + 2.575 = 7.35$. With a 0.1-hour tick, that corresponds to 0.735 stock units leaving the depot. Clinic 1 would receive $0.1(0.95)(4.775) = 0.4536$ units and clinic 2 would receive $0.1(0.8)(2.575) = 0.206$ units, up to rounding.

This calculation connects every symbol to a program field. The state vector supplies x_0, x_1, x_2 ; each Cell supplies bounds and weights; each Edge supplies M, θ , and η ; the marginal produces the local force; and all requests from source 0 must later share one P1C budget. No global search is hidden in the edge calculation.

46.11 What the graph does and does not encode

The graph says which actions are presently available. It does not prove that the chosen spatial scale is correct. If one cell represents an entire country, a large internal burden may be averaged away. If cells are too small, measurement and coordination may become impractical. Choosing cell support is part of model design and should be documented with the same care as choosing a parameter.

Likewise, the absence of an edge means “this model does not currently permit or represent that transfer.” It does not mean the two places are physically unrelated. A future long-range route can be built as a path through several local edges, with each performed segment receiving its own state-bound epoch.

Chapter recap

A cell is a local measured account. A state vector is the simultaneous ordered list of cell readings. An edge is a permitted directed transfer with mobility, threshold, and efficiency. Fluxes are rates, while Δt converts a rate into a stock change. The separable sum $V = \sum_i v_i$ connects these local objects to the global field design.

Worked questions

1. A route has $\eta = 0.75$ and accepts $q = 4$ units per hour for a 0.2-hour tick. Calculate source withdrawal, destination delivery, and loss during the tick.
2. For $L = 5$, $U = 10$, $R = 4$, and $x = 3$, identify which positive-part terms are active. Repeat for $x = 7$ and $x = 12$.
3. Explain why one bidirectional road requires two directed edges in this model.

How the software protects the scientific meaning

A program can contain the correct equation and still produce an untrustworthy experiment. It may update cells in the wrong order, read global information on a supposedly local decision path, duplicate a physical rule in two modules, or silently clip an impossible state. Architecture is therefore part of the scientific method, not merely a matter of programming taste.

47.1 The implementation stack

The relevant production modules have deliberately narrow responsibilities.

Module	Scientific responsibility
d0_v29.py	Cells, edges, penalties, marginals, local drives, loss-aware flux, certificates, and the unconstrained synchronous physical step.
p1c_v29.py	Source classification, robust aggregate safe-export budget, proportional fan-out, and the preservation-bounded physical step.
ebu_quote_v30.py	Pure local exact quote schedules, commitments, settlement envelopes, and protocol violations.
service_v30.py	A bounded service wrapper that limits service to physically available destination stock and records unmet demand.
o14_v30.py	The registered multi-edge worlds, shared action menu, arm selectors, and per-tick research records.
exp_v30_o14.py	Fail-closed preflight, the one official orchestration, strict validation, and deterministic artifact writing.

The modules use Python's standard library for the production logic. Immutable data classes make validated scientific inputs difficult to change accidentally. JSON supplies machine-readable plans and summaries; gzip compresses large JSONL traces without changing their logical rows; SHA-256 identifies locked byte sequences. Plotting is documentation work performed from committed artifacts, not part of the decision law.

47.2 Reuse the physical law; do not retype it

The P1C controller calls the released D0 edge-flux function. The quote layer calls the released penalty and marginal functions. This matters because two copies of the same formula can drift apart. One copy may receive a corrected efficiency term while the other remains old, leaving a system in which the physics and the accounting disagree.

Tests therefore inspect source code as well as outputs. They check that certain functions are imported and reused and that forked definitions do not appear on the decision path. This is a good example of a test enforcing scientific architecture rather than a single numerical answer.

47.3 Decision path and evaluation path

The *decision path* is the calculation that determines a proposed or accepted local action. It may read the permitted local views, the edge parameters, the source budget, and the registered action menu. It must not read the global V , a future trajectory, an actor's wallet, a price, or the final research label.

The *evaluation path* runs after the action. It may compute global burden, service comparisons, outcome classes, and diagnostics. Those quantities help a researcher judge the local law; they must not leak backward and secretly guide the action.

The distinction is comparable to a medical trial. A clinician may follow a declared treatment rule, while an analyst later calculates a population outcome. Allowing the treatment rule to read the future outcome would destroy the test.

47.4 Fail closed rather than repair silently

Scientific software should stop when a required invariant is violated. A non-finite number, a plan-hash mismatch, an unknown cost category, or an existing completion sentinel causes the later runners to refuse execution. They do not replace the value, choose a new plan, or overwrite the old result.

This discipline can feel inconvenient. Its benefit becomes clear when a result is surprising: the investigator can distinguish what the registered model did from what a convenience wrapper invented afterward.

47.5 Import-safe libraries and runners

Importing a Python library should define data and functions; it should not launch a 60-run study or write a result directory. The runner therefore exposes a `main()` function and performs the official work only when executed as a script. Tests verify that import is silent and side-effect free.

This separation lets a reviewer inspect and test the orchestration without accidentally performing the registered experiment. It also makes the historical statement “the official runner was invoked once” technically meaningful.

47.6 An object-by-object journey through the stack

The same physical action appears in several data objects, each adding information without changing the earlier layer.

1. A `World` contains immutable tuples of `Cell` and `Edge` objects. The mutable-in-time state is stored separately as a tuple of numbers.
2. `local_view` copies only one cell’s permitted parameters and stock into a `LocalView`. The edge decision never receives the world.
3. `edge_flux` returns a force and raw request for one edge.
4. `p1c_step` groups requests by source, creates one `SourceResult` per configured exporter and one `EdgeResult` per edge, then applies the physical update.
5. A `LocalQuoteInput` reconstructs the two endpoint views and accepted action data needed for accounting.

6. `build_quote` returns an immutable `QuoteSchedule` with a bound `QuoteEpoch`.
7. `Settlement` produces a `SettlementResult` or a separately typed `ProtocolViolation`.
8. The study wrapper adds service, unmet need, comparison fields, and the research outcome class after the physical action.

This sequence prevents a convenient but dangerous shortcut: passing one giant mutable object everywhere. Narrow objects make the information boundary visible to both a human reader and an automated source scan.

47.7 Validation happens at construction time

The data classes reject impossible declarations early. An edge cannot point from a cell to itself, mobility must be positive, efficiency must lie in $[0, 1]$, and all required numbers must be finite. A flow source must declare a flow cap and must not pretend to have a stock reserve; a regenerative stock source must declare a reserve configuration.

These checks do not guarantee that the parameters represent nature. They prevent internally meaningless objects, such as efficiency 1.4 or a NaN stock, from entering later calculations where the failure would be harder to diagnose.

47.8 Why immutability matters

An immutable object cannot have its efficiency or accepted quantity changed after its hash is calculated. To represent a new state, the program constructs a new tuple or result rather than modifying the frozen input. This mirrors the scientific distinction between observation and successor: x^n remains available for audit after x^{n+1} is calculated.

Immutability is not perfect security. It is a local design property that reduces accidental mutation and makes state transitions explicit.

Chapter recap

The code is layered so that physics, preservation, accounting, service, world construction, and official orchestration cannot silently redefine one another.

The production modules reuse one physical substrate, local decisions cannot read global outcomes, and runners fail closed before writing scientific results.

Review exercises

1. Why would copying the penalty equation into the quote module create a scientific risk even if both copies are initially identical?
2. Name two quantities that belong on the evaluation path but not on the local decision path.
3. Explain why importing a module that immediately runs a study weakens the audit trail.

Reading the experimental sequence

We now know enough about the historical engine to read its experiments without confusing their object with the current Gaussian proposal. The sequence matters. Early tests checked local mechanisms and numerical behavior. Later work exposed limitations of the questions being asked. Quote conformance and service-sensitive comparisons then required different records and different interpretations. A list of green tests would erase that development.

58.1 Begin with the question, not the result

For every experimental chapter, write down the question in one sentence before reading its outcome. Is the test about conservation, a numerical certificate, exact quotation, execution performance, preservation, service or the ability to discriminate alternatives? These are not interchangeable targets. A fast evaluator can be scientifically unhelpful; a preservation policy can correctly refuse useful service; an exact quote can be embedded in a controller that never chooses the action being quoted.

A result is meaningful when its design could have exposed the failure relevant to its question. A stress case that almost never reaches a hard boundary says little about behavior at that boundary. Conversely, a case designed to expose a boundary failure should not be discarded as unrealistic merely because it finds one. The registered scope governs the inference.

58.2 Mechanism tests and dynamical claims

The D-series belongs to a specific historical plant and response mechanism. Its first chapters teach the value of examining conservation, loss handling, directional flow and numerical behavior separately. An invariant can hold while another property fails. This is not an inconsistency: the properties have different assumptions and different failure modes.

For example, a synchronous implementation can preserve mass exactly and still overshoot the minimum of a convex potential in one large step. A soft penalty can strongly discourage a boundary crossing without making it impossible. A hard source certificate can forbid an export without solving the

destination's unmet need. These logical distinctions remain important even when the active potential family changes.

The original D9 and D10 discussions are therefore retained in full. Their purpose here is not to reintroduce old thresholds into the Gaussian state. It is to make the difference between a cost, a physical condition and an admission rule concrete. Removing their vocabulary without preserving the lesson would allow the same category error to return under new symbols.

58.3 Performance is not efficacy

Historical D5 includes software performance work. A timing measurement can support a claim about that workload on that environment. It cannot by itself show that the selected physical actions maintain function, nor can it price an unregistered future cloud campaign. The input sizes, host conditions and measured operation are part of the performance claim.

This boundary is not a reason to omit performance work. A scientifically meaningful method may still be impractical to execute or audit. The right conclusion is to report feasibility and scientific effect as separate dimensions, not to allow one to stand in for the other.

58.4 Quotation conformance asks a narrower question

Gate 1B concerns the relationship between an exact local quote and its declared mathematical and implementation contract. Such work can detect a missing factor, an inconsistent baseline or an incorrect finite schedule. It does not establish that a controller will choose useful actions over a long horizon. Its success is necessary for trusting later value records, not sufficient for believing an institutional story about them.

In the Gaussian revision, some formulae become simpler because curvature is constant. The need to bind a quote to an epoch, support, parameter version and accepted quantity does not disappear. Historical conformance remains evidence about the historical implementation; new conformance must be demonstrated for a separately accepted implementation.

58.5 When a question must change

Adversarial search can expose a defect or a limitation in the claim being tested. A subsequent study may legitimately ask a better question. The scientific obligation is to record that change prospectively, not pretend that the revised interpretation was always the original criterion.

Service alignment is an especially important example because preservation alone can favor doing nothing. An idle controller may keep every stock unchanged while providing no requested service. Whether that is success depends on the registered question, not on the fact that one physical invariant remains perfect. The new first Gaussian domain has no service objective hidden inside its random selector; service claims would require their own represented quantities and comparison.

58.6 A worksheet for every historical result

Read each result through six questions. What was fixed before execution? What implementation actually ran? What observation was recorded? Which alternative interpretation was ruled out? Which limitation remains? What later claim would exceed the evidence?

This worksheet distinguishes three situations that can otherwise look similar. A correctly completed null result can mean that the arms do not differ under the registered metric. An inconclusive result can mean the design cannot decide the claim. An invalid result can mean the execution did not satisfy its contract. None should be relabelled as another merely to make the research narrative smoother.

The following source reading preserves the experimental progression and its original qualifications. The next chapter then treats the original flat O14 outcome and the later finalized Gate 1D-C evidence as separate records. Neither is a substitute for an unperformed Gaussian experiment.

Historical reading H58: The experimental sequence from D1 to service alignment

Historical model; not Gaussian evidence

Read these six original chapters as an evidence sequence: D1-D4, D5-D8, D9, D10, Gate 1B and the move from adversarial search to service alignment. Their numbers, limitations and publication-era status are retained. Performance timing is software evidence; quotation checks are conformance; neither is a general theorem of recovery. Original Part III, PDF pages 66–100. Internal chapter, section, equation and page references retain their original meaning. The new running folio belongs to this integrated volume.

What to carry forward. The sequence changes the question as limitations become visible. That is useful research history, but it does not make the final question retroactively preregistered for every earlier run. The next chapter separates the original O14 outcome from the later finalized Gate 1D-C study and from the still-unexecuted Gaussian programme.

The first wind tunnel: D1 through D4

Before testing preservation or actor accounting, the project placed the D0 engine in small deterministic fixtures. A wind tunnel does not reproduce the entire atmosphere; it isolates forces that would be difficult to interpret in a complete aircraft flight. In the same way, D1 through D4 isolate descent, transport loss, synchronous causality, regeneration, and service.

The policy labels used in this sequence are:

- P0: no transport;
- P1: the exact synchronous loss-aware D0 law;
- P1K: a diagnostic projection wrapper, outside the theorem;
- P2: a deliberately loss-blind force;
- P3: a deliberately sequential live-state update;
- P4: a deliberately oversized timestep.

The controls are not competing economic systems. Each removes or changes one specific property so that the experiment can detect its importance.

53.1 D1: can the smallest world move downhill?

D1 contains two cells, no external drive, and a lossless edge. The initial state is (19, 2) and the initial burden is 17. Under P0 nothing moves, so the final burden remains 17. Under P1 with $\Delta t = 0.5$ and certified ratio $r_{\Delta t} = 0.5$, the field relaxes until the final burden is approximately 6.31×10^{-30} , numerically zero. There are no descent violations.

The result is simple by design. It checks that a clearly active edge sends stock in the direction predicted by the local field and that the discrete update agrees

with the expected undriven relaxation. It does not test regeneration, scarcity, service, or reserve safety.

How to read the numbers

“Final burden near zero” means both cells ended in a state where the represented penalty was essentially absent. It does not mean the stocks became zero. In fact, the observed stocks remained between 4.5 and 16.5. Burden and stock are different quantities.

53.2 D2: why route efficiency changes the correct resting point

D2 uses a lossy edge. P1 and the loss-blind P2 begin from the same state (8.5, 2) with burden 66.25. The difference is one factor: P1 uses $f = \mu_i - \eta\mu_j$, while P2 behaves as though $\eta = 1$ inside the force.

P1 ends with burden 58.243 and zero descent violations. Its final loss-aware force is approximately 0.05, equal to the activation threshold, so the edge rests. At the same state the loss-blind force is still about 5.294. P2 therefore continues moving material after the correct loss-aware law has reached its resting condition. It ends with the higher burden 61.814 and records eleven descent violations.

This fixture turns an abstract efficiency term into a physical story. Imagine a clinic reachable only through a route that destroys half the medicine. The benefit of one unit leaving the depot is not the benefit of one unit arriving. Ignoring efficiency can make continued transfer look attractive after its actual delivered benefit no longer justifies source withdrawal.

What the negative control proves

P2 shows that the fixture and diagnostics can detect omission of η . It does not prove that every real efficiency is known exactly. Measurement uncertainty is a later problem.

53.3 D3: one-hop causality

D3 places three cells in a chain $0 \rightarrow 1 \rightarrow 2$. It runs paired baseline and perturbed initial states under both synchronous P1 and sequential P3. The measured quantity is not one absolute cell value. It is the difference at the probe cell 2 between the paired runs.

Under synchronous P1, the perturbation produces zero probe difference at tick 1 and first appears at tick 2. Under sequential P3, the probe differs by 0.0162 at tick 1. The sequential loop updated the first edge and then allowed the second edge to read that newly changed relay state inside the same nominal tick.

This may sound like a small programming detail, but it changes the model's notion of time. In P1, one tick represents one simultaneous response to a common observation. In P3, stored edge order determines how far information can travel. A route listed earlier in a file can gain causal priority without any physical meaning.

53.4 D4: drive, regeneration, and useful service

D4 introduces external dynamics and demand. P0 serves only 6 of 150 requested units before the consumer leaves the declared physical domain; its minimum state reaches -144 . P1 serves all 150 requested units, remains physically admissible, and ends with burden 4.578. P1K is identical here because no projection is needed.

The burden does not decrease on every driven tick. P1 records many “descent violations” relative to an undriven monotonic expectation because demand and regeneration continuously inject and remove stock. This is not a contradiction of the undriven theorem. The correct driven inequality contains the term $\mu^T u$; natural drive can raise the field burden while transport still acts dissipatively.

Service is counted only while physically valid

The P0 destination continues to subtract demand after crossing below zero in the unconstrained equation. The study does not count that mathematical subtraction as valid delivered service. Its service total stops when the physical domain is lost. This prevents an impossible negative store from satisfying an

accounting target.

53.5 A trace-reading example

To read one D4 tick, follow this sequence:

1. inspect x^n and verify that all cells are within the declared domain;
2. read the local drive vector u ;
3. inspect the edge force and flux calculated from the frozen views;
4. multiply each rate by Δt to obtain stock changes;
5. confirm the source withdrawal, delivered amount, and transport loss;
6. inspect x^{n+1} and the ledger residual;
7. only then read burden and service diagnostics.

This order prevents a common interpretation error: starting with a final score and guessing backward which physical event produced it.

53.6 What D1–D4 established

These fixtures support the following finite implementation statements: the released P1 engine relaxed the simple lossless world; loss awareness changed the resting state and avoided the P2 violations; synchronous evaluation enforced one-hop causality; and the driven P1 world served the registered demand while remaining in domain. They also demonstrate that a no-transport baseline can be physically invalid and that undriven descent is not the correct success criterion for every driven tick.

They do not establish hard reserve preservation. That question appears only when a source can cross a dangerous regenerative boundary.

Chapter recap

D1 checks simple downhill relaxation. D2 makes efficiency observable. D3 makes synchronous causality observable. D4 adds drive and service and shows why burden, domain validity, and useful delivery must be read separately.

Together they test the physical substrate before the preservation controller is introduced.

Exercises

1. Why is $P0$'s constant burden in $D1$ not evidence that $P0$ is a good policy?
2. Describe a real scheduling system in which sequential order would create an artificial priority similar to $P3$.
3. $D4$'s $P1$ burden is not monotone on every driven tick. Explain why that does not invalidate the undriven descent theorem.

Learning from D5 through D8

The next four fixtures are valuable because they did not produce a simple series of victories. D5 failed to discriminate a reserve penalty. D6 and D7 showed recovery under shocks but did not create a hard preservation result. D8 confirmed one oversized-step failure while falsifying a stronger expectation about repeated growth. Keeping these outcomes made the later protocol better.

54.1 D5: a reserve fixture that was too forgiving

D5 compared the field with reserve weight $\chi = 1$ and $\chi = 0$. The intended question was whether the soft regenerative-danger term would protect an Allee-like source. Both configurations served all 150 requested units, remained in the declared domain, and recorded no reserve or Allee crossing. Their headline trajectories were effectively non-discriminating.

The correct conclusion is not “the reserve term is unnecessary.” The source and demand combination never forced the policies into a region where their difference mattered. A fire alarm cannot be validated in a building where no smoke reaches either sensor.

This distinction became a design rule: a study of a protection mechanism should include a fixture in which the relevant boundary is reachable under the control policy. D9 was later built to satisfy that requirement.

54.2 Why a null result can be useful

A non-discriminating world reveals weakness in the measurement instrument. It can show that initial stock is too large, demand too low, duration too short, or an outcome metric saturated. The scientific response is to preserve the null, diagnose it, and preregister a new discriminating study – not to change the old world until the desired result appears.

This lesson returns in O14, where genuine policy differences again fail to change the registered primary service outcome.

54.3 D6: a supply shock and a viability gap

D6 runs for 1000 ticks and imposes a registered supply shock. P0 leaves the physical domain, serves only 6 of 200 requested units, and does not recover. P1 serves all 200, remains in domain, and satisfies the registered recovery criterion at tick 440. Its final burden is about 1.031.

The recovery metric uses a pre-shock reference window, a threshold relative to that reference, and a sustained post-shock window. This is stronger than calling the first upward tick “recovery.” A brief rebound can occur inside an overall decline.

The fixture also reveals a *viability gap*: useful service and field burden can recover at different rates. A system may continue meeting current demand while some cells remain outside their preferred bands. One summary number cannot tell both stories.

54.4 D7: a demand shock

D7 increases demand during the run. P0 again exits the domain and validly serves only 6 of 260 requested units. P1 serves all 260, remains physically admissible, and reaches its sustained recovery condition at tick 611.

Supply and demand shocks have different local meanings. A supply shock reduces the source’s drive. A demand shock increases the destination’s draw. Both can produce a growing deficit, but they suggest different interventions in a real needs system: diversify supply in one case, improve efficiency or capacity in the other.

The model’s ability to distinguish those mechanisms depends on receipts that keep drive, transport, and service separate.

54.5 D8: deliberately crossing the timestep certificate

Discrete descent requires a sufficiently small timestep. D8 compares an unsafe P4 run with $r_{\Delta t} = 1.2$ to a certified P1 contrast with $r_{\Delta t} = 0.8$. The unsafe

run begins at burden 2 and jumps to 3.92 on the first tick, a ratio of 1.96. The contrast falls to 0.72 without a descent violation.

The unsafe result demonstrates overshoot. A large explicit step can cross the valley of a convex penalty and land higher on the other side, even though its initial force points downhill.

The preregistration also expected a stronger monotone-growth pattern after the first overshoot. That sub-claim failed: the P4 burden did not keep increasing on every subsequent tick. The correct report therefore contains both facts – the immediate counterexample succeeded, while the stronger trajectory description did not.

54.6 The dimensionless timestep ratio

The ratio

$$r_{\Delta t} = \frac{\Delta t}{\Delta t_{\text{certificate}}} \quad (54.1)$$

compares the chosen tick to a computed certificate. Values at or below 1 are inside the declared bound; values above 1 are outside it. Because the ratio is dimensionless, it can compare worlds with different time units and curvature scales.

Being inside the certificate is sufficient for the stated finite-step inequality under its assumptions, not necessary for every trajectory to behave well. Being outside does not guarantee failure on every state. D8 supplies a concrete state where failure is actually observed.

54.7 What changed after these fixtures

Three methodological decisions follow from D5–D8:

1. Protection studies require a boundary-capable positive control.
2. Shock recovery requires persistence, not a single favourable tick.
3. Timestep claims must report both certificate status and actual numerical behaviour.

Most importantly, the project stopped treating a stronger soft penalty as a substitute for a hard source-level guarantee. D5 did not show the soft term

was effective; it showed that the fixture could not answer the question. The next study therefore made the reserve cross under the reserve-blind arms and asked whether an algebraic cap prevented it.

Chapter recap

D5 preserved a non-discriminating null and motivated a harder world. D6 and D7 showed finite shock recovery under P1 while exposing the failure of the no-transport baseline. D8 made timestep overshoot visible and preserved a failed sub-claim. These are examples of experiments improving the research question, not merely confirming a model.

Exercises

1. Why might two reserve-weight settings fail to distinguish the policies?
2. Why should a recovery definition include a sustained window?
3. Explain the difference between “outside the certificate” and “observed to fail.”

D9: when a soft warning meets a hard boundary

D9 is the first preservation fixture designed so that the competing mechanisms must reveal themselves. An Allee-like source begins above its certified reserve, serves sustained demand, and can cross both the reserve and the lower regenerative threshold. The study compares a reserve-blind law, a soft reserve penalty, and the hard P1C cap.

55.1 The world and why it discriminates

The source begins at $x_0 = 13$. Its certified reserve is $R^{\text{eff}} = 11$, the Allee threshold is $A = 5$, the timestep is 0.2, and demand is 5 stock units per unit time for 200 ticks. The world is not made safe by an enormous initial buffer. If the controller continues serving without a hard cap, the source reaches the dangerous region during the registered horizon.

There are four arms:

- D9-A: reserve-blind P1, $\chi = 0$, no hard cap;
- D9-B: soft reserve penalty, $\chi = 1$, no hard cap;
- D9-C: P1C hard cap with $\chi = 1$;
- D9-D: P1C hard cap with $\chi = 0$.

The C-versus-D pair isolates the hard cap from the soft field term. If they are identical, the cap rather than χ determines the boundary behaviour in this fixture.

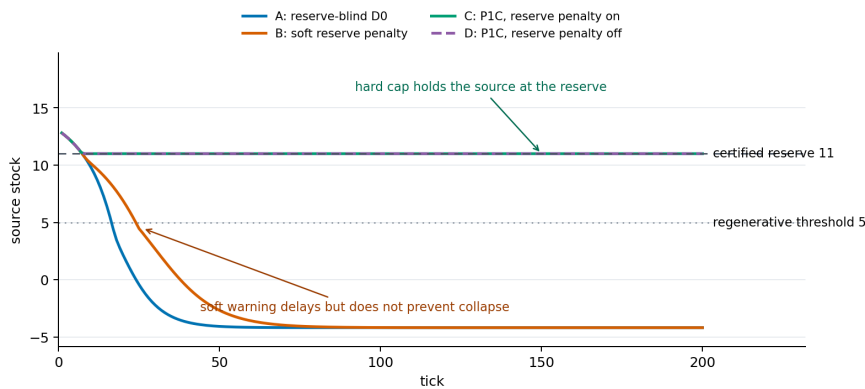


Figure 55.1: *D9 source-stock trajectories generated from the committed per-tick trace. Horizontal lines show the certified reserve and Allee threshold.*

55.2 How to read the trajectory figure

Read Figure 55.1 from left to right. All four arms begin at the same source stock. The blue reserve-blind and orange soft-penalty trajectories both cross the reserve at tick 8. The soft term changes the later path: it delays the Allee crossing from tick 17 to tick 24 and reduces cumulative delivery. It does not create invariance.

The green and dashed purple P1C trajectories lie on top of one another. They descend to 11 and remain there. P1C binds on 193 of 200 ticks. The overlap is evidence that, once the hard constraint controls quantity, switching the soft weight on or off does not change any recorded headline metric in this world.

55.3 The exact committed outcomes

Arm	Reserve tick	Allee tick	Delivered	Unmet
A	8	17	189.692	10.308
B	8	24	172.831	27.169
C	never	never	130.825	69.175
D	never	never	130.825	69.175

The reserve-blind arm delivers the most during the finite horizon, but it does so while entering the collapse basin. The soft arm sacrifices some delivery

and delays deeper damage, yet still crosses both boundaries. P1C delivers less and records the remaining need as unmet demand.

Calling only one column “success” would destroy the meaning of the study. Delivery without preservation consumes tomorrow’s source. Preservation without reporting unmet need can conceal human harm. D9 therefore classifies C and D as *safe rationing*, not as unqualified success.

55.4 A tick at the reserve

Once the P1C source reaches 11, imagine its local drive during a tick adds exactly the quantity that can be safely exported. The robust budget accepts that rate, and the successor returns to 11. If requested export is higher, proportional scaling reduces it. This repeated equality explains the flat green and purple lines: the controller is not freezing all activity; it is allowing only the flow compatible with the boundary.

The observed theorem-eligible count is 200 and the observed violation count is zero for both P1C arms. That is implementation conformance at these ticks. The algebraic theorem remains the reason the inequality should hold under its assumptions.

55.5 Comparing tick 7, tick 8, and the later plateau

The first reserve crossing in the uncapped arms occurs at tick 8. A careful trace review compares the state immediately before and after that event. Tick 7 still has the source on the non-crossed side of the boundary. At tick 8, accepted withdrawal plus local drive places the successor below 11. The crossing counter increments once; remaining below does not create a new crossing every tick.

In P1C, the same growing request meets a shrinking safe budget. The scaling factor falls below one, accepted export is reduced, and the after-state reaches 11 without passing it. On later ticks the source’s regeneration and accepted export balance at the boundary. The trace therefore records repeated binding, not repeated crossing.

This comparison helps readers interpret three different metrics:

- *crossing count* records entry through a boundary;

- *time below* records persistence in the harmful region;
- *minimum stock* records the deepest excursion.

One metric cannot replace the others. A single crossing followed by long recovery is different from a crossing followed by 193 ticks below reserve.

55.6 Why “dead source” is not the only collapse indicator

The two collapsing arms finish with negative source states in the unconstrained model, yet their registered “dead source” boolean remains false because that diagnostic used a different terminal definition. The record therefore uses reserve crossing, Allee crossing, time below thresholds, physical overuse, and domain status rather than relying on one boolean.

This is an important general lesson. A biological system can lose regenerative capacity before its stock equals zero. A hospital can remain open while losing critical service capacity. Collapse is often a process with several observable boundaries, not one dramatic final event.

55.7 What D9 proved and observed

The P1C reserve inequality is an algebraic one-step result under stated assumptions. D9 additionally observed that the implementation held the source at the reserve for this 200-tick fixture, with no observed theorem violation. It also observed that the soft penalty did not preserve the source and that the cap created explicit rationing.

D9 did not prove long-run sustainability, correct reserve certification, social fairness of proportional rationing, stochastic robustness, or a general service theorem. It demonstrated a clean distinction between warning and constraint.

55.8 A needs interpretation

Imagine the destination is a hospital rather than an anonymous sink. The 69.175 units of unmet demand are serious; preservation does not make them disappear. A responsible institution would search for another source, substitute treatment, borrow emergency capacity, improve route efficiency, or ex-

pand regenerative supply. What it should not do is erase the source boundary from the account and call destructive delivery affordable.

P1C makes the scarcity visible early enough for those institutional responses to be designed. It is a physical safety layer, not the complete ethical decision.

Chapter recap

D9 makes the reserve boundary reachable. The soft penalty delays one deeper crossing but does not prevent the reserve crossing. P1C holds the source at 11 with zero observed violations and reports 69.175 units of unmet demand. The figure and table must be read together: preservation, delivery, and scarcity are separate outcomes.

Exercises

1. Why does the C-versus-D comparison isolate the hard cap from the soft reserve weight?
2. Is arm A's larger delivery evidence that it is the best long-run policy? Explain using the threshold crossings.
3. Propose two institutional responses to D9's unmet hospital need that do not redefine the local source physics.

D10: a map of service, rationing, and collapse

One carefully chosen trajectory can still be exceptional. D10 varies demand, transport efficiency, and several secondary parameters to map where the four policies preserve, serve, ration, or collapse. The purpose is not to simulate a planet. It is to see whether the D9 mechanism persists across a registered grid and whether the outcome labels behave sensibly near analytic boundaries.

56.1 The two physical ratios

A logistic source has a maximum regeneration rate $g_{\max}$. Demand is expressed as the dimensionless ratio $d/g_{\max}$. A value below 1 means demand is below maximum source regeneration before transport loss. Useful delivered capacity is smaller when $\eta < 1$, so the more relevant service boundary is approximately

$$d \leq \eta g_{\max}. \quad (56.1)$$

This creates a region where the source could regenerate enough in gross terms but the route cannot deliver enough after loss. Transport efficiency is therefore not merely a cost modifier; it can move a need from feasible service into physical scarcity.

56.2 The registered grid

The D10 core crosses demand ratios 0.25, 0.5, 0.9, 1.0, and 1.1 and efficiencies 0.5, 0.7, 0.9, and 1.0. It compares P0, P1, soft, and P1C. Secondary slices vary activation threshold θ , reserve margin, soft weight χ , regeneration rate ρ , and timestep ratio. The complete D9/D10 study contains 144 runs; D10 contributes 140.

56.3 How to read the phase map

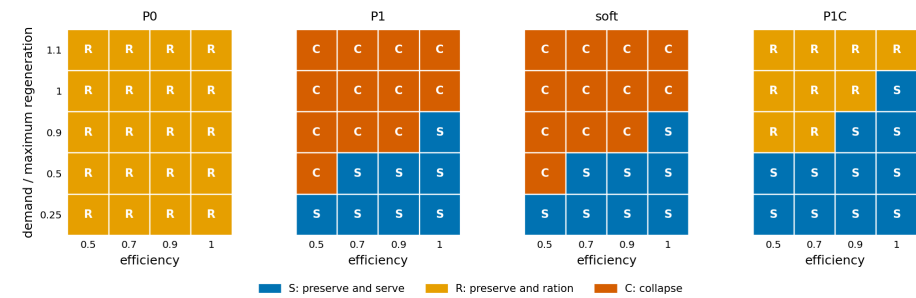


Figure 56.1: *D10 core outcome map generated from the committed summary. Each panel uses the same demand and efficiency axes. S means safe service, R safe rationing, and C collapse.*

Each small square is one registered run. Move right to improve delivery efficiency. Move upward to increase demand relative to maximum regeneration. Compare panels horizontally to ask how policy changes the outcome at the same physical point.

The P0 panel is entirely rationing because no transport serves the destination. Calling this panel “safe” would be misleading: source preservation is achieved by doing nothing for the destination. P1 and the soft arm serve the lower-demand, higher-efficiency region, but collapse through overuse at 25 of 35 registered points. P1C has no collapse points. It safely serves 25 and safely rations 10.

56.4 Outcome counts

Policy	Collapse	Safe service	Safe rationing
P0	0	0	35
P1	25	10	0
soft	25	10	0
P1C	0	25	10

P1C does not manufacture capacity. Where demand and transport make full service infeasible under preservation, the map turns from S to R. This is why

explicit unmet demand is scientifically useful. It shows the boundary between a policy failure and a physical impossibility within the declared model.

56.5 Reading one vertical and one horizontal slice

Choose the P1C panel and hold efficiency fixed at 0.9. Moving upward increases demand while route quality stays constant. Lower rows remain safe service; at higher demand the cells become safe rationing. The transition says that the same route and controller meet small needs fully but reveal scarcity when the required flow approaches or exceeds sustainable useful capacity.

Now hold demand ratio at 0.9 and move horizontally. Increasing efficiency means more of each source withdrawal reaches the destination. The service region expands because less source stock is spent per useful delivered unit. This is the kind of structural incentive an EBU account can expose: improving a route can change feasibility, not merely reduce an arbitrary charge.

Compare the same square across panels. If P1 collapses while P1C rations, the difference is not demand or efficiency; those are held fixed. It is the hard source constraint. If P0 rations while P1C serves, the difference is the ability to transport useful stock while respecting the reserve.

56.6 Why a colour map needs letters and counts

Colours provide a fast visual pattern but can be inaccessible or ambiguous. The map therefore places S, R, or C inside every cell and the text supplies exact counts. A reviewer should be able to recover the conclusion without relying on colour perception.

The count table also prevents visual area from becoming an informal statistic. The registered grid contains 35 points per policy, so “25 safe service and 10 safe rationing” is an exact inventory, not an estimate from coloured pixels.

56.7 Matching service where service is feasible

A meaningful preservation controller should not ration every world merely to make the reserve plot look safe. In the feasible region, P1C matches the serving arms’ useful service while avoiding the collapse region. At higher demand or lower efficiency it reports safe rationing.

This is a finite grid observation, not a universal optimality theorem. Yet it answers an essential diagnostic question: the hard cap is not equivalent to always refusing transport.

56.8 Secondary slices

The secondary parameter slices ask whether one headline map is hiding a fragile choice.

Activation threshold θ . A larger threshold requires a stronger field force before the route activates and can reduce service near balance.

Reserve margin. Raising the effective reserve reduces the stock available for export and can move a point from service to rationing.

Soft weight χ . Changing the warning strength modifies the field force but does not create the hard invariant seen under PIC.

Regeneration rate ρ . Faster regeneration expands the physically feasible region; slower regeneration contracts it.

Timestep ratio $r_{\Delta t}$. A step nearer the certificate probes finite-step sensitivity without pretending the ratio is a real-world safety margin.

The slices are interpretation aids. They do not exhaust the parameter space or replace an uncertainty analysis.

56.9 Domain exits remain results

Eighteen D10 trajectories end with a domain-exit status. They are not removed from the denominator or silently clipped. A domain exit tells the reader that the unconstrained numerical model left the declared physical region. Its later service values are not treated as valid physical delivery.

Dropping failed runs would make the harmful policies look better precisely where their dynamics become impossible. Keeping them is part of honest failure reporting.

56.10 The serialization incident

The first D9/D10 execution attempt failed at the artifact layer. Fifteen stability diagnostics overflowed to non-finite values that strict JSON could not represent. The trajectories themselves had already run. Under a separately documented narrow repair, those diagnostic fields were reconstructed from committed traces and represented as null; no trajectory, service total, crossing, or outcome class was rerun.

The incident distinguishes three operations that are often blurred: executing a model, interpreting a trace, and serializing a report. A failure in the third does not justify secretly repeating the first.

56.11 What the map supports

D10 supports the finite observation that P1C avoided collapse across the registered grid, served the feasible region, and exposed scarcity as rationing. It also shows that P0's preservation is socially empty in these worlds and that the soft penalty did not replace a hard cap.

The map does not prove that every real resource has logistic dynamics, that the declared reserve is correct, or that proportional sharing is fair. It does not close the theorem gap for the bounded service wrapper. A phase map is a readable summary of finite registered runs, not a law of nature.

Chapter recap

D10 maps demand and efficiency against four policies. The map must be read as a joint preservation-and-service instrument: P0 rations everything, P1 and soft serve some points but collapse at many others, and P1C moves between safe service and explicit rationing without observed collapse. Domain exits and serialization history remain visible.

Exercises

1. For $g_{\max} = 3$ and $\eta = 0.7$, what is the maximum useful delivery rate before other constraints?
2. Why is P0's all-rationing panel not a preservation success by itself?

3. Choose one D10 square and describe what additional evidence would be needed before treating it as a real resource-management recommendation.

Gate 1B: does the quote code match the equation?

The V2.9 studies established a preservation controller, but a future action account requires another object: a signed quote that is exact for the represented local transformation. Gate 1B did not ask whether actors following that quote would behave well over time. It asked a narrower question first: does the pure quote implementation conform to the preregistered mathematical definition?

No behavioural trajectory was run. The gate consisted of 21 conformance groups, Q1 through Q21, plus a read-only audit. The adversarial replay Q22 was already named in the locked plan but deliberately reserved for a later authorization.

57.1 Why equation conformance comes before behaviour

If a behavioural study produces a surprising outcome, the investigator needs to know whether the actor followed the intended quote or a coding error. Separating equation conformance from behaviour reduces that ambiguity. Gate 1B therefore tests the quote as a mathematical instrument before placing it inside a selection loop.

This is similar to calibrating a thermometer before using it in a clinical trial. A calibration result does not prove that a treatment works. It establishes that the measurement device reports the temperatures it claims to report at the tested points.

57.2 Q1–Q5: the five basic stories

The first groups use human-readable fixtures.

Q1: doing nothing

At $q = 0$, no represented state changes and $C_a(0) = 0$, so the exact quote must be exactly zero. This checks the explicit zero branch rather than a small positive limit.

Q2: a beneficial action

Worked example E1 has exact quote +7.94. The linear diagnostic reports +15.18. Both are positive, but the linear tangent substantially overstates the finite benefit. The exact value is the settlement result.

Q3: a damaging action

Worked example E2 has exact quote -5.52 and linear diagnostic -0.08 . The linear approximation misses 5.44 burden units. The test contains a negative control showing that a settlement path using the diagnostic would be detected.

Q4: a closed damage-repair cycle

In E3, a sequence returns the state exactly to $(10, 10)$. The net state-field difference therefore telescopes to zero. The remaining settlement is the negative sum of declared process costs, -0.10 . A closed physical state cycle cannot manufacture positive EBU merely by splitting the path.

Q5: natural regeneration is not actor credit

The regeneration fixture has drive $g = 2$ and no-action successor $z = 12$. If the actor does nothing, the quote is exactly zero. A deliberately naive baseline using the pre-drive state credits or debits the actor for natural regeneration; the test detects the error exactly.

Together these cases teach the quote's grammar: rest is zero, signed action values can be positive or negative, finite curvature matters, cycles pay process burden, and background drive belongs in the counterfactual baseline.

57.3 Q6–Q8: curvature and the complete schedule

Q6 compares exact and linear values across penalty branches and 200 registered random cases. For a convex state penalty and the declared cost family, the first-order tangent provides an upper diagnostic bound on the exact benefit. The group checks the numerical implementation of that relation at the sampled points.

Q7 studies schedule concavity and the discontinuity created by a fixed setup cost at $q = 0$. “Concave schedule” here means that, within the registered cost assumptions, increasing quantity has diminishing exact benefit. It does not mean that every larger quantity remains beneficial.

Q8 commits complete schedules for 100 cases and evaluates five partial-execution points on each. The test checks that settling a smaller measured quantity uses the already committed function and remains inside $[0, q^{\text{acc}}]$.

57.4 Q9 and Q12: protocol violations are not valuations

Q9 requests settlement beyond the accepted maximum. The minimum fail-closed envelope issues no positive EBU and creates a mandatory violation record. Q12 attempts duplicate settlement on ten fixtures and requires rejection.

The important design choice is that the violation layer does not invent a field value. If a shipment exceeds its authorization, the system has at least two facts: the physical action occurred, and the protocol was violated. A security or legal response may be necessary, but its penalty should not masquerade as the exact field difference.

57.5 Q10 and Q11: telescoping with and without drive

For a sequence of exact action differences evaluated on consistent successive states, intermediate state terms telescope. In an undriven closed cycle, the net state part is zero and process burden remains. Q10 checks 50 sequences and ten cycles.

With natural drive, a separate residual appears because the no-action successor changes between epochs. Q11 checks the driven identity and includes a negative control that wrongly applies the undriven theorem; the omitted residual exceeds 10^{-3} and is detected. This prevents a concise conservation slogan from being used outside its assumptions.

57.6 Q13 and Q14: the split-action trap

The registered split counterexample evaluates two withdrawals from the same frozen nonlinear source. Quoting them independently as if each were first gives a sum of 32, while the joint state change is 20. The phantom credit is 12.

The cause is curvature. Both independent lines claim the high marginal benefit available before either withdrawal. In the joint action, the first withdrawal changes the source state seen by the second part of the exact finite difference.

Q14 therefore enforces a first production discipline: one quoted action per source per micro-step. This avoids issuing independent overlapping claims while a general aggregate allocation rule is unproved. The restriction is conservative for accounting but can reduce action capability; later gates measure that cost.

57.7 Q15–Q17: permission and observation

Q15 constructs an action with a positive raw quote beyond P1C permission. The quote module refuses to extend the schedule past q^{acc} . Q16 constructs a P1C-permitted action with negative EBU, demonstrating the converse: permission does not promise a positive reward.

Q17 compares a non-binding P1C physical step with and without observational quoting. The physical tuples are exactly identical. This fixture is small, but it establishes the intended non-interference path before the same idea is tested over full trajectories.

57.8 Q18–Q21: the boundary around the equation

Q18 scans imports and forbidden identifiers, injects runtime poison, and rejects a world object where only two local views are permitted. Q19 rejects

process-cost categories that would double count state burden or mix money, labour, and security penalties into physical EBU. Q20 tests stale epochs, mismatched commitments, and the rule against reallocating a quote to another action. Q21 rebuilds 52 objects twice, checks deterministic identifiers, and requires strict finite JSON.

These groups test not only a numerical formula but its operational boundary. A quote is scientifically meaningful only if the system knows which data it may read, which state it belongs to, and whether it can be replayed.

57.9 Random fixtures without behavioural randomness

Some conformance groups use pseudorandomly generated input points under frozen ranges and seed 30001. This is not a stochastic behavioural study. The seed makes the input set exactly reproducible, while varied points reduce the chance that a branch error hides behind one hand-picked example.

The registered ranges include stocks from 0 to 24, several penalty weights, reserve values, efficiencies from 0.5 to 1, two timesteps, positive and zero setup costs, and drives between -2 and 2. Non-vacuity checks require the resulting sample to exercise more than the zero branch.

57.10 Audit independence

The separate audit recomputes plan hashes, checks the implementation import allowlist, verifies that the test does not import the adversarial runner, confirms equation-version consistency, reconstructs epoch identifiers, checks strict JSON, and verifies that released V2.9 files did not change.

This audit does not repeat every unit test. It attacks the conditions under which the test result should be believed. An implementation can return correct fixture numbers yet still be unacceptable if it imports global actor state or silently modifies the released physics.

57.11 What passed and what remains open

The conformance suite reported 184 passing checks in 21 groups, and the audit reported 19 passing checks. The maximum residual in the named worked

examples was approximately 8.9×10^{-16} , well inside the declared scale-aware tolerance.

The result supports a precise statement: at the declared fixture points, the pure quote implementation behaved consistently with the locked exact equation and its protocol boundary. It does not establish incentive compatibility, adversarial security, long-horizon alignment, robust observation under measurement error, or a complete actor economy. Those are different gates.

57.12 How the groups cover the action lifecycle

The 21 groups are easier to remember as a lifecycle than as a numbered list.

1. **Define value:** Q1–Q7 check zero, signs, cycles, drive, curvature, and schedule shape.
2. **Commit an opportunity:** Q8 binds the full schedule and permits partial execution inside its domain.
3. **Handle invalid execution:** Q9 and Q12 record overexecution and duplicates without issuing invented value.
4. **Compose time:** Q10 and Q11 distinguish undriven telescoping from driven residuals.
5. **Protect shared sources:** Q13 and Q14 expose split over-credit and enforce the conservative micro-step rule.
6. **Respect physical permission:** Q15–Q17 test P1C domain, negative permitted actions, and observational identity.
7. **Protect meaning:** Q18–Q21 enforce information, cost, epoch, and serialization boundaries.

The next gate begins only after this lifecycle is coherent. Otherwise a later behavioural failure could be caused by a malformed schedule, replayed epoch, or double-counted cost rather than the actor's selection objective.

57.13 A reader's reconstruction of E1

To reconstruct a worked quote without reading Python:

1. write the two endpoint states and local drives;
2. calculate z_i and z_j ;
3. evaluate both local penalties at z ;
4. apply $-\Delta tq$ at the source and $+\Delta tq$ at the destination;
5. evaluate both penalties again;
6. subtract the after penalties and $C_a(q)$ from the before penalties;
7. separately calculate the linear diagnostic and compare, but never replace the exact value.

This sequence is the conceptual core of the implementation. Hashes, epochs, and registries make the same calculation auditable in a multi-action program.

Chapter recap

Gate 1B calibrated the quote layer rather than testing an economy. Its fixtures cover zero, beneficial and damaging actions, cycles, drive, curvature, partial execution, violations, split-action over-credit, P1C boundaries, information boundaries, epoch integrity, and serialization. Passing results support implementation conformance at declared points; they do not answer behavioural alignment.

Exercises

1. Explain the phantom 12 in the split counterexample without using the word “bug.” Which mathematical property creates it?
2. Why does Q15 need an action that is both physically forbidden and economically attractive under the raw quote?
3. Select three Q-groups and classify each as equation, protocol, or information-boundary evidence.

From adversarial search to service alignment

After calibrating the quote, the project asked a more dangerous question: can an actor select locally profitable exact quotes while producing persistent physical harm? Gate 1C searched for such behaviour. It did not find the registered exploit, but it produced a different surprise – the quote-maximising arm served substantially less than the ordinary P1C baseline while still earning positive EBU. Gate 1D was designed to understand that surprise.

58.1 Gate 1C as a falsification search

Gate 1C translated twelve released 5-by-5 layouts into the V2.9/P1C substrate. For each layout it ran three production arms:

Arm A. P1C physical baseline, no EBU-guided selection.

Arm B. The same P1C actions with observational exact quoting.

Arm C. A production-local adversary that evaluates its permitted local menu and chooses the action with the greatest exact quote.

A separate red-team oracle used global harm as an external search objective. It was labelled as a testing instrument, never as information available to a production actor. The frozen search used depth 10, width 40, a 20-tick tail, two quantity fractions, a two-actor coalition, and layout seeds 0 through 11.

58.2 The registered exploit predicate

The standing falsifier required positive cumulative exact EBU together with persistent harm relative to a no-action baseline. This is a conjunction: profit

alone is not the exploit, and harm alone is not the exploit. The tail condition requires persistence rather than one noisy tick.

No registered production run or oracle search satisfied the predicate. Historical positive controls did. The released V2.6 naive ledger reproduced an exploit with net EBU +15.35 and positive persistent harm. The V2.6 seed-0 guarded standing falsifier reproduced +260.13 with five of five sources dead. These controls show that the search machinery could find known failures.

Under P1C and the exact quote, all 36 production runs had zero reserve crossings, zero Allee crossings, zero dead sources, and zero physical overuse. The old harm channel was physically blocked by the permission layer.

58.3 The encouraging result and its boundary

It is tempting to summarise Gate 1C as “no exploit, therefore safe.” The artifact explicitly forbids that claim. The search covered particular layouts, seeds, menu, depth, width, and predicate. It was not a proof of security.

Three limitations were especially serious:

1. all 36 production runs eventually left the physical domain because standing demand drained unconstrained consumer cells below zero;
2. the inherited unit timestep gave $r_{\Delta t} = 2.283 > 1$, outside the discrete certificate;
3. the no-action-relative harm predicate could not detect degradation relative to an actively serving baseline.

The third limitation revealed the central surprise.

58.4 The metrics answer different questions

Gate 1C reports several columns because no single one defines harm.

Cumulative EBU. Sum of settled exact action lines in the observational research account. It is not a wallet balance in this gate.

Final burden. The global diagnostic V at the last recorded state. It can compare represented field condition but does not directly measure service.

Viability percentage. Fraction of cells inside their declared viable bands. Two worlds with the same percentage can contain different cells outside the band.

Served quantity. Useful demand met before physical invalidity. It does not by itself reveal source depletion.

Reserve and Allee crossings. Entry into named source-danger regions.

Physical overuse. Accepted withdrawal beyond the P1C budget, which remained zero.

Domain exit. A state leaving its declared physical interval. This limits interpretation of all later ticks.

Persistent harm. A registered tail comparison with a baseline and margin, not a synonym for any one bad final number.

Arm C can therefore have positive cumulative EBU, zero source crossings, worse service, higher final burden, and a false persistent-harm predicate at the same time. The statements are not logically inconsistent; they refer to different comparisons. The scientific task is to decide whether the predicate captures the failure mode the gate claims to search for.

58.5 Why the no-action baseline was attractive

A no-action baseline provides a clear counterfactual: did acting make the world worse than doing nothing? It is useful for detecting destructive action in a static world. In a driven needs system, however, doing nothing can be extremely harmful because demand continues. A serving policy can degrade substantially and still remain better than total inaction.

The lesson is not that no-action baselines are always wrong. It is that baseline choice is part of the scientific claim and should be preregistered alongside a positive control showing what the predicate can detect.

58.6 Positive EBU with worse service

Averaged across twelve layouts, arm A served 247.3 units, ended with burden 1466.1, and final viability 50 percent. Quote-maximising arm C earned mean

EBU +243.60, but served only 166.0, ended with higher burden 1561.7, and final viability 42.3 percent.

The registered predicate still reported no exploit because both serving arms were far better than doing nothing. In symbolic terms, if no-action harm is very high, both A and C can satisfy

$$\text{harm}(A) < \text{harm}(C) < \text{harm}(\text{no action}), \quad (58.1)$$

so a test comparing only C with no action misses the fact that C is worse than A.

The predicate was not rewritten after seeing the result. Instead, the surprise created two open questions: a serving-baseline predicate, and whether the apparent service loss came from the quote objective or from unequal action capability.

58.7 The capability confounder

Arm A could accept many outgoing edge flows from one source during a tick. To avoid the split-action phantom-credit problem, arm C proposed only one quoted action per source per micro-step. Observational arm B could quote only a small minority of all accepted flows for the same reason. Thus the comparison changed two things at once: selection objective and available action capability.

If A serves four destinations simultaneously while C chooses one, lower service may be caused by the restriction even when C chooses the best possible single edge. Gate 1C could not separate those explanations.

58.8 Seed 0 as a concrete comparison

The standing seed-0 layout makes the ambiguity tangible. Plain P1C arm A served 120.57 units and ended with burden 3268.51. Observational arm B produced the same physics exactly and settled only one quotable action for +0.309 EBU. The quote-maximising arm C settled 140 actions for +196.564 EBU, served 108.28, and ended with the higher burden 3354.61. None crossed a reserve or killed a source.

The result contains both progress and concern. P1C prevented the catastrophic source-death exploit seen in the historical V2.6 control. Yet maximising

available quotes did not reproduce the serving baseline. Because C also had fewer action opportunities, the trace cannot assign that gap uniquely to the quote objective.

This is why a good experiment does not stop at “falsifier false.” Secondary metrics can reveal that the predicate asks too weak a question.

58.9 Coverage is part of capability

Observational arm B could quote only 0 to 43 actions per layout while hundreds of P1C edge flows occurred. The remaining flows were labelled `unquotable_multi_edge`; they were not silently assigned zero EBU. The small coverage fraction follows from the shared-source accounting restriction, not from missing trace data.

Gate 1D later constructs full quote coverage by restricting every compared arm to the same one-action menu. That removes one confounder but also changes the physical capability being compared with full P1C. A-versus-B and B-versus-D are therefore both required.

58.10 Gate 1D redesigns the service instrument

Gate 1D introduced a bounded service wrapper. Demand was removed from the unconstrained drive and applied after transport:

$$\text{available}_j = x_j + \Delta t \left(u_j + \sum_{e \rightarrow j} \eta_e q_e^{\text{acc}} - \sum_{e \leftarrow j} q_e^{\text{acc}} \right), \quad (58.2)$$

$$\text{service}_j = \min(\Delta t d_j, \text{available}_j), \quad (58.3)$$

$$\text{unmet}_j = \Delta t d_j - \text{service}_j, \quad (58.4)$$

$$x_j^{n+1} = \text{available}_j - \text{service}_j \geq 0. \quad (58.5)$$

This wrapper prevents phantom consumption. If only 0.6 stock units are available and demand is 1, service is 0.6 and unmet demand is 0.4. The state ends at zero, not negative 0.4.

The wrapper is a registered semantics extension and lies outside the original D0 theorem. Its non-negativity and ledger properties are tested; a general constrained theorem remains open.

58.11 Four capability-matched arms

Gate 1D used:

- A: full P1C capability;
- B: restricted P1C using a non-EBU force selection;
- C: the same restricted physical action plus observational quote;
- D: the same restricted menu with exact-quote-greedy selection.

B is the primary baseline for D because both may select the same number of actions from the same menu. A-versus-B measures the price of the restriction. B-versus-C tests observational identity. B-versus-D asks whether replacing force ranking with exact-total-quote ranking harms service under matched capability.

The study used seven small worlds, two certified timesteps per world, four arms, 200 ticks, and a 50-tick burn-in: 56 runs in total. All paired timesteps satisfied $r_{\Delta t} \leq 1$.

58.12 What Gate 1D observed

Bounded service removed every domain exit: zero of 56 runs had a negative state. Service never exceeded available stock or demand, unmet demand closed exactly, physical overuse was zero, and the maximum ledger residual was 8.77×10^{-15} .

B and C were byte-identical in all fourteen world-timestep pairs. B and D also selected the same physical action at every tick. Their post-burn-in service was identical, with zero service-alignment failures. D accumulated positive EBU, but the quote and force rankings agreed on the only active choice.

This was a cleaner result than Gate 1C, but it contained another instrument problem: every source had at most one outgoing edge. The one-action restriction could not bind. A and B therefore had identical capability, and the study could not measure the very confounder it was designed to quantify.

58.13 The one-ULP diagnostic defect

The first Gate 1D rendering labelled four W2 conservative runs as reserve crossings because the minimum stock was 7.999999999999999 against reserve 8. The difference was one unit in the last place, approximately 8.88×10^{-16} . The crossing counter used a strict comparison without the registered scale-aware tolerance.

The original attempt was preserved. Under a narrow new authorization, Gate 1D-A changed only the diagnostic predicate and generated a second rendering. The uncompressed traces of all 56 runs were byte-identical. Only four crossing labels, their derived outcome classes, and associated diagnostic flags changed. No material reserve crossing remained, and the B-versus-D result stayed exactly the same.

This correction is a model of how to repair labels without pretending the first attempt never happened.

58.14 What was learned before O14

Gate 1C showed that P1C blocked the historical source-destruction exploit but also revealed a service comparison confounded by capability, domain exits, uncertified time, and a blind predicate. Gate 1D removed the domain and timestep confounders, implemented a serving baseline, and found no misalignment under matched capability. Yet its worlds did not contain genuine destination choice, so it could not test whether force and exact quote rank several active outgoing edges differently.

The next study therefore needed sources with two to four simultaneously active outgoing edges, a restriction that demonstrably binds, and worlds where B and D are analytically predicted to choose different destinations. That study was O14.

Chapter recap

Gate 1C found no registered persistent-harm exploit, but its quote-maximising arm served less than the P1C baseline and exposed weaknesses in domain semantics, timestep certification, and the outcome predicate. Gate 1D bounded service, matched capability, and found no service difference, but

its one-out-edge worlds made genuine choice impossible. Gate 1D-A corrected only four floating-point labels. The sequence shows experiments refining instruments rather than moving directly from one headline to another.

Exercises

1. Draw the inequality that makes the no-action predicate blind to C's degradation relative to A.
2. Why is B, not A, the primary baseline for D in a capability-matched test?
3. Explain how the byte-identical Gate 1D-A traces distinguish a diagnostic repair from a changed scientific trajectory.

What the historical evidence actually establishes

A research programme becomes more credible when it preserves the conditions under which it failed or produced a flat result. Replacing every older chapter with a success-oriented summary would lose that information. This edition therefore retains selected original evidence chapters as historical source pages and adds a careful reading guide.

59.1 Read a result together with its model

The historical readings in Chapters 58 and 59 preserve the original account of D1–D10, Gate 1B, adversarial/service-alignment work and O14. The old potential, regeneration, preservation rules and selectors remain attached to those accounts. Their numerical results were not recomputed under the new formula.

The appropriate question is not “Did EBU pass?” without qualification. Ask which equation, physical domain, comparison, horizon and falsifier were tested. A test of quote-code conformance and a test of service alignment answer different questions even when both involve the same numerical quote.

59.2 Why the old distinctions still teach useful lessons

The original soft-versus-hard preservation comparisons distinguish a penalty from a permission boundary. That lesson survives a change in potential: assigning a large burden does not physically prevent an action. The observational-quote comparison distinguishes logging from influencing the plant. The adversarial examples ask whether a locally closing account can still coexist with unrepresented or persistent harm.

These are methodological lessons. They are not evidence that the Gaussian capacity policy has passed analogous tests. A new actor, menu or state representation changes the tested system and requires its own registered evaluation.

59.3 O14 and a later finalized gate are different records

The original O14 chapter preserves its own capability and flat-outcome analysis. It should not be relabelled as Gate 1D-C or as Gaussian evidence. Some older analytical documents also retain operational status text from before the later gate was completed. Their mathematical content can be useful without treating their historical status paragraphs as current.

The later committed Gate 1D-C manifest records a finalized study [8]. Its execution SHA is `8793bda`; its inventory contains thirty registered runs and six thousand trace rows. The recorded stdout specifies three worlds, two timestep settings and five arms, with two hundred ticks per run. Its reported outcome counts are twenty `physical_impossibility`, eight `safe_rationing_physical_scarcity`, and two `preserve_and_serve`; it reports no fired falsifier.

Those are narrow historical source facts, not thirty independent proofs of the accounting theory or a long-run Gaussian result. The revised books do not rerun, tune, extend or reinterpret the frozen campaign. The source manifest and original scientific protocol govern its actual claims.

59.4 A comparison can be invalid even when its numbers are real

An arm allowed several simultaneous deliveries has a different capability set from an arm restricted to one action per configured source. Such an arm can be informative as a capability benchmark, but it is not automatically a matched test of the contribution of the EBU selector. Historical comparison restrictions must remain visible when reading an outcome table.

Likewise, a new Gaussian arm with a different physical menu cannot claim an advantage of capacity accounting alone unless that difference is part of the registered estimand. The artifact may be genuine while the proposed cross-model inference is unsupported.

59.5 What preservation of the historical readings means

The readings are reproduced source material, not a fresh independent audit of every archived run. Its figures and statements keep their original scope and publication-era wording. The new source manifest identifies the exact PDF pages. Readers can return to the full preserved originals for cross-references outside the extract.

This treatment avoids two opposite errors: discarding useful history because the active model changed, and making the new model appear validated by repainting that history. Neither is necessary for a coherent series.

Chapter recap

Historical experiments remain evidence about their own registered systems. They guide questions and expose pitfalls for new research, but they cannot answer an unexecuted Gaussian comparison by analogy alone.

Historical reading H59: O14 and a flat outcome

Historical model; not Gaussian evidence

This reading preserves the original destination-choice account, including its limitations and flat result. It predates the later finalized Gate 1D-C record discussed above. Do not combine its run counts or conclusions with that later campaign. Original Part III, PDF pages 101–110. Internal chapter, section, equation and page references retain their original meaning. The new running folio belongs to this integrated volume.

What to carry forward. A flat result is informative only relative to a design capable of detecting the proposed difference. Preserve the original observation, distinguish implementation conformance from efficacy, and ask whether the metric and world discriminate the competing claims. This does not license tuning a frozen study after its results.

O14: genuine destination choice and a flat outcome

O14, also called Gate 1D-B, was designed around the limitation left by Gate 1D. Every registered source now had between two and four active outgoing edges. The study could observe real destination choice, a genuinely binding multi-action restriction, per-unit-versus-total quote disagreement, and aggregate shared-source over-credit.

Its headline result was not the expected decisive separation. All six worlds were non-discriminating under the registered primary outcome. Falsifier F13 fired. Understanding why requires reading behaviour, service semantics, and the outcome instrument together.

59.1 The six worlds as six questions

Each deterministic world contains one configured source and a static graph.

- W1: efficiency split.** Can route efficiency change choice, and can a stock-blind delivered-volume heuristic neglect a vulnerable destination buffer?
- W2: severity split.** Can force and exact total quote provably prefer different destinations from the first tick?
- W3: volume split.** Can a smaller material action produce more EBU than a larger delivered-volume action?
- W4: budget binding.** What happens when four simultaneous outgoing requests share a source only 0.6 stock units above reserve?
- W5: reversal.** Will the preferred edge change over time as local deficits move?
- W6: infeasibility.** Will the system expose demand greater than useful reserve-limited capacity as explicit unmet need?

Static algebra established non-vacuity before execution: at least two edges had positive raw flux at the initial state of every world, W4’s aggregate budget had to bind, W6 demand exceeded the useful maximum, and W2 contained a predicted B-versus-D destination divergence.

59.2 What would count as informative in each world

The world names describe mechanisms, but each also has a specific observable.

World	Designed tension	Key evidence to inspect
W1	high- versus low-efficiency destinations	service by destination and S-arm buffer exhaustion
W2	force prefers severe-small while quote prefers mild-large	first selected edge and accumulated B/C/D quote totals
W3	delivered volume and exact burden improvement disagree	which heuristic under-serves and whether less material earns more EBU
W4	four requests share a near-reserve source	σ_A , budget utilisation, action count, and A/B post-burn-in service
W5	local ranking reverses as destination states evolve	selected-edge switch counts and timing
W6	demand exceeds useful maximum	unmet demand, zero reserve crossings, and physical-impossibility class

This table prevents a common mistake: judging every world only by total EBU. W4 is primarily a capability test; W5 is an adaptation test; W6 is an infeasibility test. The same trace fields have different evidential roles in each fixture.

59.3 Five arms and the primary comparisons

The arms were:

- A. Full simultaneous multi-edge P1C. It measured physical capability and recorded an aggregate quote diagnostic, but settled no EBU.
- B. Restricted matched non-EBU force selector. This is the primary baseline.
- C. The same physical selector as B with observational quotes.
- D. Restricted exact-total-quote greedy selector. It rests unless the best exact quote is strictly positive.
- S. A restricted local service-volume comparator, intentionally stock-blind and used only for secondary attribution.

Every restricted arm consumes the same candidate menu. Selectors do not build their own menus. Request shaping makes the selected accepted quantity equal to the quantity executed by the unchanged P1C step. These details prevent a hidden capability difference from returning through implementation.

59.4 Why total quote, not EBU per unit

The registered heuristic ranks complete candidate actions by $\Delta_e(q^{\text{acc}})$. It does not divide by quantity. Per-unit ranking answers “which unit is most efficient?”; total ranking answers “which available action produces the largest positive field improvement?” Those are different objectives.

O14 observed 4,798 candidate-ranking ticks where per-unit and total rankings diverged. The selector nevertheless followed the exact total rule. This makes the distinction operational rather than a footnote.

Under the registered costs and certified timesteps, the exact schedule was strictly increasing over the feasible four-level quantity menu in every world. Quantity misalignment was therefore structurally excluded in this design; the scientific choice lay mainly between destinations. The continuous schedule vertex was recorded only as a diagnostic and never used by the selector.

59.5 One restricted-arm tick from menu to service

At the beginning of a restricted-arm tick, the implementation performs the following sequence:

1. freeze the world state and calculate local drives;
2. calculate the authoritative P1C budget of the configured source;
3. calculate raw D0 flux on every outgoing edge;
4. form four registered quantity fractions for each edge, capped by the same budget logic;
5. present the identical candidate dictionary to B, C, D, and S;
6. let the arm selector choose one candidate or rest;
7. shape the active world so unchanged `p1c_step` executes the selected accepted quantity exactly;
8. apply bounded destination service and record unmet demand;
9. calculate quotes and global diagnostics only in their registered layers;
10. append one complete tick record.

Arm B orders candidates by loss-aware force, then accepted quantity, with identifier tie-breaking. Arm D evaluates each complete exact quote and chooses the largest strictly positive value, also using identifiers only for exact ties. Arm S maximises useful delivered volume toward a demanding destination without reading destination stock. Arm C selects exactly as B and merely observes the quote.

A test drops one candidate from one arm's menu and requires detection. Another test constructs a tie in both identifier directions. These controls ensure that the observed W2 choice divergence comes from the registered score, not from menu construction or iteration order.

59.6 Why selected quantity had to equal executed quantity

If the selector quotes one quantity but P1C later rescales it differently, the study would compare a planned score with another physical action. O14 avoids that confounder by using the same arithmetic as P1C when constructing candidates and by shaping the request so the selected q^{acc} is the executed value.

Tests prove bit-identical equality on all W1 candidates at both timesteps. An unscaled negative control executes a different quantity and is detected. The identity is also asserted on every official tick before the run can complete.

59.7 The registered study and execution integrity

The Cartesian product was six worlds, five arms, and two timesteps: 60 runs. Every run lasted 200 ticks with a 50-tick burn-in and 20-tick persistence window. The trace contains exactly 12,000 ordered tick rows. Every row carries the canonical plan hash. No seed, retry, filter, subset, or rerun was used.

All scientific numbers were finite. There were zero domain failures, no negative state, no material reserve or Allee crossing, and no dead source. Aggregate service, EBU, unmet demand, and crossings reconstruct from trace rows within 10^{-9} ; the maximum ledger residual was approximately 1.03×10^{-14} .

The official scientific execution occurred only after the earlier runner self-match defect had failed closed, been recorded, corrected, re-tested, and separately authorized.

59.8 What behaviour actually differed

B and C were physically identical in all twelve world-timestep pairs, with maximum state difference exactly zero. This is strong evidence that observational quoting remained inert.

B and D were not behaviourally identical. In W2 they selected different destinations from tick 1 at both timesteps. In W5 D switched selected edges 37 and 33 times, while B switched 39 and 35 times. D's accumulated EBU differed from C's in W2, confirming that D did not merely quote B's actions.

Arm A acted simultaneously on multiple edges during 2,399 ticks. W4 and W6 had binding aggregate budgets, with very small scaling factors at their most constrained ticks. The one-action restriction was therefore real.

These observations are crucial because F13 did not fire from absence of choice. Choice and capability differences were present; the primary service outcome did not respond to them.

59.9 The primary null and F13

The primary discriminator marked a world as informative if either A-versus-B capability cost was materially non-zero or D-versus-B post-burn-in service ratio differed materially from 1 at either timestep. None did. The list of discriminating worlds was empty, so F13 fired exactly as registered.

D and B had equal post-burn-in service within the preregistered tolerance in all twelve comparisons. One W6 conservative-timestep ratio was 0.9999999999999999, one ULP below one and inside that tolerance. A and B also had zero post-burn-in capability cost everywhere, including W4 and W6 where the restriction bound. H3, the prediction of a non-zero capability cost somewhere, was not supported.

This is a negative result about the sensitivity of the chosen world-and-outcome instrument. It is not proof that quote-greedy selection and the restriction are harmless.

59.10 How destination buffering creates a plateau

The bounded service wrapper lets a destination store delivered stock. For a non-exporting destination j ,

$$\text{available}_j(t) = x_j(t) + \delta_j(t), \quad (59.1)$$

$$\text{service}_j(t) = \min(\Delta t d_j, \text{available}_j(t)), \quad (59.2)$$

$$x_j(t+1) = \text{available}_j(t) - \text{service}_j(t), \quad (59.3)$$

where $\delta_j(t)$ is efficient delivery during the tick.

If available stock never falls below tick demand during the measurement window, service is exactly the demand cap:

$$S_j(W) = |W| \Delta t d_j. \quad (59.4)$$

Inside this buffered region, changing the delivery schedule does not change cumulative service. Service responds only when a running buffer reaches zero.

The equivalent running condition is

$$x_j(t_0) + \sum_{s=t_0}^t (\delta_j(s) - \Delta t d_j) \geq 0 \quad \text{for every } t \text{ in the window.} \quad (59.5)$$

B and D produced different delivery patterns but stayed inside this inequality. The service functional was flat over the region they occupied.

59.11 Temporal multiplexing

The restriction limits simultaneous actions, not time-averaged throughput. B can serve destination 1 on one tick, destination 2 on the next, and return before either buffer empties. In W4, B selected the four destinations in an exact 50/50/50/50 round-robin. The per-selected-tick useful delivery rate was 2.4075, slightly greater than the total demand rate 2.4. One action at a time was enough to maintain all four stored buffers.

Both B's force and D's exact quote are deficit-seeking. When a destination is not served, its stock falls, its deficit becomes more urgent, and it tends to win a later selection. This feedback behaves like a longest-deficit-first polling loop. The worst B buffer margin in W1 was only 0.368 stock units, so the plateau was thin rather than vacuous.

59.12 Walking through a W4 multiplexing cycle

W4 has four outgoing destinations, each demanding 0.6 stock units per unit time. Arm A can act on all four during a tick. Arm B chooses one. At the initial selected edge, useful delivery rate is approximately

$$\eta J = 0.9(2.675) = 2.4075, \quad (59.6)$$

while total demand rate across all four destinations is

$$4(0.6) = 2.4. \quad (59.7)$$

Suppose B visits destinations 1, 2, 3, and 4 in four successive ticks. A selected destination receives slightly more than four ticks of its individual demand rate, so it can bank enough stock to wait through the other three selections. The trace shows an exact 50/50/50/50 selection count across 200 ticks and 199 switches. The restriction binds instantaneously but not cumulatively at the service read-out.

This explains a result that initially looks contradictory: W4 can have $\sigma_A < 1$, one action per tick in B, and still zero A-versus-B service cost. Instantaneous capability and long-window demand satisfaction are different quantities.

59.13 Hypotheses and falsifiers as two ledgers

The O14 summary reports hypotheses H1–H10 and falsifiers F1–F15 separately. A supported hypothesis records a predicted pattern such as B/C identity, restriction binding, W5 switching, or explicit W6 scarcity. A fired falsifier records a condition declared capable of invalidating or limiting the study.

Nine hypotheses were supported; H3, non-zero capability cost somewhere, was not. F13 fired because no primary world discriminated. The remaining falsifiers did not fire: menus stayed matched, all worlds had several active outgoing edges, information boundaries held, timesteps remained certified, no material reserve harm occurred, and no forbidden post-hoc world was added.

“Not fired” does not mean universally impossible. It means the registered condition was absent from this completed study.

59.14 The S arm reaches the boundary

The stock-blind S comparator served W1 differently. It under-served by 4.315 and 5.000 post-burn-in units at the two timesteps and was classified as distributive-or-policy under-service. S is not part of the F13 primary predicate, so this result does not contradict F13.

It strengthens the buffering diagnosis. The service instrument could respond in the registered worlds when a policy drove a buffer to zero. B and D stayed on the plateau because they tracked deficit; S left it because its score ignored destination stock.

Figure 59.1 should not be read as “all policies are equal.” The left panel shows equality of one saturated outcome for B and D. The right panel shows that another registered heuristic left the saturated region. Behavioural traces are needed to understand why.

59.15 The aggregate quote diagnostic

Arm A permitted simultaneous withdrawals from one nonlinear source. Independent per-edge quotes are not conservative because each line treats the source as if no other withdrawal occurs. Across 2,399 multi-action ticks, the naive independent sum was +3910.750 while the exact group quote was

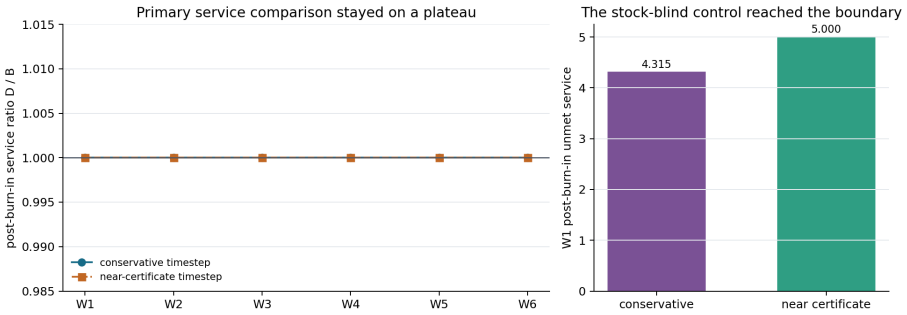


Figure 59.1: *O14 outcome instrument. Left: D/B post-burn-in service remained equal within the registered tolerance in every world; one plotted ratio was one ULP below one. Right: the stock-blind S control produced unmet service in $W1$, showing that the service boundary was reachable.*

+3894.460. The measured over-credit was 16.289.

Nothing was settled or allocated in arm A. The diagnostic quantifies O3 rather than solving it. Proportional, marginal, source-group, and Aumann–Shapley style allocations each introduce incentive and fairness questions. An arithmetic split that closes is not automatically strategy-proof or scientifically justified.

59.16 Honest infeasibility in W6

W6 demand exceeded useful reserve-limited capacity. All ten W6 runs recorded post-burn-in unmet demand, zero reserve crossings, and the outcome class physical impossibility. Near the reserve the source scaling factor reached zero at the near-certificate timestep.

This is the correct meaning of a controller saying no. The system does not claim that the need is unimportant. It records that the present local field and infrastructure cannot meet it without violating the declared source boundary. That information can motivate another source, storage, route improvement, or capacity construction.

59.17 What O14 changes and what it cannot close

O14 directly measured multi-out-edge capability and genuine destination choice. It found no reserve harm, no B-versus-D service-alignment failure,

and a clear aggregate over-credit diagnostic. It also revealed that the primary outcome was analytically flat over the relevant buffered regime.

O11 therefore remains open. Zero difference on a flat functional is not evidence of alignment outside that plateau. A future outcome-discrimination study needs a positive control and worlds in which delivery timing can exhaust one policy's buffer but not another's. Candidate mechanisms include smaller initial buffers, perishable destination stock, pulsed need, useful flow below multiplexed demand, or service drawn directly from current delivery.

Any such world is a new preregistered study, not a correction to O14. If static algebra cannot construct a discriminating region without post-hoc search, the honest outcome is to stop that line and proceed only with the independent accounting-theory programme.

Chapter recap

O14 created genuine multi-edge choice and binding restrictions. B and D behaved differently but delivered post-burn-in service equal within the registered tolerance because banked destination stock placed the outcome on a plateau. F13 fired as registered. The S control showed that the boundary was reachable, and the arm-A diagnostic measured 16.289 of naive shared-source over-credit without settling it. The result narrows the next question; it does not prove harmlessness.

Exercises

1. Use the running-buffer inequality to explain how alternating two destinations can match simultaneous service.
2. Why does S's W1 under-service strengthen, rather than contradict, the F13 interpretation?
3. Propose a positive-control world in which one selection policy exhausts a buffer and another does not. State which parameter is changed and why the choice is made before running.

Measurement, calibration and the loss boundary

A perfectly implemented equation can return a misleading number if its inputs do not represent the physical event. This is why a book about code must return to measurement: what quantity changed, where was it measured, what uncertainty remains, and who justified the reference against which the change is evaluated?

60.1 References and scales need an interpretation

The Gaussian reference x_i^* describes a declared functional condition. The scale σ_i describes how deviations are weighted. Neither is automatically an observed mean, a measurement uncertainty or a just allocation. A physical unit conversion rescales stock, reference and scale together; changing only the scale is a substantive change in valuation.

An empirical programme would need a parameter record explaining the functional evidence, measurement domain, validity period and uncertainty of each declaration. That work is not supplied by choosing convenient numbers for a textbook example. The examples here are explicitly illustrative.

60.2 Measurement and account boundaries

Source withdrawal, useful destination delivery and loss are different measurements. A pipeline meter at the source does not establish delivery at the destination. A receipt should retain those distinctions even in a first lossless model, where the declared values coincide.

Physical time matters too. An endpoint observation does not necessarily identify the intermediate path of several simultaneous actions. If the account uses endpoint interpolation, record that convention. If it claims a model-realized constant-rate path, identify the physical assumption and the evidence required to use that reading outside the model.

60.3 Conservation does not choose a valuation

An explicit loss coordinate can close total physical accounting. It does not automatically become an argument of the homeostatic potential, and excluding it from that potential does not automatically authorize an additional charge. The represented consequence, process burden and physical carrier must be reconciled without double counting.

The historical Definition 6.4 discusses a dissipated-stock charge with a rationale that no destroyed-stock coordinate is present [4]. Later candidate work includes a loss ledger coordinate. The statements “absent from state” and “present in a ledger but absent from V ” are not literally identical. A broader loss-aware rule needs an explicit authority/wording reconciliation; this book does not silently choose one.

The active first domain has $\eta = 1$ and $C = 0$. Loss-aware runtime is not certified by that choice. An aggregate conservation residual can hide an invalid sink update if the declared conservation profile is not enforced in the actual transition. Carrier-specific validation remains necessary.

60.4 Do not invent burdens to complete a story

Fuel conversion, electricity delivery, pumping and water transfer can be represented as linked atomic transformations when they are in scope. Their existence is not a reason to add an arbitrary extra cost to every downstream action. Each physical effect needs a declared carrier and one justified valuation location. The chain must neither disappear inside a mega-action nor be charged repeatedly under several names.

Monetary price, inconvenience, sanctions and physical process burden also have different meanings. A payment may explain how an essential service is financed without becoming a physical term in the EBU equation.

60.5 Governance and privacy are not residual variables

Selecting references, identifying owners and deciding who may access measurements have institutional consequences. A model that calls itself local does not automatically settle consent, privacy, auditability or access. Those questions require an explicit governance layer and appropriate domain review; they cannot be solved by adding a coefficient to V .

The practical benefit of keeping records distinct is transparency. A necessary action can have negative represented field value without the account declaring that its recipient should be denied service. The limited experimental capacity

policy must not be presented as a ready rule for real-world access to essential resources.

Chapter recap

Measurement and calibration determine what a precise calculation means. The first lossless world allows focused mathematics while leaving, rather than pretending to solve, the broader loss and institutional boundaries.

Historical reading H60: What a field account must actually observe

Historical model; not Gaussian evidence

The original measurement chapter develops the practical difference between a mathematical coordinate and an observed physical quantity. Its examples remain scoped to their own plant. New loss carriers or new calibration rules require new declarations rather than the relabelling of an old sensor field. Original: Part III, PDF pages 111–118. Internal chapter, section, equation and page references retain their original meaning. The new running folio belongs to this integrated volume.

What to carry forward. A hash authenticates a record's bytes, not the truth of a reading. The Gaussian potential still requires credible stocks, references and scales. With that distinction in place, the final chapters can state what long-run research would need to establish without claiming those results already exist.

What a real field account would have to observe

The executable studies use exact model states. A real system would not receive a perfect vector x from nature. It would assemble resource-specific accounts from sensors, inventories, ecological surveys, maintenance records, route telemetry, and certified models. This is not a minor engineering detail. The meaning of an EBU quote depends on what the field variables measure, how uncertain they are, and who can challenge the calibration.

60.1 There is no single planetary EBU sensor

The phrase “the EBU field” can suggest one universal instrument analogous to a thermometer. The model is more structured. Each resource account declares a state variable and its units: usable water volume, available medicine doses, soil functional capacity, grid flexibility, repair capacity, or another measurable quantity. The global state is a collection of those local accounts.

An EBU quote is meaningful only relative to that declared model. A water action cannot be calculated from a biodiversity sensor alone. A medicine route cannot be evaluated from fuel consumption while ignoring whether usable medicine arrived. Cross-resource combination requires explicit conversion or weighting rules and their own scientific justification.

60.2 What must be measured at a cell

At minimum, a mature local account may require:

- present stock or functional capacity x_i and its unit;
- the spatial and temporal support of the cell;
- viable-band references L_i and U_i ;

- a separately certified reserve R_i^{eff} where preservation is claimed;
- regeneration, decay, supply, demand, and leakage rates;
- uncertainty bounds and calibration date;
- the equation and parameter version under which a quote is produced.

The word *stock* should be read functionally. A wetland's useful capacity may not be captured by water volume alone. A hospital's medicine stock depends on expiry and cold-chain integrity, not merely the number of boxes. The measurement model must represent the capacity relevant to the declared need.

60.3 What must be measured along an edge

An edge requires more than geographic distance. Its account may include:

- source and destination identifiers;
- the resource or service carried;
- measured withdrawal and delivered quantity;
- route efficiency η_e and how it was estimated;
- mobility or capacity coefficient M_e ;
- activation threshold θ_e ;
- time, duration, and any process burden not already represented in the endpoint state change;
- provenance of the measuring devices and route model.

Distance can influence energy use, loss, delay, and infrastructure wear, but no inverse-square law follows automatically. Gravity depends on masses and geometry through a specific physical law. EBU depends on the measured state transformation and the declared route. Two equally distant destinations can have different efficiency, urgency, buffers, and process burden.

60.4 Needs must be represented without turning desire into physics

The current bounded wrapper uses a demand rate and records service and unmet demand. A future needs layer must distinguish essential need, preferred service, and optional desire through explicit social and domain definitions. EBU itself should not infer moral worth from price or willingness to pay.

Consider three requests:

1. an antibiotic needed by a remote patient;
2. a replacement bearing needed to keep a water pump operating;
3. a luxury object requested for status.

Their physical routes can all be quoted. Institutional priority may differ because the first two protect health and essential infrastructure. That priority layer should be named rather than hidden in the physical field weights. A large negative EBU for the antibiotic does not prohibit treatment. It says that meeting this genuine need presently imposes a large verified burden, which society may support while investing to lower future cost.

60.5 Uncertainty has units

Suppose a reservoir sensor reports 13.0 units with error at most 0.2, and the forecast drive is 0.5 units per hour with downward error at most 0.1. Those margins enter the robust budget as $\varepsilon_x = 0.2$ and $\varepsilon_u = 0.1$. They are not chosen because a simulation looks unstable; they come from calibration and model validation.

Other uncertainties may not fit a simple interval. Regeneration can depend on temperature, age structure, or delayed ecological feedback. Route efficiency can vary with weather. A mature system may require interval quotes, confidence sets, or robust optimisation. Gate 1E is reserved for delay and uncertainty questions; the current exact-observation studies make no claim about them.

60.6 Latency and stale state

An action quoted from a ten-minute-old stock can be unsafe even if the equation is exact. Another actor may have withdrawn material during the delay. A quote epoch therefore needs an expiry rule and a state commitment. The shorter the permissible latency, the stronger the infrastructure requirement.

The problem is especially important for high-frequency shared sources. In a slow forest account, daily measurement may be sufficient for one class of action. In an electrical flexibility market, milliseconds may matter. “Local” does not mean “timeless.” Spatial support and observation time are both part of the field definition.

60.7 Reserve certification is a scientific institution

P1C can preserve a declared reserve without knowing whether the reserve was ecologically correct. Certification must therefore answer:

- What loss of function is the reserve intended to avoid?
- Which data and model support it?
- How conservative is the margin?
- When must it be reviewed?
- Who can contest the value?
- What happens to old receipts after recertification?

Changing a reserve closes an accounting epoch. Historical receipts retain the equation and parameters under which they were calculated; they should not be retroactively rewritten. Otherwise an institution could create apparent EBU by changing boundaries after actions occur.

60.8 Calibration governance

Weights α_i , β_i , and χ_i determine how represented departures contribute to burden. They must be versioned, reviewable, and linked to domain evidence. A settlement operator should not be able to adjust them unilaterally to favour one participant.

A plausible governance separation includes:

1. domain scientists define measurable resource accounts and uncertainty;
2. independent calibration bodies verify instruments and parameter versions;
3. transaction infrastructure records action receipts;
4. auditors reproduce calculations from committed data;
5. affected communities can contest measurements and priorities;
6. institutional decision-makers handle essential-need guarantees and emergency exceptions without rewriting physical history.

This is not a complete constitutional design. It shows why a formula alone cannot govern measurement power.

60.9 Observability can reveal hidden inefficiency

Money is excellent at coordinating claims, but price does not reconstruct the physical history of a product. Complete action receipts could reveal recurring patterns that prices obscure:

- emergency air shipments caused by the absence of regional medicine stock;
- repeated pump replacement caused by unrepairable design;
- food loss caused by a poor cold chain;
- traffic and fuel burden caused by the location of essential services;
- simultaneous half-empty deliveries that could share one route;
- avoidable extraction caused by missing recovery and reuse systems.

This calculability may be economically valuable even before a complete E-BU economy exists. Procurement, infrastructure planning, and maintenance can use physical receipts to find interventions that reduce both burden and operational risk.

60.10 Repeated needs can reshape infrastructure

Imagine a region repeatedly importing one critical medicine by air. Each action is necessary, but the receipts show a persistent negative route burden. Several responses can reduce future EBU:

- move a buffer stock closer to the patients;
- build reliable refrigerated rail capacity;
- license regional production;
- reformulate the medicine for longer shelf life;
- coordinate orders among hospitals;
- relocate a specialised clinic only if doing so improves access without shifting burden to patients.

The system does not need to prohibit the original shipment. By accumulating comparable receipts, it identifies where investment can reduce the verified burden of meeting the same genuine need. In this sense EBU can influence structure, not only individual choices.

Demonstrating that adaptive actors actually make such investments requires a new model with capital, construction lead time, multiple actors, and learning. The current code does not contain that economy. The receipts provide the measurement substrate for testing it.

60.11 Measurement power and privacy

Detailed action accounting can become surveillance if every physical observation is tied permanently to a personal identity. A defensible system should minimise personal data while preserving physical auditability. Possible principles include purpose limitation, short retention for identifiable data, separation of identity from telemetry, aggregation where action-level identity is unnecessary, cryptographic proofs, and independent access controls.

The problem is particularly sensitive for medicine and health. The physical system may need to know that a verified dose was delivered under cold-chain conditions. It need not publish the patient's diagnosis to every network participant. "More observable" is not automatically "better governed."

60.12 Sensor failure and strategic reporting

Actors may benefit from understating source stock before receiving aid, overstating route loss to claim higher cost, or reporting a false restoration. A real design needs calibration checks, redundant measurements, tamper evidence, and penalties in a security layer.

Those penalties must remain distinct from EBU. If falsifying a sensor is assigned an arbitrary negative physical field value, the account loses interpretability. The receipt can contain both the corrected physical settlement and a separate fraud or protocol record.

60.13 A hospital measurement walk-through

Consider a rural hospital whose oxygen plant has failed. Cylinders must arrive from a city 300 kilometres away.

1. The need account declares required oxygen service for the next treatment window.
2. The source account records usable cylinder stock and its own emergency reserve.
3. The route account records loaded quantity, delivered usable oxygen, fuel and refrigeration or compression burden not already represented, and delay.
4. P1C or another physical permission layer determines what the source can release without violating its reserve.
5. The exact quote evaluates the permitted, measured shipment from the frozen local field state.
6. The service layer records oxygen delivered to treatment and any unmet need.
7. The receipt preserves all versions and uncertainty.

If the quote is strongly negative, the shipment still occurs because the need is essential and an institution guarantees access. Repeated receipts support a later decision: repair local production, maintain regional backup, or redesign the supply network. The patient is neither priced out nor blamed for the physical cost.

60.14 What the current implementation actually observes

The present studies observe exact cell stocks, exact deterministic drives, registered efficiencies, exact action quantities, and zero latency. They do not implement sensors, uncertainty distributions, calibration authorities, privacy, strategic reporting, or essential-need adjudication.

That boundary should not be expressed as “the idea is unusable.” It means the exact local core is ready for carefully bounded prototypes, while measurement and institutional responsibility require their own gates before broader authority is granted.

Chapter recap

A real EBU field would be assembled from resource-specific, versioned, uncertain measurements. Cells require stock, viable ranges, reserve evidence, and local drives; edges require measured withdrawal, delivery, efficiency, loss, and process burden. Essential needs guide institutional priority without changing physical history. Observability can reveal powerful efficiency opportunities, but it also creates governance, calibration, privacy, and manipulation risks.

Exercises and design workshop

1. Choose one resource account and define x , its unit, a plausible viable band, and one uncertainty source.
2. For the hospital oxygen example, list which facts belong in physical EBU, which belong in need priority, and which belong in security governance.
3. Explain why geographic distance alone cannot determine the EBU of a route.
4. Design one privacy-preserving receipt field and one field that should not be public.
5. Describe an infrastructure investment that could reduce the EBU of a recurring essential transaction without reducing service.

What a long-run study could show

The future study should be capable of finding that the proposed mechanism does not help. If every unfavorable outcome is reclassified as an invalid world or repaired by changing the actor, the programme ceases to test its own question. Interpretation must therefore be designed before the outcomes are visible.

61.1 A single disturbance and continuing forcing

The roadmap distinguishes a response after one declared disturbance from a continuously driven stochastic regime [7]. After one shock, forcing stops and the external audit remains fixed. The physical potential and total capacity can exchange subject to the conditional invariant. The process may recover, cycle, become trapped or refuse all available actions.

Continuing forcing changes the question. Even if physical deviation appears to stabilize, external audit and accumulated capacity may drift. A claim of stationarity must name its object: the physical state, potential, selected observables, balance distribution, or the entire augmented process. A flat-looking plot of one coordinate does not establish stationarity of all.

61.2 Boundedness is not the benefit metric

Finite conserved mass and positive fixed scales already bound the Gaussian potential on its domain. The physical-random control has that same bound. Therefore “the EBU arm did not diverge to infinity” is not a discriminating success claim for this first closed model.

Potential occupancy, time away from reference, recovery after disturbance, refusal, cycling and distribution are possible families of observables. They are not frozen metrics here. Their definitions, aggregation, uncertainty treatment and falsifiers belong to a later preregistration. Service belongs in the interpretation only if the plant actually defines and measures it.

61.3 Physical validity precedes performance

A trajectory that violates conservation or declared physical constraints cannot support a performance claim merely because its potential is low. Similarly, an invalid comparator undermines a comparative claim differently from a valid comparator that misses its own target.

It is helpful to retain raw facts separately: physical validity, each arm's measured outcomes, and any oracle or feasibility verdict. Derived labels should not force unresolved cases into success or failure by default. If a claim requires an oracle that was not resolved, report that inference as inconclusive while preserving the observed facts.

61.4 No-action is a scientifically meaningful possibility

At reference with zero genesis, the proposed account blocks nonzero-net negative-valued departures. After a disturbance, however, menu or ownership structure may still leave no affordable action. A refusal can therefore be an intended consequence of the declared rules, a reachability limitation, or a specification defect. The records must distinguish them.

Adding a free initial balance, refill, loan or target-directed rescue to avoid refusal would change the experiment. Such alternatives can be studied later, but must not be inserted after seeing an inconvenient result and reported as though the original policy succeeded.

61.5 From finite evidence to broader claims

A finite registered study can support statements about its tested regimes and uncertainty. It does not establish unlimited-horizon safety or economic validity. A theorem may establish an invariant for all permitted events, yet leave the quality of the resulting dynamics unresolved. Evidence and proof can complement one another without substituting for one another.

The strongest report includes adverse and null comparisons, actual exposure, software violations, missing data and protocol deviations. It distinguishes what was observed from the causal explanation offered for it. Where the design cannot separate explanations, the report should say so.

Chapter recap

The long-run question remains open because accounting identities constrain but do not choose behavior. An honest study needs registered observations that can distinguish mechanisms, not a success label guaranteed by geometry.

A worked audit and the next research boundary

We end by reviewing a complete illustrative record. The point is not to execute the future engine on paper. It is to identify exactly which facts a reader would need in order to check one event, and which conclusions remain unavailable even after the arithmetic is correct.

62.1 The declaration

Use two cells, reference $(10, 10)$ and scales $(2, 2)$. Begin at reference with zero capacity and zero external audit. A declared conservative forcing increment $(4, -4)$ creates baseline $(14, 6)$. Its potential is four. Two accepted child actions each transfer one unit from the first cell to the second. The aggregate increment is $(-2, 2)$ and endpoint $(12, 8)$.

This is an illustrative declaration, not a registered world. It does not select a random law, timestep, horizon or performance target. The children are fixed mathematical objects for the audit.

62.2 The physical and field checks

Total stock is twenty in every listed state and every stock is nonnegative. The accepted source withdrawal is two, within its fourteen-unit baseline. If there were another declared physical constraint, it would also need checking; the absence of such a field from this example is not permission to ignore it in another model.

At baseline the marginals are $(1, -1)$. Each child has increment $(-1, 1)$, and the aggregate curvature term is $\frac{1}{2}(-2, 2)^T H(-2, 2) = 1$. The tangent group benefit is four, so the exact group value is three. Endpoint evaluation agrees: $4 - 1 = 3$.

For the straight common path, each child attribution is $3/2$. Two standalone singleton values would instead be $7/4$ each and sum to $7/2$, overcounting the group by one-half. This discrepancy is not a floating-point artifact; it is the exact shared-field cross term.

62.3 The account check and its missing authority

Suppose, conditionally, the adopted policy assigns both children to one owner. That owner's capacity change would be three. If each child belongs to a separate owner, each would receive $3/2$ under the same declared path-to-capacity policy. Both pass zero-genesis affordability in this example. Neither assignment is derived from the sum identity; the owner rule is a separate input that the actual programme still must adopt.

The external audit is four. With total capacity three and endpoint potential one, $V + \sum B - J = 1 + 3 - 4 = 0$. This verifies the conditional aggregate account. It does not verify the provenance of the owner declaration, the sensor readings or the random selection that supposedly chose the group.

62.4 A list of claims the record cannot support

It cannot show that the action was sampled uniformly, because no random stream or menu record has been supplied. It cannot show that the process will keep recovering, because only one fixed event is described. It cannot show fairness, because arithmetic closure does not define fairness. It cannot show causal entitlement to the child amounts, because the path decomposition is not an intervention study.

It also cannot establish that a runnable Gaussian engine exists. A future implementation must faithfully create, execute and preserve these objects under its accepted profile, not merely print the same numbers for one example.

62.5 The claim ledger

Claim	Current status in this edition
Finite Gaussian expansion	Derived exactly under fixed quadratic geometry.
Local affected-factor cancellation	General conditional identity.
Common-path sum	Exact decomposition; no ownership conclusion.
Onsager-type dissipation	Conditional on the specified flux law and domain.
Capacity invariant	Conditional on the proposed signed owner/update policy.
Gaussian runtime acceptance	Not established by this book.
Gaussian scientific outcomes	No new study executed.
Historical Gate 1D-C completion	Recorded by its separate committed manifest.

62.6 The next separately authorized stages

First resolve the load-bearing event, owner/path, menu/locality, forcing coupling and numerical-policy contracts. Then implement reviewed generic reuse and isolated Gaussian evaluation. Validate pure foundation properties before separately authorized transition fixtures. Only after that should a scientific packet freeze the study configuration, comparators and interpretation rules. Execution requires packet-specific permission.

This sequence does not require the author to decide every routine coding detail. It identifies the decisions that change the scientific object. Routine implementation can proceed within an accepted contract; it must stop where a missing choice would create a different process.

Stage A and Stage B are separate study questions, not automatic continuation flags. A single-shock outcome, including nonrecovery, needs a disposition before a continuously forced regime is treated as the next approved study. Neither stage is automatically SD-01, and no AWS launch or cost is authorized by a manuscript explaining the programme.

62.7 What this revision accomplished

The first two parts now have a consistent active Gaussian vocabulary and a visible connection to the original force–mobility–flux mathematics. The

historical record remains identifiable rather than being rewritten as new evidence. The derivations and worked arithmetic can be reviewed now while implementation and empirical claims remain at their actual status.

That is not a delay in doing science. It is the distinction that lets the next experiment answer a clear question. A model can be implemented, tested and found wanting without destroying the exact identities it was built to respect. Conversely, correct identities do not guarantee the model deserves institutional adoption. Both possibilities must remain open.

Sources and authority map

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LOCAL_GAUSSIAN_EBU_PROGRAMME_RECONCILIATION.md, inspected at 05f3cb0. Sections 3–8 supply the prospective domain and conditional accounting identities; Section 10 retains the open decisions. Not an executed study.
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V3.0_LOCAL_EBU_FOUNDATION_DRAFT.md. Sections 6–8: exact finite quotation, cost categories, local cancellation and scoped sequential identities. Historical D0/P1C assumptions remain attached to their statements.
- [5] Energy Balance Project. *The Sequential–Parallel Bridge of EBU*, analytical checkpoint v0.2. SEQUENTIAL_PARALLEL_BRIDGE.md, Sections 3–7. Joint state changes, telescoping and comparator-relative interaction. Its operational status passages are historical, not a current Gate 1D-C completion record.
- [6] Energy Balance Project. *Local Gaussian Programme: Framework Decision*. LOCAL_GAUSSIAN_EBU_FRAMEWORK_DECISION.md.
Source-locked F2 reuse decision for planning, with separate event, ownership, capacity and locality dependencies. Does not authorize scientific runtime.
- [7] Energy Balance Project. *Local Gaussian EBU: Deferred Implementation Roadmap*. LOCAL_GAUSSIAN_EBU_IMPLEMENTATION_ROADMAP.md.
Implementation, transition fixtures, preregistration and execution are separate stages. Historical book-stage wording is superseded only by the author’s later book-revision instruction.

- [8] Energy Balance Project. *Gate 1D-C Finalized Execution Manifest*, `results/v3.0/gate1dc/MANIFEST.md`; accompanying registered protocol, JSON plan, summary and stdout. Historical finalized evidence: 30 runs and 6,000 trace rows. Execution identity `8793bda`. No Gaussian inference is made.
- [9] Konrad Grzyb. *The Energy Balance Project: Unified Explanatory Edition*, Parts II and III. Author-supplied PDFs, 160 and 153 pages. Content/style baselines and sources of the explicitly marked historical readings integrated by subject. Original artifacts remain unchanged.
- [10] Konrad Grzyb. *Physical Intuition and the EBU Idea*, revised Gaussian explanatory Part I. Author-review edition. The shared Ridge–Vale examples and typographic language continue here; examples are mathematical teaching constructions, not experimental evidence.

How to verify a source claim

The companion source manifest records full file hashes, the checkout identity, the original PDF identities and the exact original pages retained in each historical reading. The outline disposition records subjects rewritten rather than claiming every original sentence survives. All new derivations state their assumptions in the chapter itself. A citation identifies a source; it does not transform a prospective design into an accepted theorem or experiment.

The eight-part series

The series now has eight parts. Its first two form a continuous course: understand the physical account, derive it carefully, inspect how it can be implemented, and learn how evidence limits a claim. Later volumes deepen specific research questions; their position in the list is not execution permission or a guarantee of a positive result.

New	Responsibility	Former
I	Physical intuition, motivation and the EBU idea	I
II	Mathematics, implementation and evidence	II+III
III	Measurement, evidence and falsification	IV
IV	Homeostasis, recovery and stochastic dynamics	V
V	Multiple actions, interaction and local structure	VI
VI	Across distance	VII
VII	Coordination, memory and system dynamics	VIII
VIII	Institutions and an action-accounted economy	IX

Part III’s methods and Part V’s simultaneous foundations support the stochastic questions in Part IV. Part V supports the route and boundary work in Part VI. Parts IV–VI inform Part VII’s explicit coordination policies. Part VIII depends on surviving scientific and social evidence, including adverse results, and on normative arguments that no accounting identity can supply.

The introductory group treatment in this Part II is sufficient to explain exact finite accounting and its boundaries. The later Part V has a different job: deeper interaction diagnostics, admissible subset structures, topology, canonical motifs and certified compression. Likewise, this volume’s evidence chapters teach how to read the existing record; Part III develops measurement and falsification methods in greater depth. These are deliberate levels of treatment, not reasons to repeat the same chapter in two places.

Historical references keep the numbering printed in their original edition. The renumbering changes the forward editorial map only. It neither rewrites a theorem citation nor changes an experimental programme’s stage identifiers.